# *Virtual University Of Pakistan*

*Subject Bt301*

*Introduction of Biotechnology*

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Create By

*VU Bio Mates Team*

https://www.facebook.com/groups/vubiomates/

## 01 Introduction to Biotechnology

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.



**Traditional biotechnology** refers to the conventional techniques that have been used for many centuries to produce beer, wine, cheese etc.

**Modern Biotechnology** embraces all methods of genetic modification by recombinant DNA & cell fusion techniques together with the modern Developments of traditional biotechnological processes.




### White Biotechnology

Development of processes and microorganisms for Industrial processes.

### Key concept

Scale of operation Low volume high price

### White Biotechnology

Example

Enzyme production image1






### Red Biotechnology

It is concerned with the discovery and development of innovative drugs and treatments. Image2

### Green Biotechnology

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones.

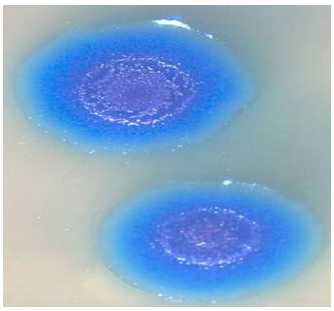
Example

Bt corn from Syngenta

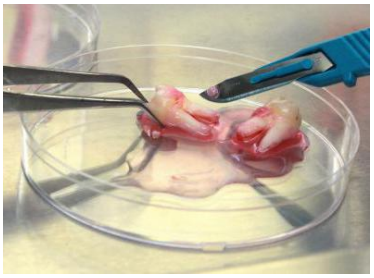
Bt cotton from Monsanto

## 02 Modern Biotechnology (Principles & Applications)

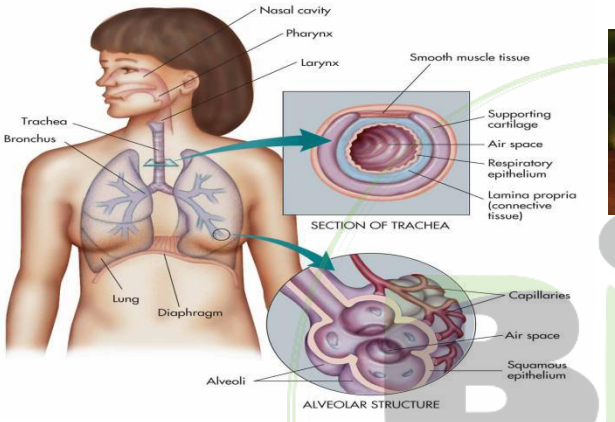
### Biotechnology in the 21st Century



Growing Teeth



Claudia's Trachea






Growing Blood Vessels

Cicken with an extra leg



Growing Meat





## 03 Roots of Biotechnology

6000 BC

Sumerians & Babylonians used anaerobic respiration of yeast to make beer.

4000 BC

Egyptians used aerobic respiration of yeast to leaven or rise bread and used mold to flavour cheese.

1350 BC

1350 BC

**Pregnancy test**



Barley grows – male child
Wheat grows – female child

400 BC
Use of salts to preserve food specially meat.



1676 AD
Antonie Van
Leeuwenhoek
Observation of microscopic single celled organisms



1828 AD
Synthesis of Urea
Fredrick Wohler
Pioneer in inorganic chemistry for synthesizing urea. This discovery became refutation of vitalism, the hypothesis that living things are alive because of some special "vital force"



1834 - 1839 AD
Cell Theory
Theodore Schwann
"All living things are composed of cells and cell products"



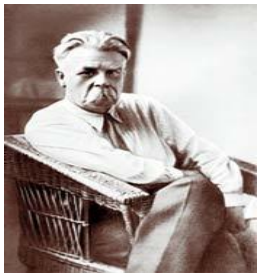

1857 AD
Rudolf Virchow
Omnis cellula e cellula
"Every cell arise from another



cell"

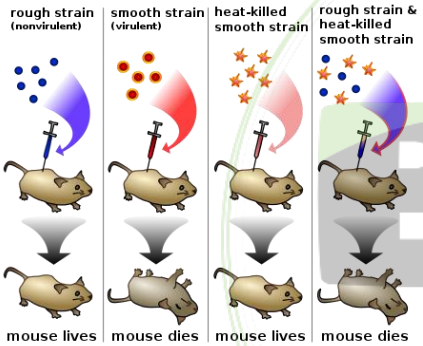
03 Discovery of DNA
1869
Friedrich Miescher
discovered nuclein
1878

Albrecht Kossel
Isolated non-protein component of nuclein & discovered nucleic acids & isolated 5 bases: A, T, G, C & U



1927
Nikolai Konstantinovich Koltsov
Inherited traits would be inherited via a "giant hereditary molecule" which would be made up of "two mirror strands that would replicate in a semi-conservative fashion using each strand as a template"

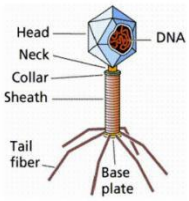
1928Frederick Griffith Discovered transformation, proof that DNA contains genetic information, different strains of streptococcus pneumoniae, Type 3S and Type 2R




1944
Avery, MacLeod & McCarty Experiment

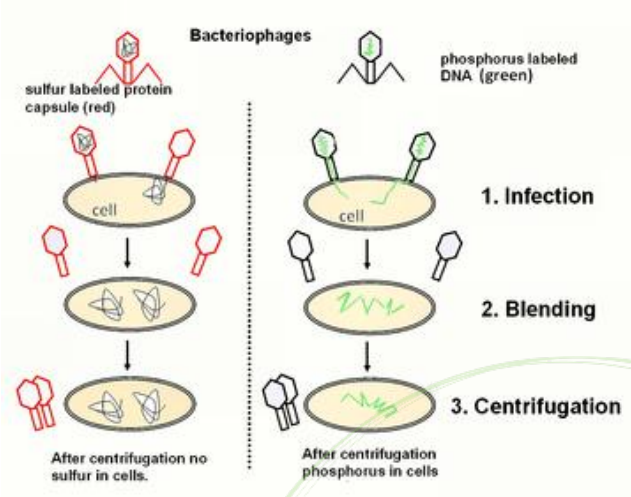


1952
Hershey-Chase



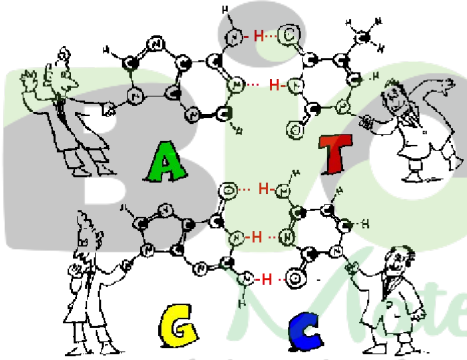

Experiment

1952
Hershey-Chase Experiment



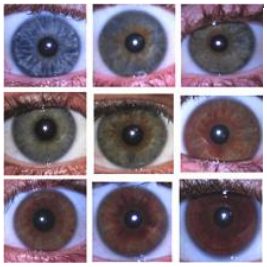

Watson & Crick 1953






Watson & Crick 1953

## 06 Dawn of Biotechnology





Eye colour curly hair toung roller

1973

Boyer brought in restriction enzymes

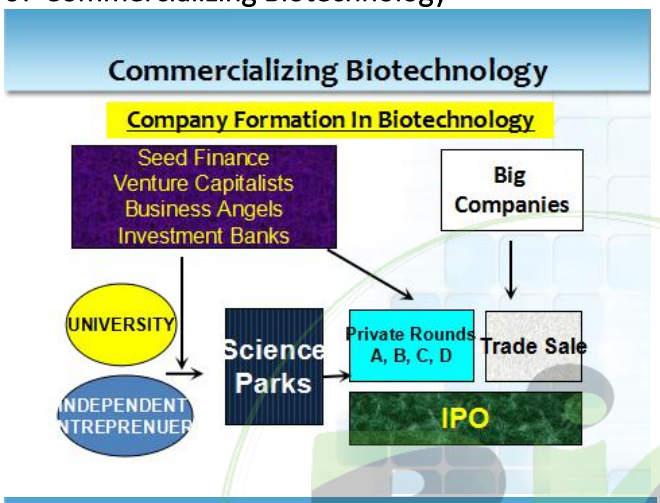
Cohen contributed with plasmid (pSC101)

African Clawed Frog gene into a bacterium

**Negative Perception**

"Genetic Engineering can give rise to something which can crawl out of the laboratory such as Frankenstein"

07 Commercializing Biotechnology




## Commercializing Biotechnology

First generation of Biotech Companies

| Company | Date of IPO | Amount Raised |
|---|---|---|
| Genentech | 10/80 | $35M |
| Cetus | 03/81 | $107M |
| Genetic Systems | 04/81 | $6M |
| Ribi Immunochem | 05/81 | $1.8M |
| Genome Therapeutics | O5/82 | $12.9M |
| Centocor | 12/82 | $21M |
| Scios | 01/83 | $12M |

## Commercializing Biotechnology

First generation of Biotech Companies

| Company | Date of IPO | Amount Raised |
|---|---|---|
| Biotechnology General | 09/83 | $8.9M |
| Immunex | 03/83 | $16.5M |
| Amgen | 06/83 | $42.3M |
| Biogen | 06/83 | $57.5M |
| Chiron | 08/83 | $17M |
| Immunomedics | 11/83 | $2.5M |
| Repligen | 04/86 | $17.5M |

08 Commercializing Biotechnology

Roche acquired Genentech for $46.8 billion

Current employees = 12,895

19 Breakthrough Biopharmaceutical Products



AMGEN

Revenue $20 billion
Total Assets $69 billion
net Income $ 5 billion
Current employees = 17,900
11 Breakthrough Biopharmace

Biogen




Revenue $9.7 billion
Total Assets $15 billion
Net Income $ 3 billion
Current employees = 7,550
10 Breakthrough Biopharmaceutical Products

genzyme




Net Income $ 421 million
Current employees = 12,000
16 innovative Biopharmaceutical Products

## Gaucher's Disease

Genetic disease in which fatty substances accumulate in cells and certain organs

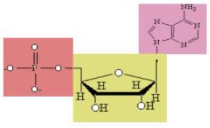
## Molecular Reason:

Hereditary deficiency of the enzyme glucocerebrosidase
Symptoms:
Bruising, Fatigue, Anemia, low blood platelets, enlargement of the liver and spleen

Prior to this, they had to collect 22,000 placentas to treat Gaucher's patients

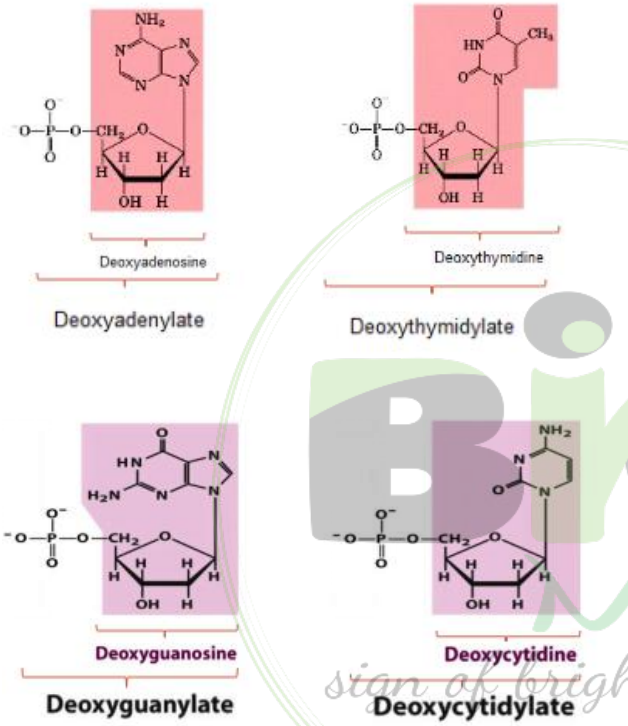
## 11 Nucleoside & Nucleotid

A molecule containing all these three components is called a nucleotide.
While a molecule without the phosphate group is called a nucleoside



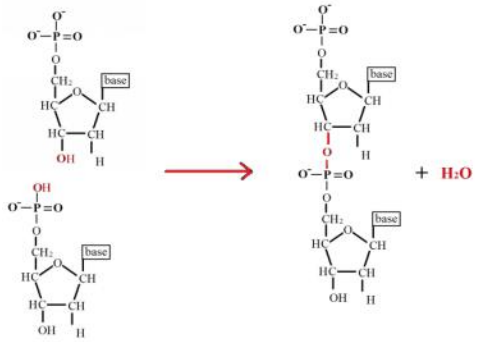
Nucleotide = Nucleoside + Phophoric acid
Nucleoside = Nucleotide – Phoshoric acid

## 12 Types of Deoxyribonucleotides




## 13 How do Deoxyribonucleotides Join?

The successive nucleotides of DNA are joined together through phospho-diester linkages.




## 14 Structure of DNA

### Work of Chargaff (Late 1940s)

The discovery of the structure of DNA is one of the greatest events in the history of science.

Erwin Chargaff and his colleagues provided a most important clue to the structure of DNA.

The work of Chargaff led him to following conclusions, also called "Chargaff Rules":-

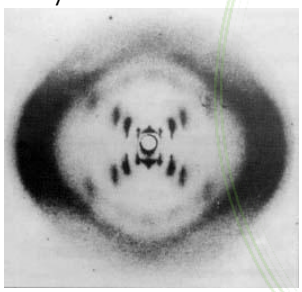
1.Base composition of DNA varies from one species to another.

2. The DNA isolated from different tissues of the same species have the same base composition.
3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.

4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C).
5. It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

## 15 Work of Franklin & Wilkins (1950s)

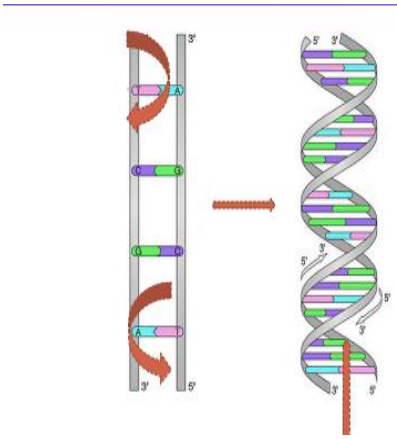
Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibers.
They showed that DNA produces a characteristic x-ray diffraction pattern.



From this pattern, they made two important findings:-
1. DNA molecules are helical.
2. The helices have two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å.

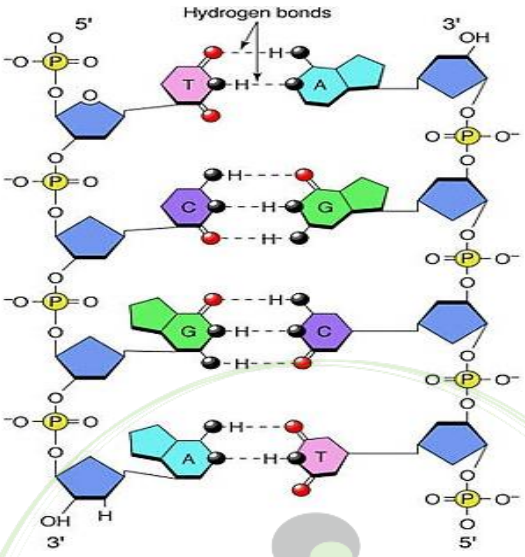
## 16 Work of Watson & Crick

James Watson and Francis Crick postulated a three dimensional model of DNA structure in 1953. The major features of this model are as follow:-



1. DNA consists of two helical polynucleotide strands which are wound around the same axis to form a right handed double helix.
2. The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water.

3. The nitrogenous bases of both strands are stacked inside the double helix lying perpendicular to the long axis of the helix.
4. The pairing and coiling of the two strands create a major groove and minor groove on the surface of the helix.

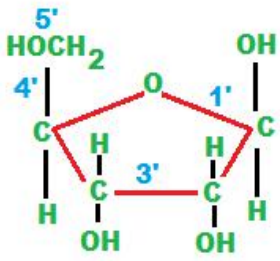


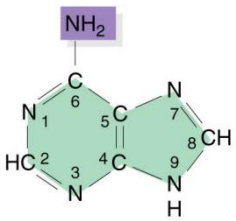

17 Chemical composition of RNA

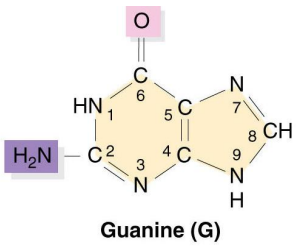
RNA (Ribonucleic acid) is a polymer of ribonucleotides.

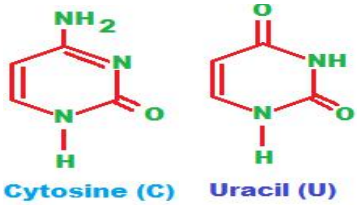
Each ribonucleotide is composed of three components:

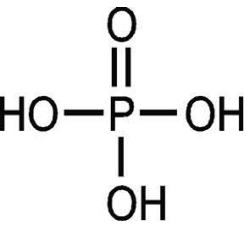
A ribose sugar
A Nitrogenous Base
A Phosphoric acid



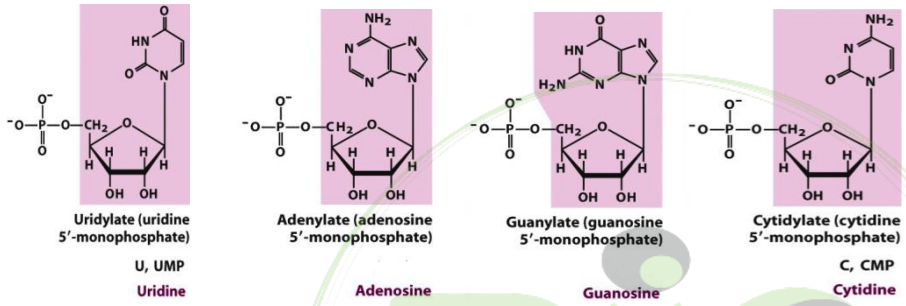



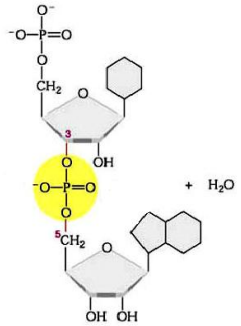
## 18 Types of Ribonucleotides

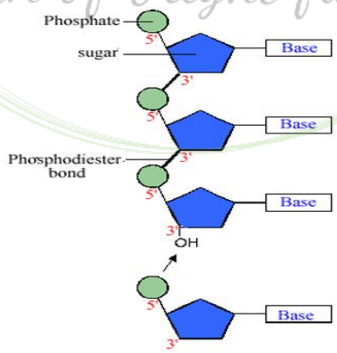
There are mainly four types of ribonucleotides depending upon the types of nitrogenous bases present in RNA.




How do Ribonucleotides Join?

a Poly-Ribonucleotide




## 19 Types of RNAs

There are mainly three types of Ribonucleic acids (RNAs) present in the cells of living organisms.

Messenger RNA (mRNA)
Transfer RNA (tRNA)
Ribosomal RNA (rRNA)

### Messenger RNA (mRNA)

It is the type of RNA that carries genetic information from DNA to the protein biosynthetic machinery of the ribosome.
It provides the templates that specify amino acid sequences in polypeptide chains.
The process of forming mRNA on a DNA template is known as transcription.
It may be monocistronic or polycistronic.

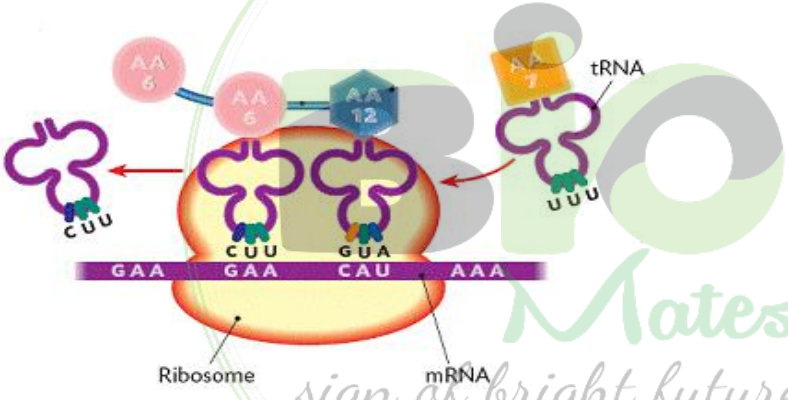
The length of mRNA molecules is variable and it depends on the length of gene.

### Transfer RNA (tRNA)

Transfer RNAs serve as adapter molecules in the process of protein synthesis.
They are covalently linked to an amino acid at one end.
They pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence.

### Ribosomal RNA (rRNA)

Ribosomal RNAs are components of ribosomes.
rRNA is a predominant material in the ribosomes constituting about 60% of its weight.
It has a number of functions to perform in the ribosomes.




### 20 Structures of RNAs

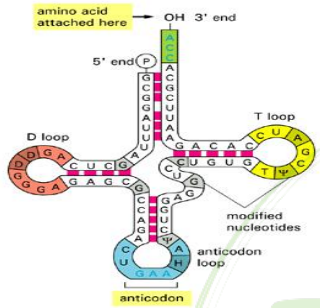
mRNA is always single stranded when it is formed from DNA.
But this single strand assumes a double helical conformation soon after its formation.
This confirmation is achieved mainly due to base stacking interactions.
Messenger RNA (mRNA)



Self-complementary sequences may occur in the RNA molecules which produce more complex structures.
So RNA can base-pair with complementary regions of either RNA or DNA.

RNA has no any regular secondary structure that serves as a reference point. The three-dimensional structures of many RNAs are complex and unique.

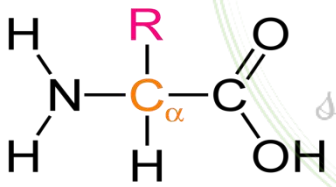
Breaks in the helix caused by mismatched or unmatched bases in one or both strands are common and result in bulges or internal loops.
Hairpin loops form between nearby self-complementary sequences.

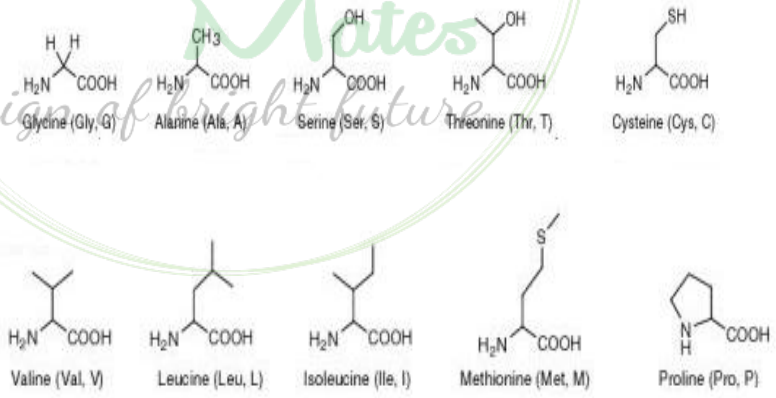



transfer RNA

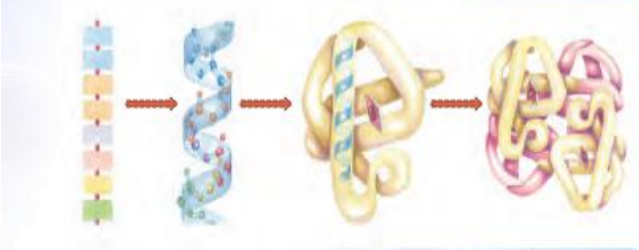
## 21 Chemical composition of proteins

Proteins are polymers of amino acids.
They range in size from small to very large.
All the proteins are made up of Twenty different types of amino acids. So these amino acids are called standard a



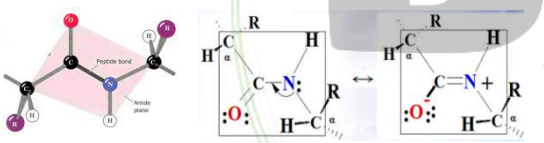




In a protein molecule, each amino acid residue is joined to its neighbour by a specific type of covalent bond which is called Peptide Bond. Amino acids can successively join to form dipeptides, tripeptides, tetrapeptides, oligo peptides and polypeptides.



## 22 Primary structure of proteins

Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain.
Each type of protein has a unique amino acid sequence.
Linus Pauling and Robert Corey carefully analyzed the peptide bond.
Their findings laid the foundation for our present understanding of protein structure.
They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine.
The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other.

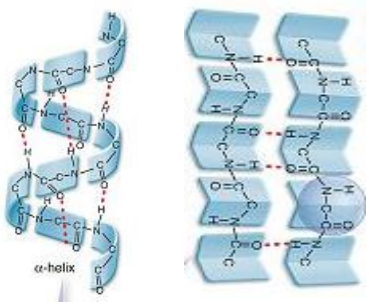


Pauling and Corey concluded that the peptide C - N bonds are unable to rotate freely because of their partial double-bond character.
Rotation is permitted about the N - C and the C - C bonds.
The bond angles resulting from rotations at C are labelled (phi) for the N - C bond and (psi) for the C - C bond.
In principle, and can have any value between +180 & -180

## 23 Secondary structure of proteins

Secondary structure of proteins refers to the local conformation of some part of a polypeptide.
A few types of secondary structures are particularly stable and occur widely in proteins.
The most prominent are:-

- α-helix
- β- conformations.

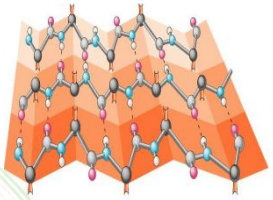



### 24 α- Helix

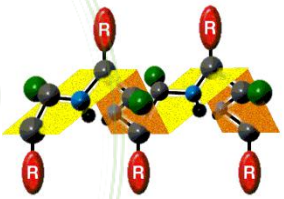
The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the α-helix.
The helical twist of the α-helix found in all proteins is right-handed.
The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid

residues) along the long axis.
The amino acid residues in an helix have conformations with psi = –45 to –50 and phi = – 60.
An helix makes optimal use of internal hydrogen bonds.
About one-fourth of all amino acid residues in polypeptides are found in α-helices while in some proteins it is the predominant structure.

## 25 β- Pleated Sheets

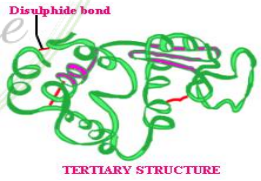


Pauling and Corey predicted a second type of secondary structure which they called β-sheets.
This is a more extended conformation of polypeptide chains.
The backbone of the polypeptide chain is extended into a zigzag structure.
The zigzag polypeptide chains are arranged side by side to form a structure resembling a series of pleats.




The R groups of adjacent amino acids protrude from the zigzag structure in opposite directions.
Hydrogen bonds are formed between adjacent segments of polypeptide chain.
The adjacent polypeptide chains in a sheet can be either parallel or antiparallel.

## 26 Tertiary Structure of Proteins

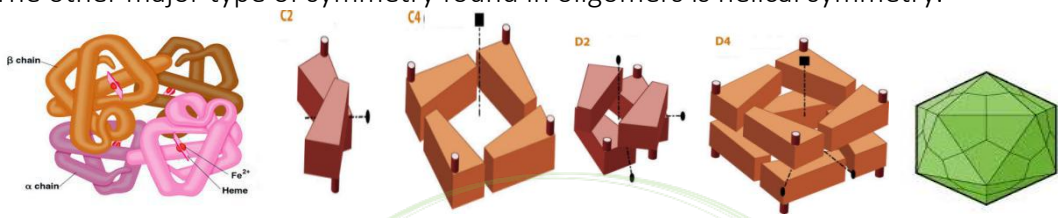



The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's tertiary structure.
It includes longer-range aspects of amino acid sequence.
Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein.
Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments.
Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi- or multilobal appearance.

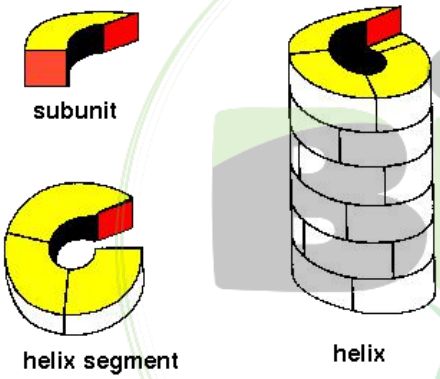
## 27 Quaternary Structure of Proteins

Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different.
The spatial arrangement of these subunits is known as a protein's quaternary structure.
A multi-subunit protein is also referred to as a multimer.
A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer.
Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.
Oligomers can have either rotational symmetry or helical symmetry.
Quaternary Structure of Proteins
There are several forms of rotational symmetry. The simplest is cyclic symmetry, involving rotation about a single axis.

A somewhat more complicated rotational symmetry is dihedral symmetry, in which a twofold rotational axis is present.
More complex rotational symmetries include icosahedral symmetry.
An icosahedron is a regular 12-cornered polyhedron having 20 triangular faces.
The other major type of symmetry found in oligomers is helical symmetry.

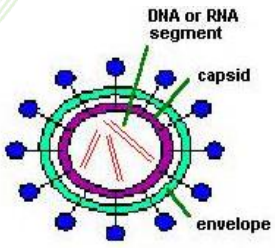



**Helical Symmetry**

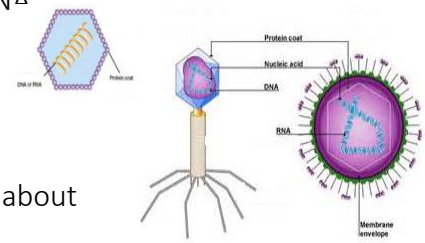



(adapted from Voet and Voet, 1990)

## 28 Genetic Materials in Viruses



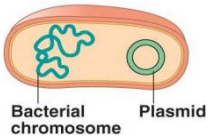

Viruses are exceptionally simple and extremely small microorganisms.
They have a very simple structural organization consisting of a molecule of nucleic acid and a protein coat.
Viruses can have either DNA or RNA as a genetic material but never both.
The nucleic acid of a virus can be single-stranded or double-stranded.
There are viruses with the familiar double-stranded DNA, with single-stranded DNA, with double-stranded RNA and with single-stranded RNA
The viral nucleic acid can be linear or circular.
Some viruses may contain coiled RNA.
In some viruses , the nucleic acid is in several separate segments.



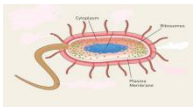

The percentage of nucleic acid in relation to protein is about 1% for the influenza virus and about 50% for some bacteriophages.
The total amount of nucleic acid varies from a few thousand nucleotides to as many as 250,000 nucleotides.
E. coli's chromosome consists of approx. 4 million nucleotide pairs.

## 29/30/31 Organization of Genetic Material in Bacteria

Bacteria typically have a single circular chromosome consisting of a single circular molecule of DNA with associated proteins.

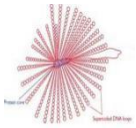


The bacterial chromosome is a very long (up to 1mm).
It is looped and folded and attached at one or several points to the plasma membrane.



Specific proteins interact with the bacterial DNA to form a highly condensed nucleoprotein complex called the nucleoid.
Bacterial chromatin can be released from the cell by gentle lysis of the cell.
Electron micrograph of the chromatin reveals that it consists of multiple loops which emerge from a central region of the chromatin.





Some of the loops are super-coiled while some are relaxed.
Relaxed loops are formed as a result of a nick introduced into super-coiled loops by a cellular DNase.

If a super-coiled DNA molecule receives a nick, the strain of under-winding is immediately removed, and all the super-coiling is lost.
Studies confirm that continued nuclease treatment increases number of relaxed loops.
The bacterial DNA is arranged in super-coiled loops that are fastened to a central protein matrix, so that each loop is topologically independent from all the others.
So a nick that causes one super-coiled loop to relax would have no effect on other super-coiled loops.
The super-coiled loops are dynamic structures which change during cell growth & division.
An E. coli chromosome is estimated to have about 400 super-coiled loops.
Each loop has an average length of about 10-20 kbp.
The DNA compaction in a bacterial cell is contributed by super-coiling of loops, macromolecular crowding and DNA-binding proteins.

## 32 Genetics and Genomics

Genetics and Genomics - Difference

### *Genetics*

Genetics is the study of heredity, or how the characteristics of living organisms are transmitted from one generation to the next generation through DNA.
Genetics involves the study of specific and limited numbers of genes that have a known function.
Genetics deals that how genes guide the body's development, cause disease or affect response to drugs.

### *Genomics*

Genomics, in contrast, is the study of the entirety of an organism's genes – called the genome.
Using high-performance computing and math techniques known as bioinformatics, genomics analyzes enormous amounts of DNA sequence data to find variations.
Genomics particularly deals with genetic variants that affect health, disease or drug response.

In humans that means searching through about 3 billion units of DNA across 23,000 genes.
Genomics is a much newer field than genetics and became possible only in the last couple of decades due to technical advances in DNA sequencing and computational biology.
Genetics: How the characteristics of living organisms are transmitted from one generation to the next generation.
Genomics: study of the entirety of an organism's genes – called the genome.

## 33 Genomics, Proteomics and Metabolomics

Genomics, Proteomics, Metabolomics

*Genome and Genomics*

The complete set of DNA found in each cell is known as the genome and study is called as genomics.

*Proteome and Proteomics*

The complete set of proteins found in each cell is known as the proteome.
Proteins concentration (and activity) may be different than gene expression due to post-translational modification

*Metabolomics*

The complete set of metabolites found in each cell is known as the metabolome.
Use of high-throughput mass spectrometry to analyze the metabolic components of cell.
Useful for determining the effects of the environment or gene transformation on the metabolism of the plants/animals.

*Conclusion*

Genomics, proteomics and metabolomics will give an integrated, wholistic view of the cell.
Can be used to monitor or modify organisms in a comprehensive way.
Bioinformatics - the key to understand the plethora of information and modeling the cell.

## 34 Why Sequence Genomes

To identify gene numbers, their locations on genomes, and to study their functions.
Genes regulation
DNA sequence
Genome organization
Chromosomal structure and organization

Noncoding DNA types, amount, distribution and functions.
Coordination of gene expression, protein synthesis, and post-translational events.
Interaction of proteins in complex molecular machines
Predicted vs experimentally determined gene function
Evolutionary conservation

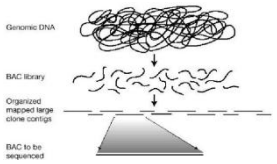
- Proteins structure and function.
- Proteomes (total protein content and function) in organisms.
- Correlation of SNPs with health and disease

Disease-susceptibility prediction based on gene sequence variation
Genes involved in complex traits and multigene diseases

*Novel Diagnostics*
Complex systems biology, developmental genetics.
To provide platform for microchips and DNA microarrays.
Gene expression - RNA
Complex systems biology, developmental genetics and genomics
Drug target discovery
Rational drug design
Molecular docking
Gene therapy
Stem cell therapy
*Understanding Metabolism*
To understand the metabolism of cells and tissues within different organisms.
*Understanding mechanism of diseases*
Inherited diseases
Infectious diseases
Pathogenic bacteria
Viruses
Conclusions
Better understanding of the genomes would be possible by sequencing of the genomes.

35 Major Techniques used for Genomes Characterization

*Genomes Characterization -Techniques*
*Major Techniques used for Genomes Characterization*
Cloning
Hybridization
PCR amplification
Sequencing
Computational tool
Genomes digested with restriction enzymes and inserted in vectors to produce genomic libraries.
BACs
YACs
Genomes Characterization Techniques - Hybridization
To arrange large contigs of genomes to produce genetic maps and physical maps of genomes. To arrange large contigs of genomes to produce genetic and physical maps




Genomes Characterization Techniques – PCR
Technique to amplify the DNA. Different variants of the technique used

Genomes Characterization Techniques – DNA Sequencing
One of the important technique used to characterize the genomes
To study structure and function of genomes.
DNAs are amplified and sequenced

Genomes Characterization Techniques – Computational Tools
Used to align the sequenced DNA to produce physical maps of the genomes.

Genomes Characterization Techniques – Conclusion
Different techniques used for genomes characterizations.

## 36 Genomes Analysis - Steps

Steps of Genomes Analysis

- Genome sequence assembled
- Identify repetitive sequences – mask out
- Gene prediction – train a model for each genome
- Look for EST and cDNA sequences
- Genome annotation
- Microarray analysis
- Metabolic pathways and regulation
- Protein 2D gel electrophoresis
- Functional genomics
- Gene location/gene map
- Self-comparison of proteome
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling

## 37 Benefits of Genomes Research

Genomes Research – Molecular Medicine
Improve diagnosis of disease
Detect genetic predispositions to disease (cancer, diabetes etc)
Create drugs based on molecular information
Use gene therapy and control systems as drugs

Genomes Research – Risk Assessment
Evaluate the health risks faced by individuals who may be exposed to radiations and to cancer causing chemicals and toxins.

Genomes Research – Bioarchaeology, Anthropology, Evolution and Human Migration
Study evolution through genetic variants in lineages.
Study of migration of different populations
Study mutations on the Y chromosome to trace lineage and migration of males
Evolution of mutations with ages of populations

Genomes Research – DNA Forensics
Identify potential suspects whose DNA may match evidence left at crime scenes.
Exonerate persons wrongly accused of crimes.
Identify catastrophe victims.
Establish paternity and other family relationships.
Identify endangered and protected species as an aid to wildlife officials.

Genomes Research
Detect bacteria and other organisms that may pollute air, water, soil and food.
Match organ donors with recipients in transplant programs
Determine pedigree for seed or livestocks

Genomes Research – Disease-resistant crops and disease-resistant animals
Grow disease/insect resistant and drought-resistant crops.
Breed healthier, more productive, disease-resistant farm animals.
Develop biopesticides.
Incorporate edible vaccines incorporated into food products.

Genomes Research – Microbial Genomics
Rapidly detect and treat pathogens (disease-causing microbes).
Develop new energy sources (biofuels)
monitor environment to detect pollutants.
Protect populations from biological and chemical warfare
Clean up toxic waste safely and efficiently

38 Size of Genomes
Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb).
Genomes of eukaryotes are usually larger

Size of Genomes
Most plants and animals have genomes greater than 100 Mb.
Humans have genome size of 3,000 Mb

Within each domain there is no systematic relationship between genome size and phenotype

| Organism | Haploid Genome Size (Mb) | Number of Genes | Genes per Mb |
|---|---|---|---|
| **Bacteria** | | | |
| *Haemophilus influenzae* | 1.8 | 1,700 | 940 |
| *Escherichia coli* | 4.6 | 4,400 | 950 |
| **Archaea** | | | |
| *Archaeoglobus fulgidus* | 2.2 | 2,500 | 1,130 |
| *Methanosarcina barkeri* | 4.8 | 3,600 | 750 |

| Organism | Haploid Genome Size (Mb) | Number of Genes | Genes per Mb |
|---|---|---|---|
| **Eukaryotes** | | | |
| *Saccharomyces cerevisiae* (yeast, a fungus) | 12 | 6,300 | 525 |
| *Caenorhabditis elegans* (nematode) | 100 | 20,100 | 200 |
| *Arabidopsis thaliana* (mustard family plant) | 120 | 27,000 | 225 |
| *Drosophila melanogaster* (fruit fly) | 165 | 13,700 | 83 |
| *Oryza sativa* (rice) | 430 | 42,000 | 98 |
| *Zea mays* (corn) | 2,300 | 32,000 | 14 |
| *Mus musculus* (house mouse) | 2,600 | 22,000 | 11 |
| *Ailuropoda melanoleuca* (giant panda) | 2,400 | 21,000 | 9 |
| *Homo sapiens* (human) | 3,000 | <21,000 | 7 |

Bio Mates sign of bright future

## Conclusion

Although most eukaryotes have large size of genomes.
Within each domain there is no systematic relationship between genome size and phenotype

## 39 viral Genomes

Genomes of Viruses
Viral genomes can be

- ssRNA
- dsRNA
- ssDNA
- dsDNA
- Linear
- Ciruclar

In Virus

- A viral genome is the genetic material of the virus.
- Also termed the viral chromosome.
- Viral genomes vary in size -few thousand to more than a hundred thousand nucleotides.

Virus with RNA

- Viruses with RNA Genomes
- Almost all plants viruses and some bacterial and animal viruses
- Genomes are rather small (a few thousands nucleotides)

Viruses with DNA Genomes

- Often a circular genome
- lambda = 48,502 bp

Replicative form of Viral Genomes

- All ssRNA viruses produce dsRNA molecules
- Many linear DNA molecules become circular

Viruses and Kingdoms

- Many plants viruses contain ssRNA genomes.
- Many fungal viruses contain dsRNA genomes.
- Many bacterial viruses contain dsDNA genomes.

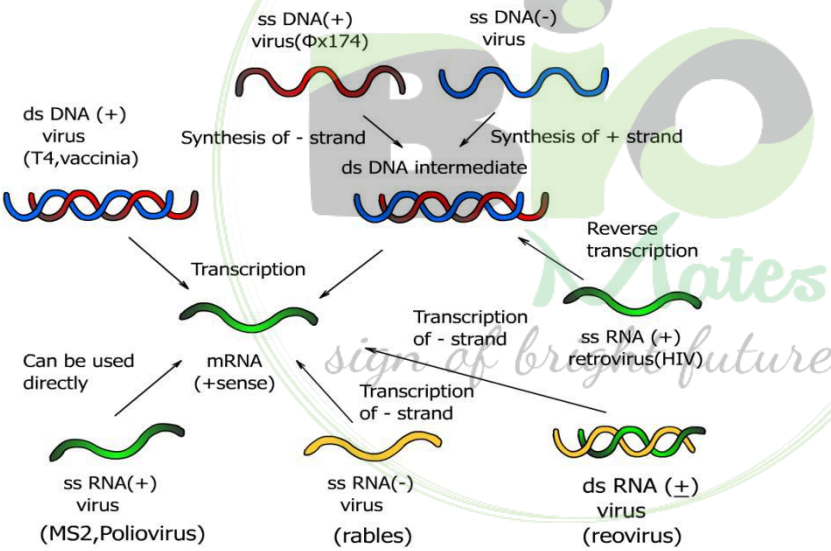
Genomes in Virions: The genomes of viruses can be composed of either DNA or RNA, and some use both as their genomic material at different stages in their life cycle. However, only one type of nucleic acid is found in the virion of any particular type of virus.

Virus and number of Gene

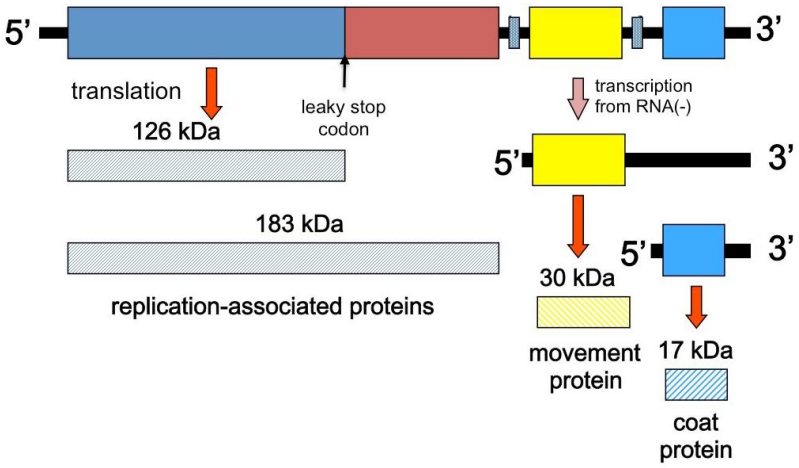
| Virus | Host | Type of Nucleic Acid | Number of Genes |
|---|---|---|---|
| Parvovirus | Mammals | ssDNA | 5 |
| Phage fd | *E. coli* | ssDNA | 10 |
| Lambda | *E. coli* | dsDNA | 36 |
| T4 | *E. coli* | dsDNA | >190 |
| Qβ | *E. coli* | ssRNA | 4 |
| TMV | Many plants | ssRNA | 6 |
| Influenza virus | Mammals | ssRNA | 12 |

| Virus | Genome structure | Genome size (kb) | Number of genes |
|---|---|---|---|
| Adenovirus | Double-stranded linear DNA | 36.0 | 30 |
| Hepatitis B | Partly double-stranded circular DNA | 3.2 | 4 |
| Influenza virus | Single-stranded segmented linear RNA | 22.0 | 12 |
| Parvovirus | Single-stranded linear DNA | 1.6 | 5 |
| Poliovirus | Single-stranded linear RNA | 7.6 | 8 |
| Reovirus | Double-stranded segmented linear RNA | 22.5 | 22 |
| Retroviruses | Single-stranded linear RNA | 6.0–9.0 | 3 |
| SV40 | Double-stranded circular DNA | 5.0 | 5 |
| Tobacco mosaic virus | Single-stranded linear RNA | 6.4 | 6 |
| Vaccinia virus | Double-stranded circular DNA | 240 | 240 |

## Genomes in Virions

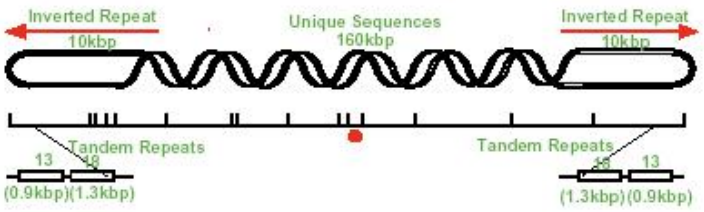



Genome of Tobacco Mosaic Virus
Single, 6400 nucleotides RNA, 3 Essential Genes

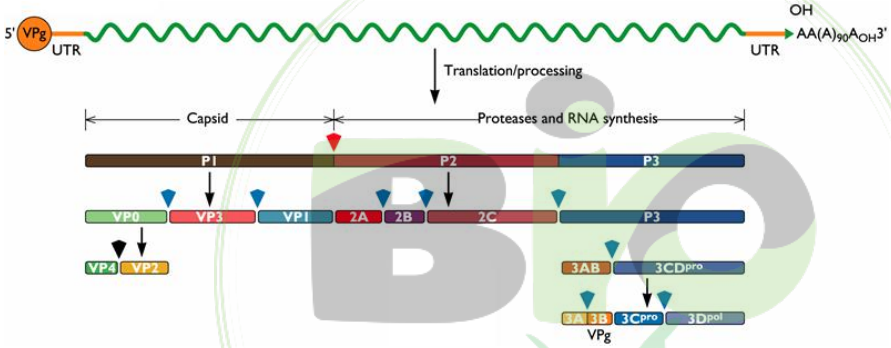



Genome of Poxvirus –A typical large dsDNA Virus

180 kb DNA, >100 Essential Genes




Genome of Polio Virus: Single-stranded positive-sense RNA genome that is about 7500 nucleotides long




## Genome of Pox Virus

Linear dsDNA 130-375 kbp; covalently closed termini.
Large hairpin structure at each terminus - up to 10 kb total at each end is repeat sequence.
Encode 150-300 proteins.
Coding regions are closely spaced, no introns.
Coding regions are on both strands of genome, and are not tightly clustered with respect to time of expression or function.

Viral genomes can be

- ssRNA
- dsRNA
- ssDNA
- dsDNA
- Linear
- Ciruclar

## 40 Bacterial Genomes

Small organisms carry high coding density (85-90%)
1 gene per 1000 bases in prokaryotes
Large variation in genome size between bacteria

Genomes of Bacteria – Large Variation
Tremblaya princeps 140kb, 121 coding

**Comparison of regulatory genes in bacterial genomes**

| Microorganism | # Genes in the Genome | # Regulatory Proteins | % of Total |
|---|---|---|---|
| *Pseudomonas aeruginosa* | 5570 | 468 | 8.4 |
| *Escherichia coli* | 4289 | 250 | 5.8 |
| *Bacillus subtilis* | 4100 | 217 | 5.3 |
| *Mycobacterium tuberculosis* | 3918 | 117 | 3.0 |
| *Helicobacter pylori* | 1566 | 18 | 1.1 |

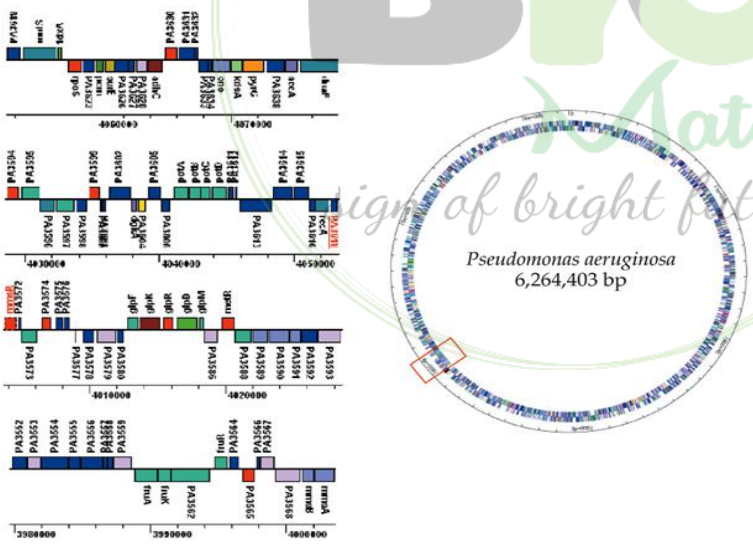
sequences
Sorangium cellulosum
14000kb
11599 coding sequences

| Organism | Genome Size (Mbp) | No. of ORFs (% coding) | | Unknown Function | | Unique ORFs | |
|---|---|---|---|---|---|---|---|
| *Methanococcus jannaschii* | 1.66 | 1,783 | (87%) | 1,076 | (62%) | 525 | (30%) |
| *M. tuberculosis* CSU#93 | 4.41 | 4,275 | (92%) | 1,521 | (39%) | 606 | (15%) |
| *M. genitalium* | 0.58 | 483 | (91%) | 173 | (37%) | 7 | (2%) |
| *M. pneumoniae* | 0.81 | 680 | (89%) | 248 | (37%) | 67 | (10%) |
| *N. meningitidis* MC58 | 2.24 | 2,155 | (83%) | 856 | (40%) | 517 | (24%) |
| *Pyrococcus horikoshii* OT3 | 1.74 | 1,994 | (91%) | 589 | (42%) | 453 | (22%) |
| *Rickettsia prowazekii* Madrid E | 1.11 | 878 | (75%) | 311 | (37%) | 209 | (25%) |
| *Synechocystis* sp. | 3.57 | 4,003 | (87%) | 2,384 | (75%) | 1,426 | (45%) |
| *T. maritma* MSB8 | 1.86 | 1,879 | (95%) | 863 | (46%) | 373 | (26%) |
| *T. pallidum* | 1.14 | 1,039 | (93%) | 461 | (44%) | 280 | (27%) |
| *Vibrio cholerae* El Tor N1696 | 4.03 | 3,890 | (88%) | 1,806 | (46%) | 934 | (24%) |
| | 50.60 | 52,462 | (89%) | 22.358 | (43%) | 12,161 | (23%) |

| Organism | Genome Size (Mbp) | No. of ORFs (% coding) | | Unknown Function | | Unique ORFs | |
|---|---|---|---|---|---|---|---|
| *Aeropyrum pernix* K1 | 1.67 | 1,885 | (89%) | | | | |
| *A. aeolicus* VF5 | 1.50 | 1,749 | (93%) | 663 | (44%) | 407 | (27%) |
| *A. fulgidus* | 2.18 | 2,437 | (92%) | 1,315 | (54%) | 641 | (26%) |
| *B. subtilis* | 4.20 | 4,779 | (87%) | 1,722 | (42%) | 1,053 | (26%) |
| *B. burgdorferi* | 1.44 | 1,738 | (88%) | 1,132 | (65%) | 682 | (39%) |
| *Chlamydia pneumoniae* AR39 | 1.23 | 1,134 | (90%) | 543 | (48%) | 262 | (23%) |
| *Chlamydia trachomatis* $MoP_n$ | 1.07 | 936 | (91%) | 353 | (38%) | 77 | (8%) |
| *C. trachomatis* serovar D | 1.04 | 928 | (92%) | 290 | (32%) | 255 | (29%) |
| *Deinococcus radiodurans* | 3.28 | 3,187 | (91%) | 1,715 | (54%) | 1,001 | (31%) |
| *E. coli* K-12-MG1655 | 4.60 | 5,295 | (88%) | 1,632 | (38%) | 1,114 | (26%) |
| *H. influenzae* | 1.83 | 1,738 | (88%) | 595 | (35%) | 237 | (14%) |
| *H. pylori* 26695 | 1.66 | 1,589 | (91%) | 744 | (45%) | 539 | (33%) |
| *Methanobacterium thermotautotrophicum* | 1.75 | 2,008 | (90%) | 1,010 | (54%) | 496 | (27%) |

Distribution of genes among selected bacterial genomes and their sizes

Genes in a portion of a bacterial genome




Bacterial Genomes - Conclusion
Small organisms carry high coding density.
Large variation in genome size between bacteria.

## 41 Yeast Genome

The nuclear genome consists of 16 chromosomes.
In addition, there is a mitochondrial genome and a plasmid, 2 micron circle.

The haploid yeast genome consists of ~ 12.1 Mb
Yeast genome was completely sequenced by 1996

Yeast Genome - Characteristics
Small and compact
Small intergenic sequences

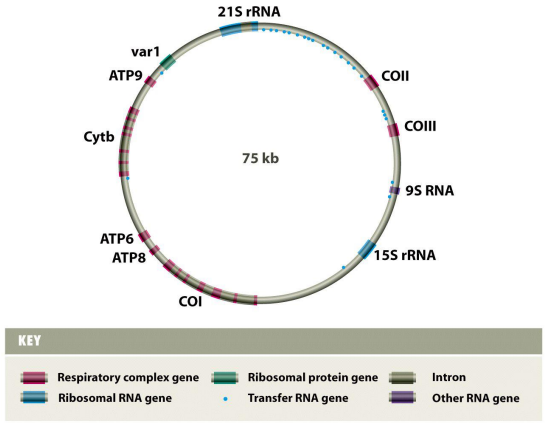
Few transposable elements
Few introns
Limited RNA interference
The yeast genome is predicted to contain about 6,200 genes
274 tRNA
287 introns
Small percentage of yeast genes have introns
The intergenic space between genes is only between 200bp - 1,000bp
Genome of Yeast Cell

| Characteristic | Chromosomes | Plasmid | Mitochondria |
|---|---|---|---|
| Relative amount (%) | 85 | 5 | 10 |
| Number of copies | 2 x 16 | 60-100 | ~50 (8-130) |
| Size (kbp) | ~ 12,100 | 6.318 | 70-76 |

The largest known regulatory sequences are spread over about 2,800bp
MUC1/FLO11

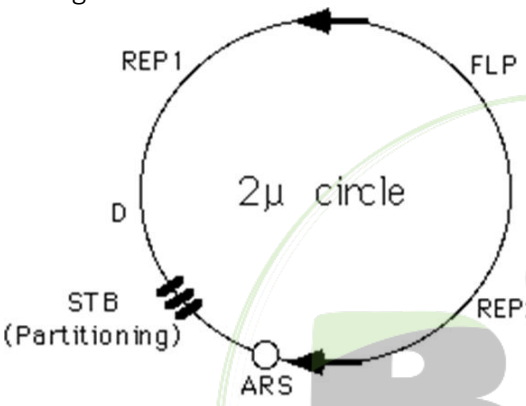
Yeast genes have names consisting of three letters and up to three numbers
GPD1, HSP12, PDC6
Usually they are meaningful
Yeast Genome – Genes Nomenclature
Wild type genes are written with capital letters in italics: TPS1, RHO1, CDC28
Recessive mutant genes are written with small letters in italics: tps1, rho1, cdc28

Yeast Genome – Genes Nomenclature
Three letters provides information about a function, mutant phenotype, or process related to that gene.
CDC - Cell Division Cycle ; ADE-ADEnine biosynthesis
Yeast – Mitochondrial DNA




Yeast – Plasmid DNA

The 2u circle is a 6.3 kb
50 to 100 copies per haploid genome of the yeast cells
ARS, the FLP gene, the three genes which encode proteins required for regulation of FLP expression (REP2, REP1, and D)
Set of small direct repeats (called "STB") required for partitioning into daughter cells during mitosis and meiosis.




Yeast nuclear genome has 16 chromosomes.
A mitochondrial genome.
A plasmid.

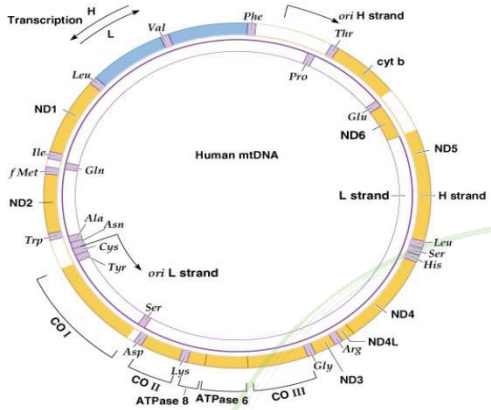
## 42 Mitochondrial Genome

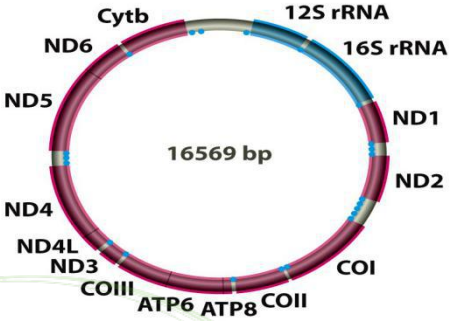
- Multiple identical circular chromosomes
- ~15-16 Kb in animals
- ~ 200 kb to 2,500 kb in plants
- Over 95% of mitochondrial proteins are encoded in the nuclear genome.
- Often A+T rich genomes.

Human Mitochondrial Genome

- Circular, double stranded, 16.6 kb
- The two strands are notably different in base composition, leading to one strand being heavy (H strand) and the other light (L strand).
- Both strands encode genes, although more are on the H strand.
- A short region (1121 bp), the D loop is a DNA triple helix: two overlapping copies of the H strand.
- The D loop is also the site where most of replication and transcription is controlled.
- Genes are tightly packed, with almost no non-coding DNA outside of D loop.
- Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants)
- 37 Genes
- 22 tRNAs
- 2 rRNAs
- 13 polypeptides
- tRNA: only 60 of the 64 codons code for amino acids.

Mitochondrial Genome Human

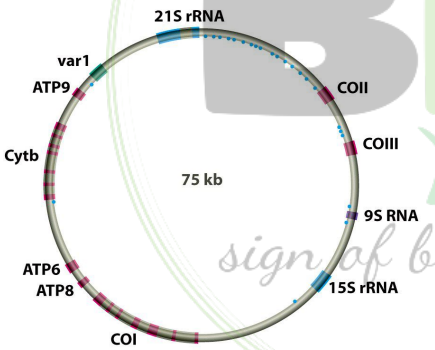






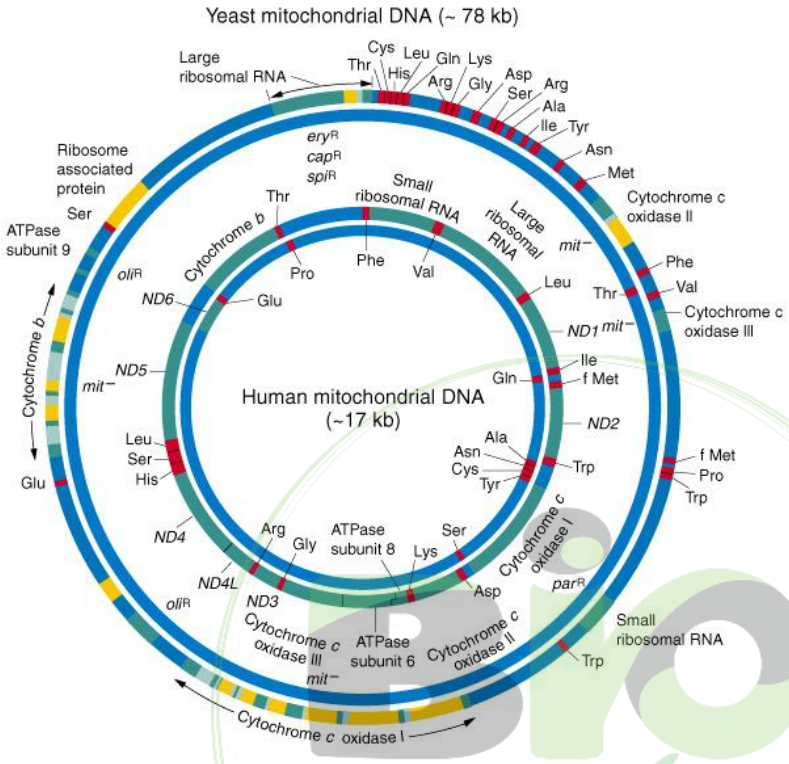
Mitochondrial Genome – Yeast







Mitochondrial Genome – Human and Yeast




| Species | Type of organism | Genome size (kb) |
|---|---|---|
| **Mitochondrial genomes** | | |
| *Plasmodium falciparum* | Protozoan (malaria parasite) | 6 |
| *Chlamydomonas reinhardtii* | Green alga | 16 |
| *Mus musculus* | Vertebrate (mouse) | 16 |
| *Homo sapiens* | Vertebrate (human) | 17 |
| *Metridium senile* | Invertebrate (sea anemone) | 17 |
| *Drosophila melanogaster* | Invertebrate (fruit fly) | 19 |
| *Chondrus crispus* | Red alga | 26 |
| *Aspergillus nidulans* | Ascomycete fungus | 33 |
| *Reclinomonas americana* | Protozoa | 69 |
| *Saccharomyces cerevisiae* | Yeast | 75 |
| *Suillus grisellus* | Basidiomycete fungus | 121 |
| *Brassica oleracea* | Flowering plant (cabbage) | 160 |
| *Arabidopsis thaliana* | Flowering plant (vetch) | 367 |
| *Zea mays* | Flowering plant (maize) | 570 |
| *Cucumis melo* | Flowering plant (melon) | 2500 |

| Feature | *Plasmodium falciparum* | *Chlamydomonas reinhardtii* | *Homo sapiens* | *Saccharomyces cerevisiae* | *Arabidopsis thaliana* | *Reclinomonas americana* |
|---|---|---|---|---|---|---|
| **Total number of genes** | 5 | 12 | 37 | 35 | 52 | 92 |
| **Types of genes** | | | | | | |
| Protein-coding genes | 3 | 7 | 13 | 8 | 27 | 62 |
| Respiratory complex | 3 | 7 | 13 | 7 | 17 | 24 |
| Ribosomal proteins | 0 | 0 | 0 | 1 | 7 | 27 |
| Transport proteins | 0 | 0 | 0 | 0 | 3 | 6 |
| RNA polymerase | 0 | 0 | 0 | 0 | 0 | 4 |
| Translation factor | 0 | 0 | 0 | 0 | 0 | 1 |
| Functional RNA genes | 2 | 5 | 24 | 27 | 25 | 30 |
| Ribosomal RNA genes | 2 | 2 | 2 | 2 | 3 | 3 |
| Transfer RNA genes | 0 | 3 | 22 | 24 | 22 | 26 |
| Other RNA genes | 0 | 0 | 0 | 1 | 0 | 1 |
| **Number of introns** | 0 | 1 | 0 | 8 | 23 | 1 |
| **Genome size (kb)** | 6 | 16 | 17 | 75 | 367 | 69 |

## Mitochondrial code

Second letter

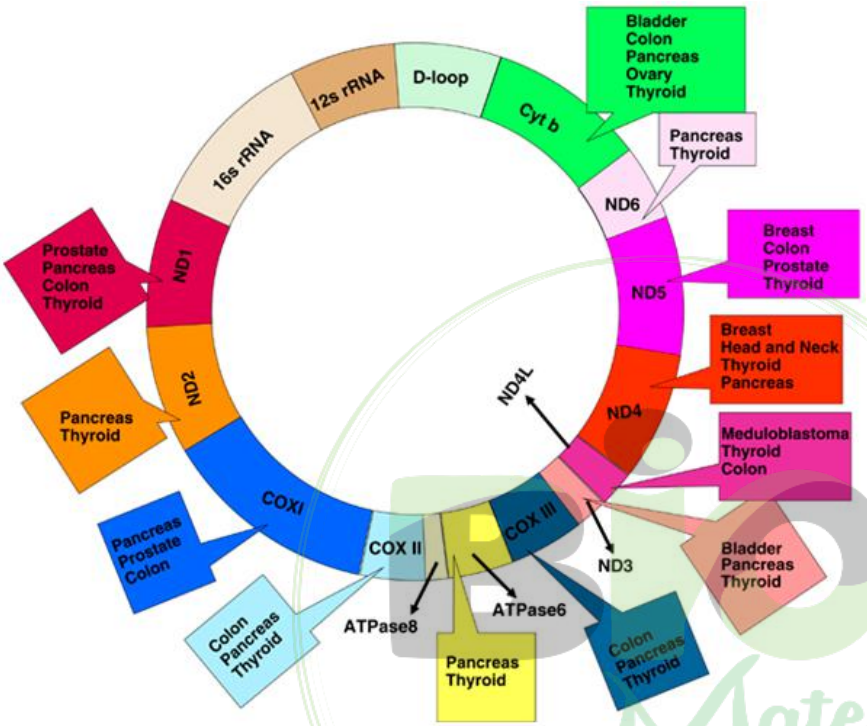
| First letter | U | C | A | G | Third letter |
|---|---|---|---|---|---|
| U | Phe | Ser | Tyr | Cys | U |
| | Phe | Ser | Tyr | Cys | C |
| | Leu | Ser | *Stop* | (*Stop*) Trp | A |
| | Leu | Ser | *Stop* | Trp | G |
| C | Leu | Pro | His | Arg | U |
| | Leu | Pro | His | Arg | C |
| | Leu | Pro | Gln | Arg | A |
| | Leu | Pro | Gln | Arg | G |
| A | Ile (Met) | Thr | Asn | Ser | U |
| | Ile | Thr | Asn | Ser | C |
| | (Ile) Met | Thr | Lys | (Arg) *Stop* | A |
| | Ile | Thr | Lys | (Arg) *Stop* | G |
| G | Val | Ala | Asp | Gly | U |
| | Val | Ala | Asp | Gly | C |
| | Val | Ala | Glu | Gly | A |
| | Val | Ala | Glu | Gly | G |

## Universal code

Second letter

| First letter | U | C | A | G | Third letter |
|---|---|---|---|---|---|
| U | UUU Phe | UCU Ser | UAU Tyr | UGU Cys | U |
| | UUC Phe | UCC Ser | UAC Tyr | UGC Cys | C |
| | UUA Leu | UCA Ser | UAA Stop | UGA Stop | A |
| | UUG Leu | UCG Ser | UAG Stop | UGG Trp | G |
| C | CUU Leu | CCU Pro | CAU His | CGU Arg | U |
| | CUC Leu | CCC Pro | CAC His | CGC Arg | C |
| | CUA Leu | CCA Pro | CAA Gln | CGA Arg | A |
| | CUG Leu | CCG Pro | CAG Gln | CGG Arg | G |
| A | AUU Ile | ACU Thr | AAU Asn | AGU Ser | U |
| | AUC Ile | ACC Thr | AAC Asn | AGC Ser | C |
| | AUA Ile | ACA Thr | AAA Lys | AGA Arg | A |
| | AUG Met | ACG Thr | AAG Lys | AGG Arg | G |
| G | GUU Val | GCU Ala | GAU Asp | GGU Gly | U |
| | GUC Val | GCC Ala | GAC Asp | GGC Gly | C |
| | GUA Val | GCA Ala | GAA Glu | GGA Gly | A |
| | GUG Val | GCG Ala | GAG Glu | GGG Gly | G |

Universal Code and Mitochondrial Code




Mitochondrial Genome – Tumors due to mutations

## Conclusion

Multiple identical circular chromosomes
15-16 Kb in animals
200 kb to 2,500 kb in plants

## 43 Chloroplast Genome

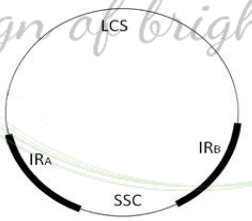
- Multiple circular molecules
- Size ranges from 70 kb to 2000 kb
- Land plants typically 120 – 170 kb
- ~ 70 kb – Epifagus
- ~2,000 kb – Acetabularia
- Similar to mtDNA
- Many chloroplast proteins are encoded in the nucleus (separate signal sequence)
- Double stranded DNA molecule
- Chloroplasts genomes are relatively larger
- Multiple copies of genome
- Large enough to code 50-100 proteins as well as rRNAs & tRNAs
- cpDNA regions includes Large Single Copy & Small Single-Copy (SSC) regions, and Inverted Repeats (IRA & IRB).

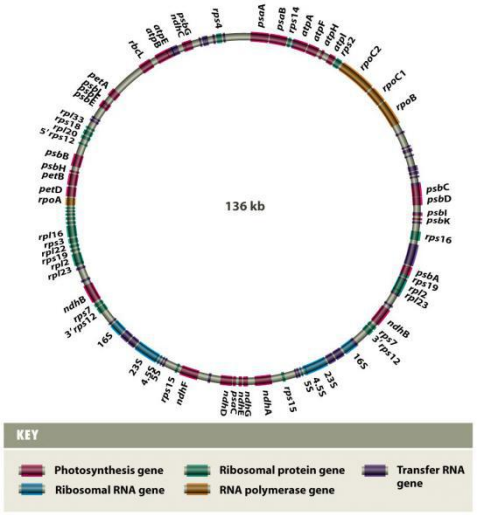
| Taxa | Genome size (In kb) | Inverse duplication (In kb) |
|---|---|---|
| **Angiospermae** | | |
| *Nicotiana tabacum* | 156 | 25 |
| *Spinacia oleracea* | 150 | 24 |
| *Pelargonium hortorum* | 217 | 76 |
| *Pisum sativum* | 120 | Not present |
| *Epifragus virginiana* | 70 | 22 |
| *Oryza sativa* | 134 | 21 |
| **Gymnospermae** | | |
| *Pinus* | 120 | Not present |
| *Ginkgo biloba* | 158 | 17 |
| **Pteridophyta** | | |
| *Osmunda cinnamomea* | 144 | 10 |
| **Bryophyta** | | |
| *Marchantia polymorpha* | 121 | 10 |
| **Chlorophyta** | | |
| *Codium fragile* | 85 | Not present |
| *Chlamydomonas reinhardtii* | 195 | 22 |
| *Chlamydomonas moewusii* | 292 | 41 |
| **Rhodophyta** | | |
| *Cyanophora paradoxa* | 127 | 10 |
| **Chromophyta** | | |
| *Dictyota dichotoma* | 123 | 5 |

| Species | Type of organism | Genome size (kb) |
|---|---|---|
| **Chloroplast genomes** | | |
| *Pisum sativum* | Flowering plant (pea) | 120 |
| *Marchantia polymorpha* | Liverwort | 121 |
| *Oryza sativa* | Flowering plant (rice) | 136 |
| *Nicotiana tabacum* | Flowering plant (tobacco) | 156 |
| *Chlamydomonas reinhardtii* | Green alga | 195 |

sign of bright future

- Large Single Copy Region
- Small Single Copy Region
- Inverted Repeats




Chloroplast Genome of Rice

## Functions of Chloroplast Genes

- Most cp genes fall into two functional groups:
- Genes involved in replication, transcription, translation
- Genes involved in photosynthesis

## Genes Nomenclature

- Based on bacterial naming system, which uses lower case letters, and a descriptive prefix, based on the probable function. If the gene product is part of a multi-subunit complex, a letter of the alphabet is used to denote different subunits.
- psa for genes of photosystem I (psaA, psaB, etc.)
- psb for genes of photosystem II (psbA, psbB, etc.)
- Non-mendelian inheritance
- Self replication

- Somatic segregation in plants
- Inherited independently of nuclear genes
- Conservative rate of nucleotide substitution enables to resolve plant phylogenetic relationships at deep levels of evolution.
- Multiple circular molecules
- Size ranges from 70 kb to 2000 kb

## 44 Eukaryotic Genomes

Located on several chromosomes
Relatively low gene density
Carry organelles genome in addition to nuclear genome
Contains repetitive sequences
SINEs (short interspersed elements)
LINEs (long interspersed elements)

## LINES and SINES

### LINES

1-5 kb
10-10,000 copies

### SINES

200-300 bp
100,000 copies

Highly repetitive

- Minisatellites
- Microsatellites
- Telomeres

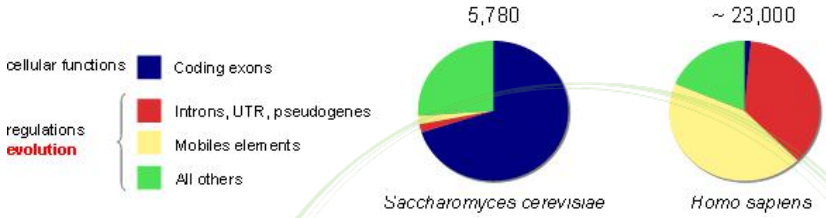
Minisatellites

- Repeats of 14-500 bp
- 1-5 kb long
- Scattered throughout the genome
- Repeats up to 13 bp

Telemeres

- Short repeats (6 bp)
- 250-1,000 at ends of chromosomes

**Chromosomes:** linear, centromeres, telomeres, origins of replication, replicons
**Protein-coding genes** and **spliceosomal introns**
**Genes for non coding RNAs:** rRNAs, tRNAs, snoRNAs, snRNAs, microRNAs
**Mobile genetic elements:** and their remnants
**Pseudogenes:** and processed pseudogenes
**Satellite DNAs:** micro-, minisatellites, repeated sequences
**Fragments of organellar DNAs:** NUMTs and NUPTs




| Prokaryotic | Eukaryotic |
|---|---|
| • Usually circular | • Complex with a large amount of protein to form chromatin |
| • Smaller | • Highly extended and tangled during interphase |
| • Found in the nucleoid region | • Found in the nucleus |
| • Less elaborately structured and folded | |

Conclusion

- Located on several chromosomes
- Relatively low gene density
- Chromosomes vary in length
- Carry organelles genome

Genomes Comparisons
Genomes vary in size
Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb)
Most plants and animals have genomes greater than 100 Mb; humans have 3,000 Mb

Genomes Comparisons
Genomes vary in genes numbers
Free-living bacteria and archaea have 1,500 to 7,500 genes
Fungi have about 5,000 genes and multicellular eukaryotes upto 40,000 genes

Number of genes is not correlated to genome size
Nematode
C. elegans has 100 Mb and 20,000 genes, while Drosophila has 165 Mb and 13,700 genes
Genomes vary in genes numbers
Vertebrate genomes can produce more than one polypeptide per gene because of alternative splicing of RNA transcripts
Humans and Mammals have low gene density
Humans and other mammals have the lowest gene density, or number of genes, in a given length of DNA. Multicellular eukaryotes have many introns within genes and noncoding DNA between genes
Multicellular eukaryotes have much noncoding DNA and multigene families
Most of eukaryotic genomes neither encodes proteins nor functional RNAs
Evidence indicates that noncoding DNA plays important roles in the cell

Genomes Comparisons
Human Genome: Distribution of coding and non-coding DNA
Significant similarity between genomes of "distant" species (Man – Yeast 23%)
Similarity increases for taxonomically close species.
Closely related species help us understand recent evolutionary events
Distantly related species help us understand ancient evolutionary events
Comparing Genomes: Bacteria, archaea, and eukaryotes diverged from each other between 2 and 4 billion years ago
Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions

## 45 Comparing distantly/closely related species

Comparing distantly related species

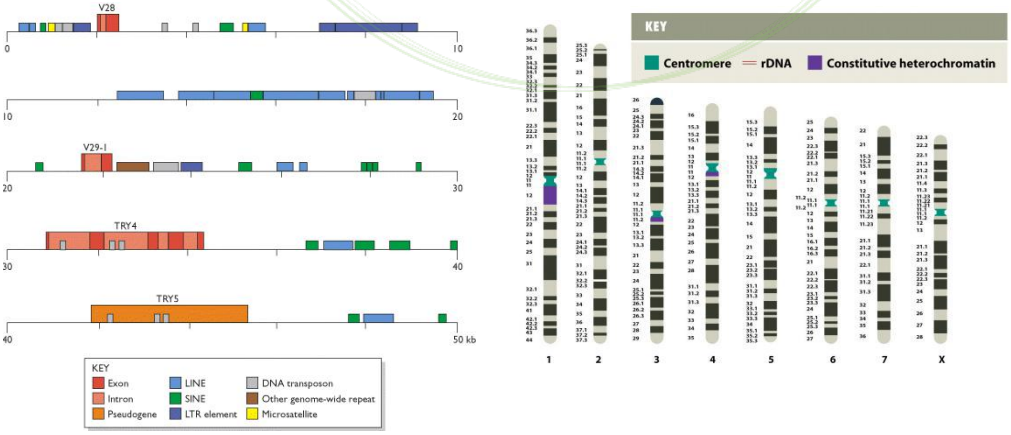
- Highly conserved genes have changed very little over time
- These help to clarify relationships among species that diverged from each other long ago
- Bacteria, archaea, and eukaryotes diverged from each other 2 and 4 billion years ago
- Highly conserved genes can be studied in one model organism.
- Genetic differences between closely related species can be correlated with phenotypic differences
- Genetic comparison of several mammals with non-mammals helps to identify what make mammals
- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions
- Several genes are evolving faster in humans than chimpanzees
- Genes involved in defense against malaria and tuberculosis and in regulation of brain size, genes code for transcription factors

- Humans and chimpanzees differ in the expression of the FOXP2 gene, whose product turns on genes involved in vocalization
- Differences in the FOXP2 gene may explain why humans but not chimpanzees communicate by speech

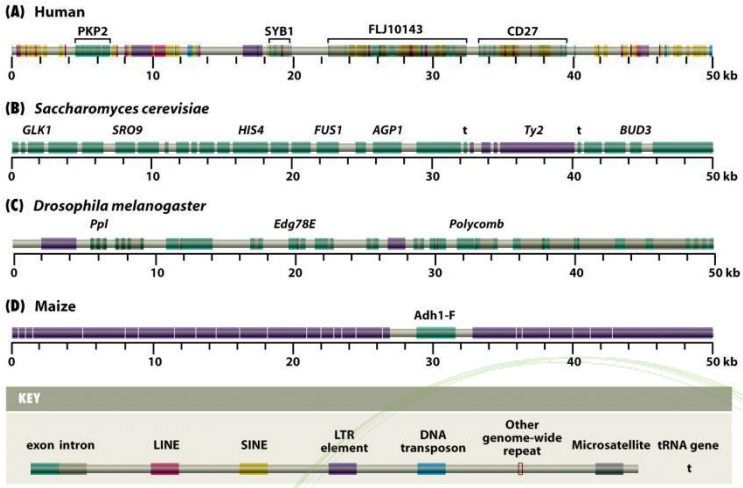
Conclusion
Highly conserved genes have changed very little over the time
These help to clarify relationships among species that diverged from each other long ago

## 47 Anatomy and Organization of Genomes

Genome Anatomy
Anatomy of different genomes differ from each other.
Eukaryotes and prokaryotes genomes differ very significantly.
Size of genomes - 1000 fold difference between eukaryotes and prokaryotes.
~ 30 fold between genomes of different eukaryotes.
In humans ~ 23,000
Bacterial genomes ~ 1,500 – 2,000 genes.
Genome Anatomy - Eukaryotes
Eukaryotic genomes are full of simple repeats, numerous types of transposable elements and other sequences.
Genome Anatomy - Prokaryotes
Prokaryotes have a few repeats and transposable elements and their genomes consist of mainly the genes.
Genome Anatomy: Segment of human chromosome 7




Genome Organization: Human, Yeast, Fruit Fly , Maize




Compactness of Genomes

Conclusion

- Anatomy of different genomes differ from each other.
- Eukaryotes and prokaryotes genomes differ very significantly.

## 48 Gene Anatomy

What is Gene

A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

Entire nucleic acid sequence necessary for the synthesis of a functional polypeptide (protein chain) or functional RNA

Genetic information is stored in DNA

Segments of DNA that encode proteins or other functional products are called genes.

Gene sequences are transcribed into messenger RNA (mRNA).

mRNA is translated into Proteins

mRNA intermediates are translated into proteins that perform most of the life functions.

Three components

Open Reading Frame: From start codon (ATG) to stop (TGA, TAA, TAG)

Upstream region with binding site. (e.g. TATA box, GC box, CAAT box).

Poly-a tail

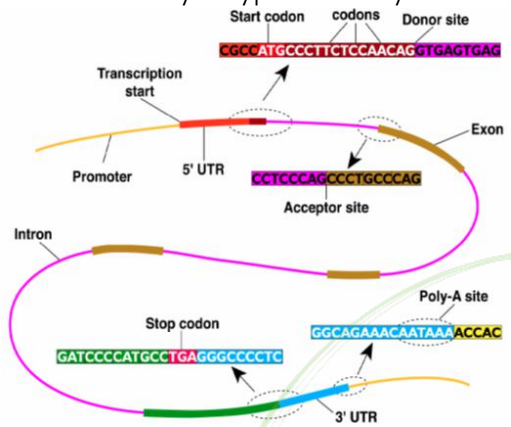
Gene Anatomy

Gene Anatomy – Typical Prokaryotic Gene



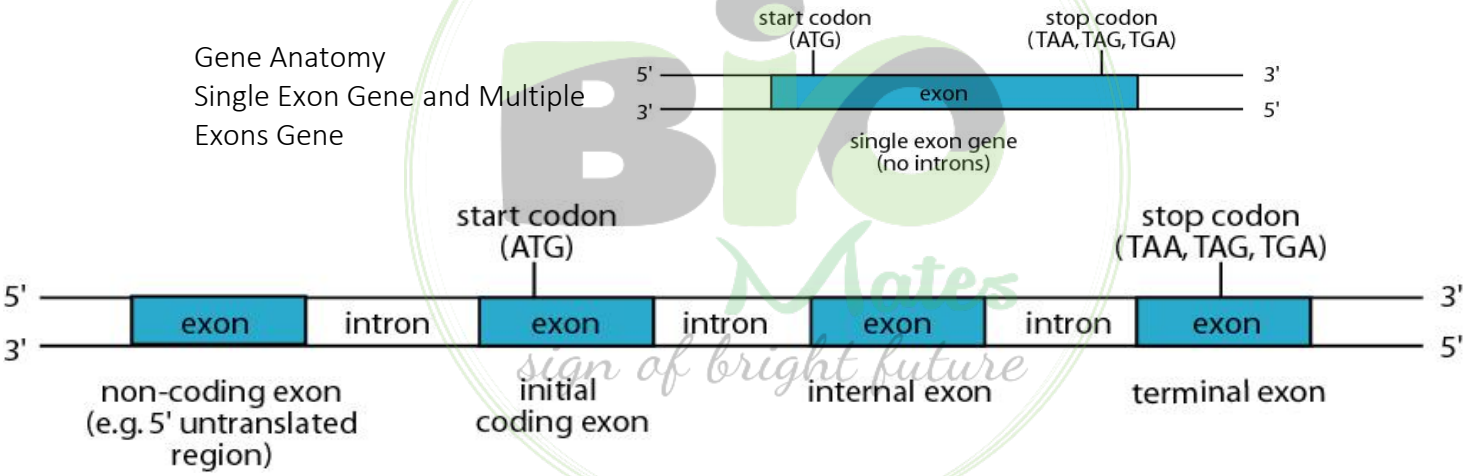
Gene Anatomy

Gene Anatomy – Typical Eukaryotic Gene




Gene Anatomy

Single Exon Gene and Multiple Exons Gene




A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

49 Prokaryotic Gene and Eukaryotic Gene

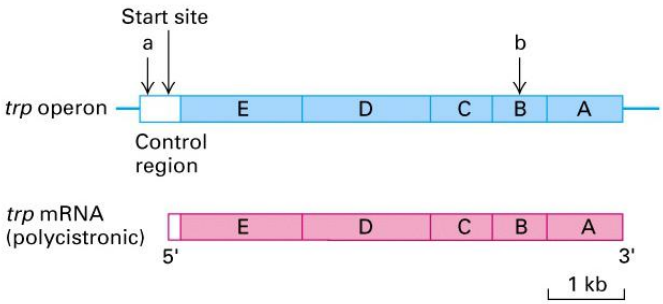
Prokaryotic Gene/Eukaryotic Gene

Bacterial Gene

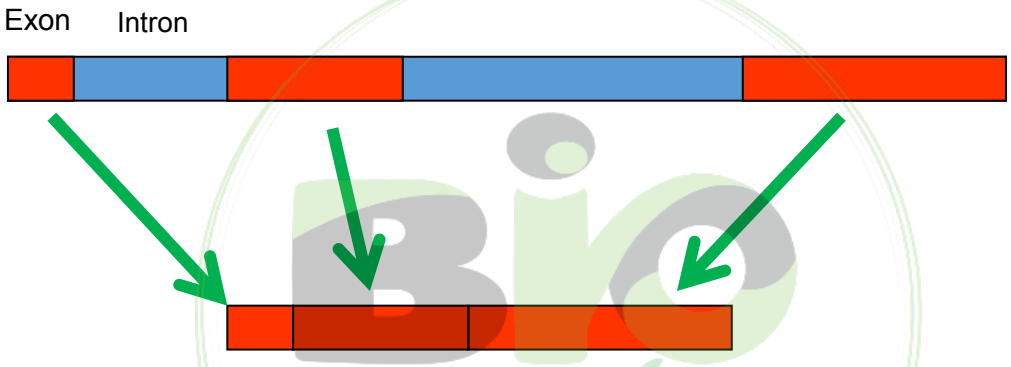
Most do not have introns

Many are organized in operons: contiguous genes, transcribed as a single polycistronic mRNA, that encode proteins with related functions

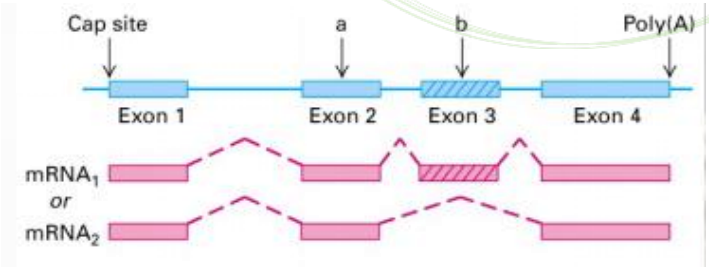
Polycistronic mRNA encodes several proteins



Bacterial Gene: Polycistronic mRNA encodes several proteins
Eukaryotic Gene: Exons and Introns
Introns: intervening sequences within a gene that are not translated into a protein sequence.
Exons: sequences within a gene that encode protein sequence.
Eukaryotic Gene
Splicing: Removal of introns from the mRNA molecule
Prokaryotic Gene/Eukaryotic Gene
Splicing: Removal of introns from the mRNA molecule




Prokaryotic Gene/Eukaryotic Gene
Organize expression of genes' (function calls)
Promoter region (binding site), usually near coding region
Binding can block (inhibit) expression
Alternative Splicing in Eukaryotic Genes




Most have introns
Produce monocistronic mRNA
Large in size

Eukaryotic Gene: Alternative Splicing

Eukaryotic Gene
Computational challenges
Identify binding sites
Correlate sequence to expression
Conclusion
Most of prokaryotic genes are without introns and in the are polycistronic.
Eukaryotic genes have introns and alternative splicing.

## 50 Types of Eukaryotic DNA

Protein coding genes

- Tandemly repeated genes
- Repeated DNA
- Unclassified spacer DNA
- Protein coding genes can also be in the form of:
- Duplicated and diverged genes
- Functional gene families and non-functional pseudo-genes
- Tendemly repeated genes encoding rRNA, 5sRNA, tRNA and histones.
- Repetitive DNA
- Simple sequence DNA
- Moderately repeated DNA (mobile DNA elements)
- Tranposones
- Retrotransposons

- Types of Eukaryotic DNA
- Repetitive DNA
- Long interspersed elements
- Short interspersed elements
- Unclassified spacer DNA

Protein-coding genes
Solitary genes
Duplicated and diverged genes (functional gene families and nonfunctional pseudogenes)
Tandemly repeated genes encoding rRNA, 5S rRNA tRNA, and histones
Repetitious DNA
Simple-sequence DNA
Moderately repeated DNA (mobile DNA element
Transposons
Viral retrotransposons
Long interspersed elements (LINES; nonviral retrotransposons)
Short interspersed elements (SINES; nonviral retrotransposons)
Unclassified spacer DNA

| Class | Length | Copy Number in Human Genome | Fraction of Human Genome, % |
|---|---|---|---|
| Protein-coding genes | | | |
| Solitary genes | Variable | 1 | ≈15* (0.8)† |
| Duplicated or diverged genes in gene families | Variable | 2—≈1000 | ≈15* (0.8)† |
| Tandemly repeated genes encoding rRNAs, tRNAs, snRNAs, and histones | Variable | 20–300 | 0.3 |
| Repetitious DNA | | | |
| Simple-sequence DNA | 1–500 bp | Variable | 3 |
| Interspersed repeats | | | |
| DNA transposons | 2–3 kb | 300,000 | 3 |
| LTR retrotransposons | 6–11 kb | 440,000 | 8 |
| Non-LTR retrotransposons | | | |
| LINEs | 6–8 kb | 860,000 | 21 |
| SINEs | 100–300 bp | 1,600,000 | 13 |
| Processed pseudogenes | Variable | 1—≈100 | ≈0.4 |
| Unclassified spacer DNA | Variable | n.a.‡ | ≈25 |

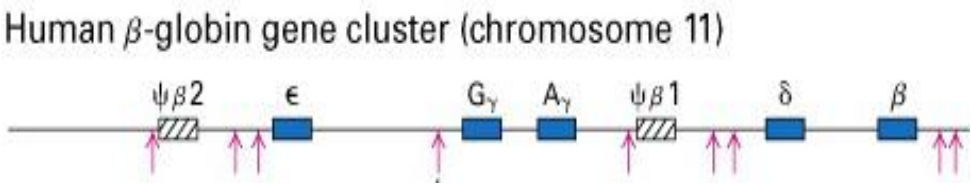
Major Classes of Eukaryotic DNA in Human Genome
Protein coding genes
Tandemly repeated genes
Repeated DNA
Unclassified spacer DNA

## 51 Duplicated Genes/Pseudo-Genes

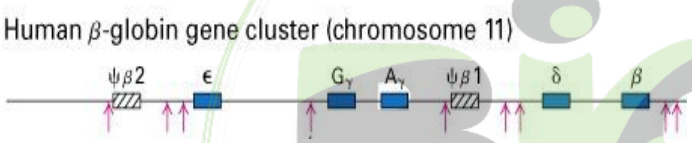
Duplicated Genes
Encode closely related(homologous) proteins
Clustered together in genome
Formed by duplication of an ancestral gene followed by mutation

Duplicated Genes/Pseudo-Genes
Pseudo-Genes

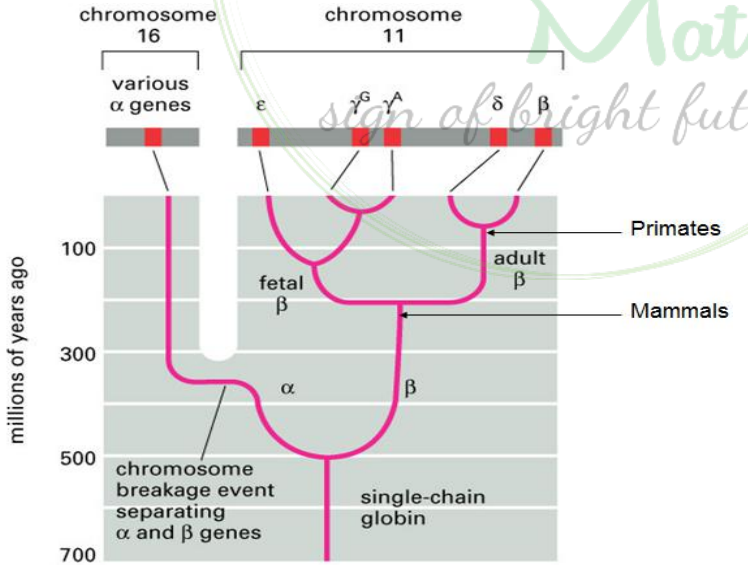
Nonfunctional copies of genes
Formed by duplication of ancestral gene, or reverse transcription and integration.
Five functional genes and two pseudo-genes



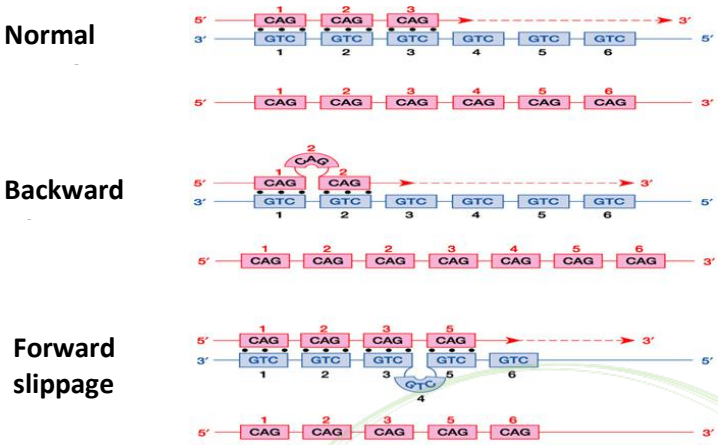

Not expressed due to mutations that produce
A stop codon
Prevent mRNA processing
Due to lack of regulatory sequences
Five functional genes and two pseudo-genes



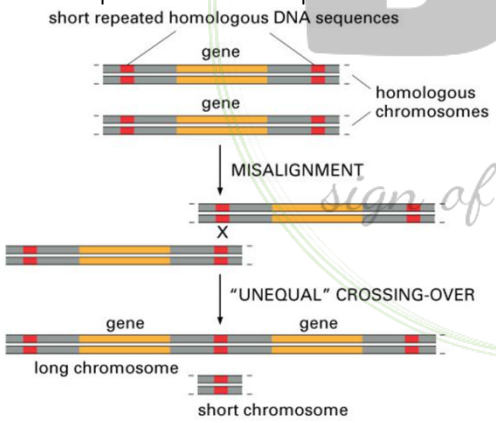

Duplicated Genes/Pseudo-Genes
Genes duplication: Globin family



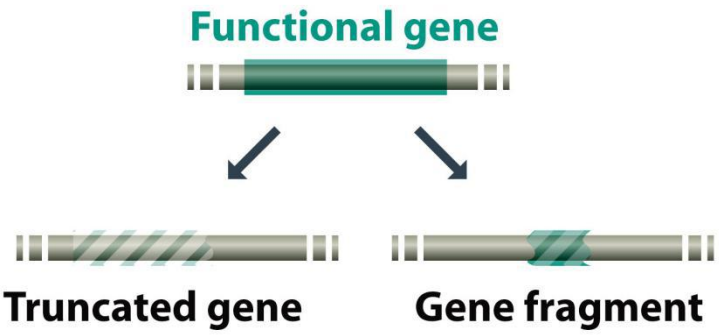

Ancestral globin gene was duplicated ~ 500 million years ago.
Mutations in both genes to differentiate them - α and β present in all higher vertebrates
Further gene duplications produced alternative forms in mammals and in primates
DNA Duplications:
Two phenomenon are common in DNA duplications:
Slipped strand mispairing



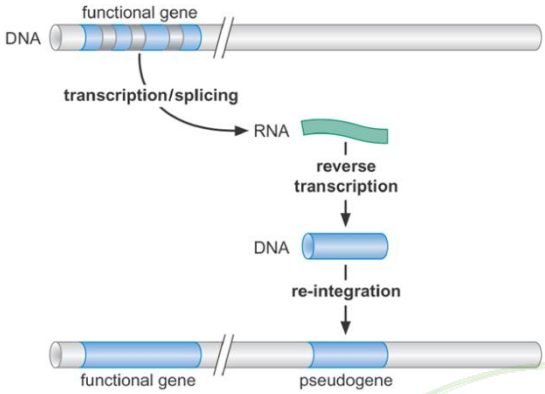

Unequal crossover during recombinatinations
DNA Duplications: Slipped strand mispairing
DNA Duplications: Unequal Cross Over




Functional Genes: Process of evolution produced truncated genes and gene fragments




Duplicated Genes/Pseudo-Genes




Duplicated Genes and Pseudo Genes
Clustered together in genome.
Formed by duplication of an ancestral gene
Nonfunctional copies of genes

## 52 Repetitive DNA

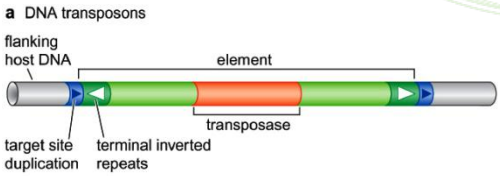
Whole genome duplication
The genomes of two distinct species can merge
An individual can acquire an extra copy of a chromosome
Chromosomes can fuse; e.g. human chromosome 2 derived from a fusion of two ancestral primate chromosomes
Chromosomal regions can be inverted or deleted
Segmental and other duplications can occur
Five main classes of Repetitive DNA
Interspersed repeats (RNA/DNA transposon-derived)
Approximately 45% of human genome (e.g. LINES, SINES, Alu)
Retrotransposons constitute over 40% of the human genome and consist of several millions of family members.
They play important roles in shaping the structure/evolution of the genome.
Processed Pseudogenes
These genes have a stop codon, frameshift mutation, or loss of promoter activity and do not encode a functional protein.
They commonly arise from retrotransposition, or following gene duplication and subsequent gene loss.
Repetitive DNA
Simple Sequence Repeats
Simple sequence repeats
Microsatellites (1-12 bp)
Minisatellites (12-500 bp)
Segmental Duplications
Segmental duplications blocks of about 1 kb to 300 kb that are copied within or between chromosomes
5% of human genome.
Blocks of Tandem Repeats
Blocks of tandem repeats includes telomeric and centromeric repeats and can span millions bp
Often species specific.

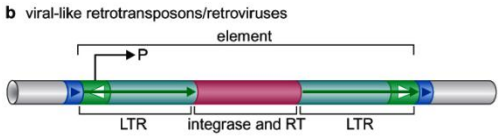
Repeats in Mouse genome

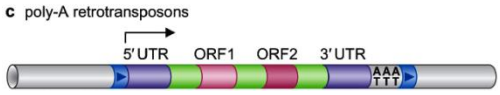
| Type | No. of Repeats | Size | Percent of genome |
|---|---|---|---|
| Highly repetitive | > 1 Million | < 10 bp | 10 % |
| Moderately repetitive | > 1000 | ~ 150 - ~300 bp | 20 % |

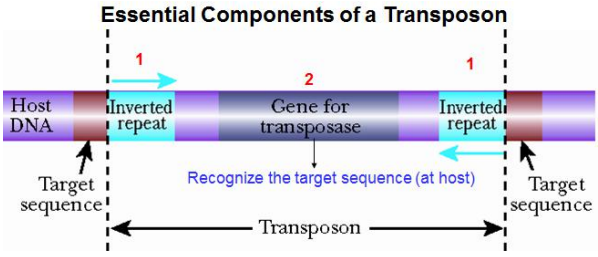
Conclusion
Main classes of Repetitive DNA
Simple sequence of DNA
DNA transposons
LTR retrotransposons
Non LTR retrotranposons like LINE, SINE

**53 Mobile DNA**
Move within genomes
Most of moderately repeated DNA sequences found throughout higher eukaryotic genomes
Most mobile DNA -transposable elements
Includes DNA-based transposons and retro-transposons.
Some encode enzymes that catalyze movement.

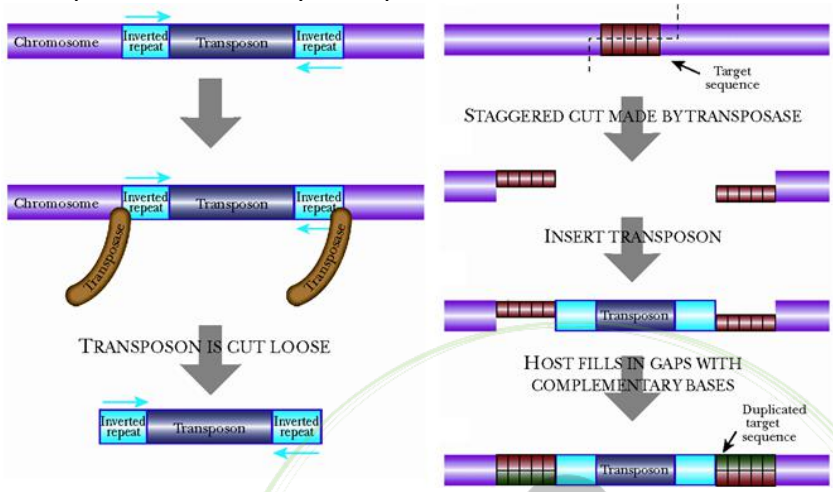


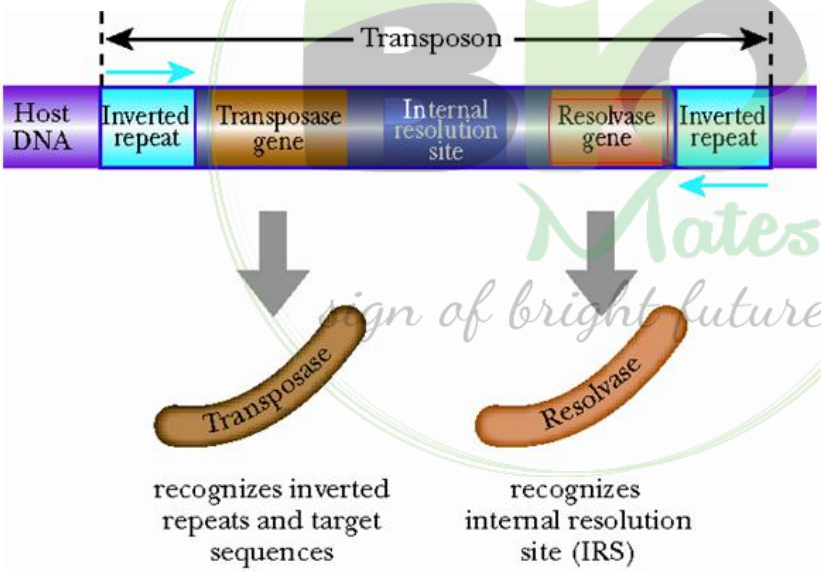



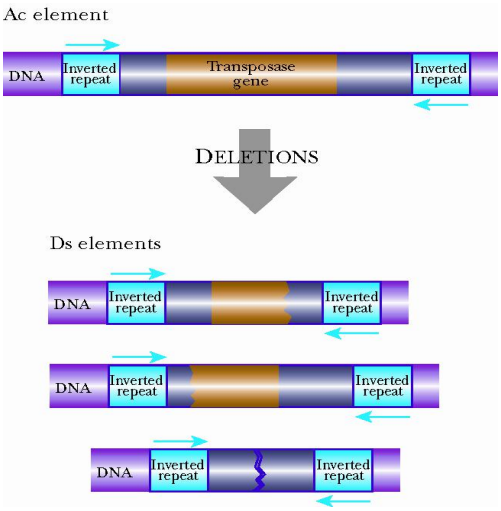
Transposon is cut off by transposase and inserted on another site



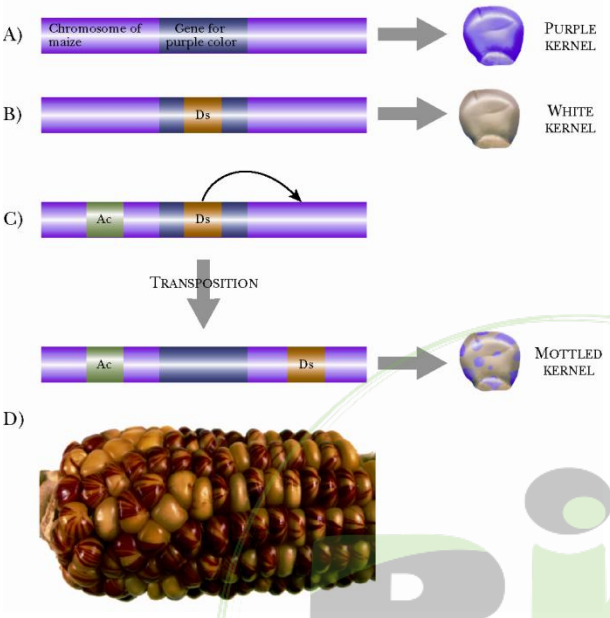

Components of a Complex Transposon




Ac and Ds Elements of Maize – Higher Life Form



Movement of Ds Elements Gives Mottled Corn




**Mobile DNA Transpositions**

Transposition is movement of mobile DNA

Involves copying of mobile DNA element and insertion into new site in genome

Molecular parasite: "selfish DNA"

**Mobile DNA - Retrotransposition**

Moving in the form of RNA by element coding for reverse transcriptase

- LINEs
- SINEs
- retrovirus-like elements (long terminal repeat)

Mobile DNA – Interspersed Repeats in Human Genome

| Element | Transposition | Structure | Length | Copy number | Fraction of genome |
|---|---|---|---|---|---|
| LINEs | Autonomous | ORF1 ORF2 (*pol*) AAA | 1–5 kb | 20,000–40,000 | 21% |
| SINEs | Nonautonomous | A B AAA | 100–300 bp | 1,500,000 | 13% |
| Retrovirus-like elements | Autonomous | *gag* *pol* (*env*) | 6–11 kb | 450,000 | 8% |
| | Nonautonomous | *gag* | 1.5–3 kb | | |
| DNA transposons | Autonomous | *transposase* | 2–3 kb | 300,000 | 3% |
| | Nonautonomous | | 80–3000 bp | | |

Mobile DNA – Fuel for Evolution

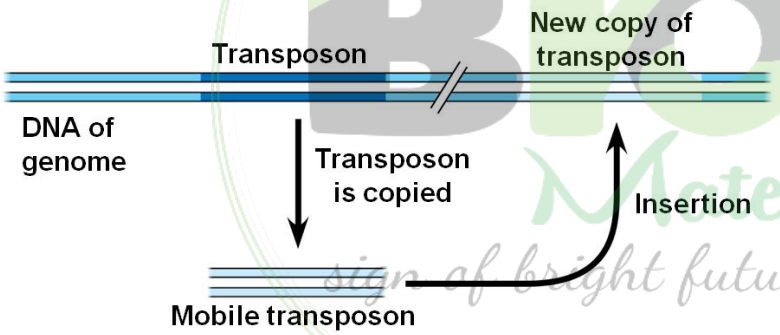
Probably have significant effect on evolution by facilitating gene duplication, which provides the fuel for evolution, and exon shuffling

## 54 Movement of Transposons and Retro-transposons

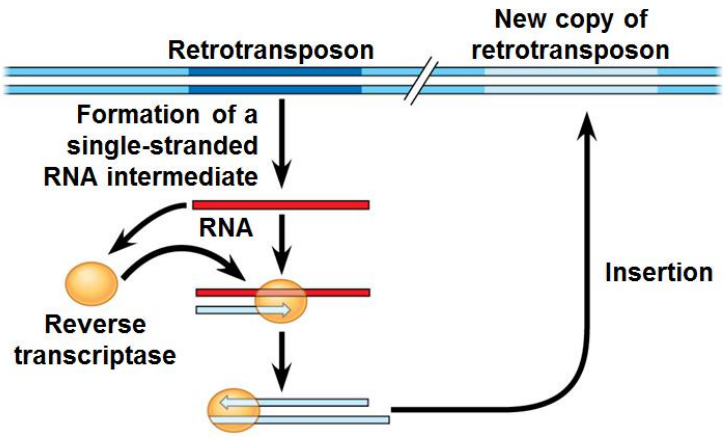
**Transposons/Retrotransposons-Movement**
**Movement of Transposons**
Eukaryotic transposable elements - two types
Transposons, which move by means of a DNA intermediate
Retrotransposons, which move by means of an RNA intermediate

**Transposons** - major content of eukaryotic genomes
~50% of Human/Mouse
~75% of the maize genome
~85% of the barley genome
~98% of the iris genome

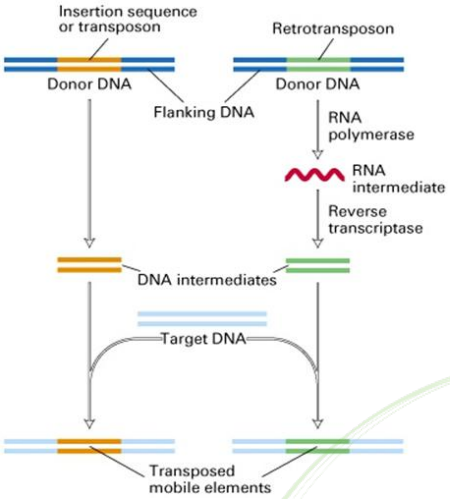
Movement of Transposons




Movement of Retrotransposons




Transposons moves using DNA intermediate




Conclusion
Eukaryotic transposable elements - two types
Transposons, which move by means of a DNA intermediate
Retrotransposons, which move by means of an RNA intermediate

**55 Transposons in Prokaryotes and Eukaryotes**

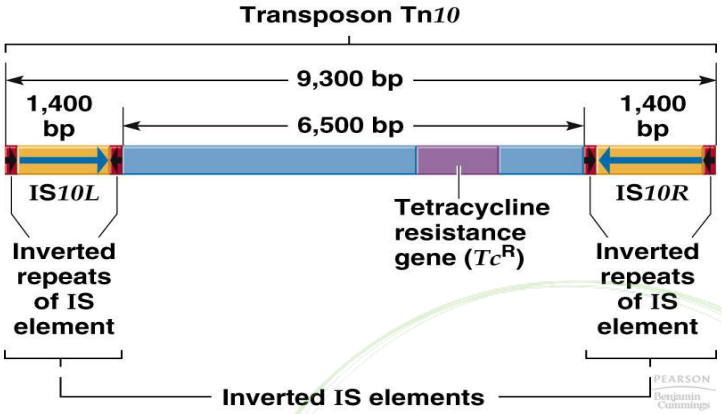
Transposons in Prokaryotes
Two types of transposons in prokaryotes i.e.

- Insertion Sequence (IS)
- Transposons (Tn)
- Bateriophage Mu

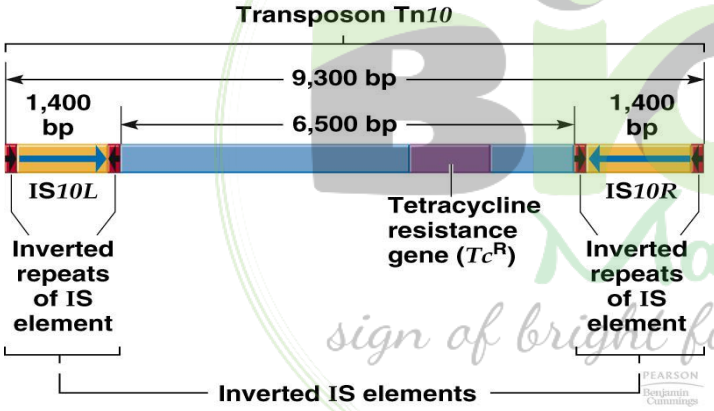
**IS Elements**
Prokaryotic IS elements range in size from 768 bp to over 5 kb.
IS1 is 768 bp long, and present in 4–19 copies on the E. coli chromosome.
IS2 has 0–12 copies on the chromosome, and 1 copy on the F plasmid
IS10 is found in R plasmids
The ends of all sequences (IS elements) show inverted terminal repeats (IRs) of 9–41 bp
(e.g., IS1 has 23 bp of nearly identical sequence).

**Transposons in Prokaryotes**
Transposons are similar to IS elements, but carry additional genes, and have a more complex structure.
Two types of prokaryotic transposons:
Composite transposons carry genes (e.g., antibiotic resistance) flanked on both sides by IS elements (IS modules). Example Tn10
Non-composite transposons also carry genes (e.g., drug resistance) but do not terminate with IS elements.

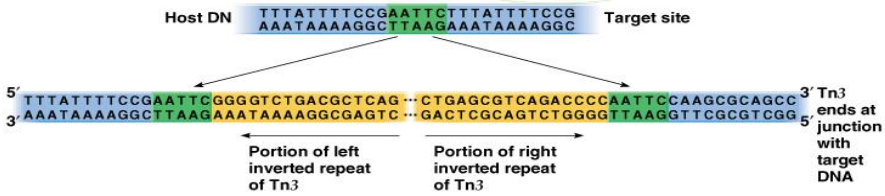
Transposons in Prokaryotes: Composite Tn10



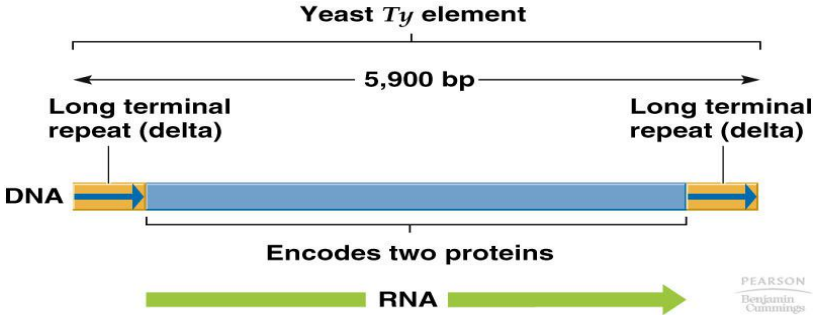

Transposons in Prokaryotes: Non-composite Tn3



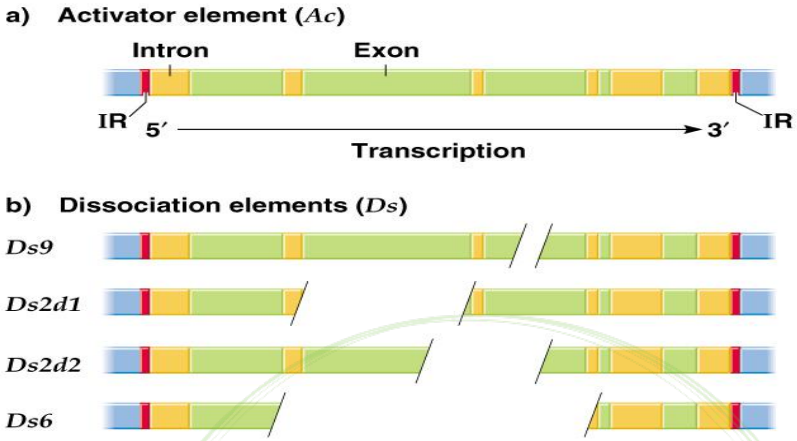

Transposons in Prokaryotes: Non-composite Tn3




Transposons in Eukaryotes: Ty Elements in Yeast



Transposons in Eukaryotes: Ac and Ds Elements

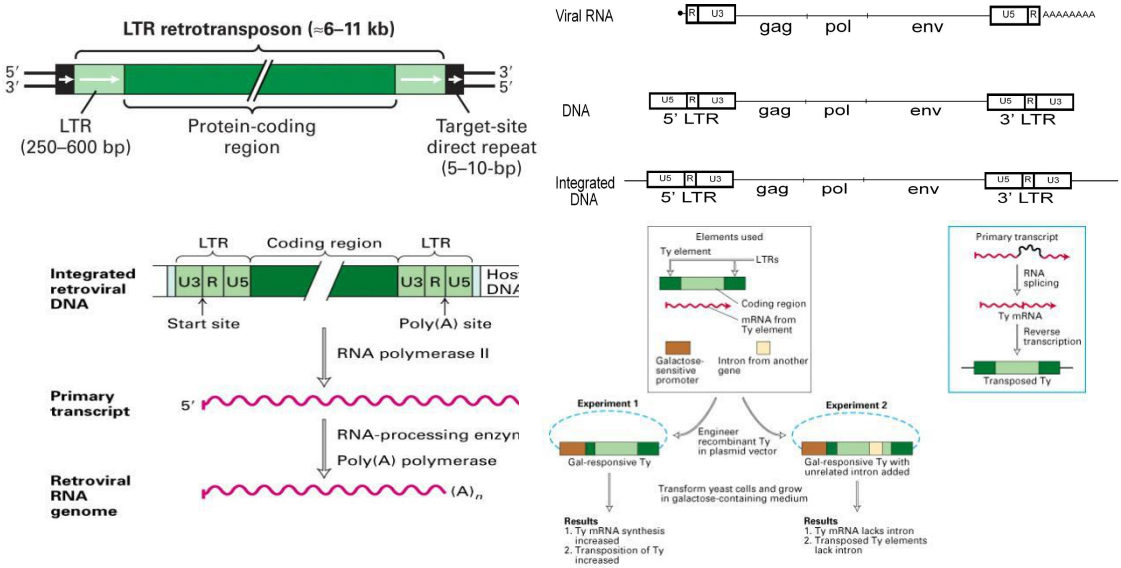



**Transposons in Eukaryotes: Ac and Ds Elements in Maize**

Ac is 4,563 bp, with 1 1-bp imperfect terminal IRs and 1 transcription unit producing a 3.5 kb mRNA encoding an 807 amino acid transposase.
Ac activates Ds to transpose or break the chromosome where it is inserted.
Ds elements vary in length and sequence, but all have the same terminal IRs as Ac, and many are deleted or rearranged versions of Ac

Transposons
Transposons are found in both Prokaryotes and Eukaryotes.

**56 Long Terminal Repeats**
Eukaryotic retrotransposons fall into two major groups:
LTR retrotransposons
Non-LTR retrotransposons
LTRs stand for Long direct Terminal Repeats.
LTRs consist of 250-600 bp direct repeat sequences located at the ends of the retrotransposon coding region
Flank viral retrotransposons and retroviruses



Contain regulatory sequences like transcription start site and poly (A) site

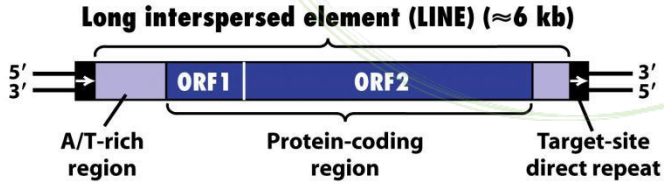
Conclusion
Eukaryotic retrotransposons fall into two major groups:
LTR retrotransposons
Non-LTR retrotransposons

**57 Long Interspersed Nuclear Elements**
**and**
**Short Interspersed Nuclear Elements**

LINEs and SINEs
Non-LTR retrotransposons consist of two sub-types;
Long Interspersed Nuclear Elements (LINEs)
Short Interspersed Nuclear Elements (SINEs)
High copy numbers, in the plants species
Widespread in eukaryotic genomes
LINEs possess two ORFs, which encode all the functions needed for retrotransposition

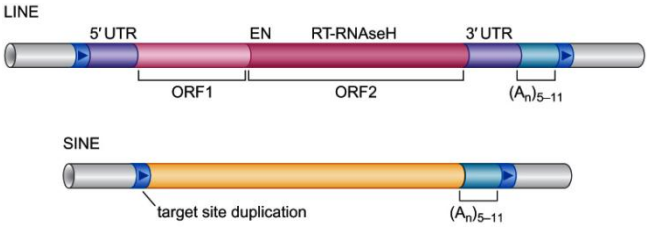
**LINEs**

Functions of ORFs include
Reverse transcriptase
Endonuclease activities, in addition to a nucleic acid-binding property needed to form a ribonucleoprotein




**SINEs**

SINEs, on the other hand, use the LINE machinery and function as non-autonomous retro-elements




**LINEs**

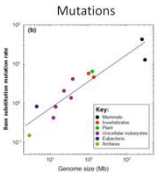
Several subgroups, such as L1, L2 and L3.
Human coding L1 begin with an untranslated region (UTR) that includes an RNA polymerase II promoter, two non-overlapping open reading frames (ORF1 and ORF2), and ends with another UTR.
ORF1 encodes an RNA binding protein and ORF2 encodes a protein having an endonuclease
The 5' UTR contains the promoter sequence, while the 3' UTR contains a polyadenylation signal (AATAAA) and a poly-A tail

Human genome contains about 500,000 LINEs
LINEs and SINEs
Short Interspersed Elements
Short Interspersed Elements
~300 base pairs
Alu
~ 11-13% of human genome
Short Interspersed Elements
Short DNA sequences ~ 300 bases that represent reverse-transcribed RNA molecules
SINEs do not encode a functional reverse transcriptase protein and rely on other mobile elements for transposition
In some cases they may have their own endonuclease that will allow them to cleave their way into the genome
Short Interspersed Elements
The most common SINEs in primates are called Alu sequences.
Alu elements are approximately 350 base pairs long, do not contain any coding sequences, and can be recognized by the restriction enzyme AluI
SINEs make up ~ 11-13% of human genome
Conclusion
LINES
6-7 kb long
SINES
~ 300bp long
Example: Alu ~ 350 bp long

**58 Genetic Variations**

Variation between individuals of a population (within species) can be due to differences in the nucleotide sequence
How genetic variations emerge ?

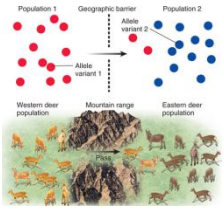
Genetic Variations can be due to the Mutations



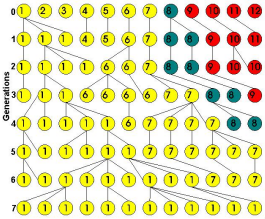

Genetic Variations: Recombinations



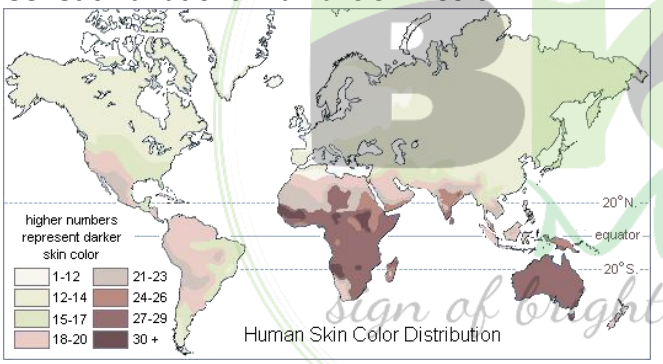

Genetic Variations: Genes Flow




Genetic Variations: Genetic Drift: Change in the frequency of a gene variant (allele) in a population due to random sampling of organisms

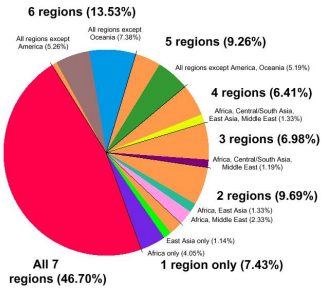



Genetic Variations: Humans Skin Color




Genetic Variations: Yellow Brown Hair Australian natives

Genetic Variations: Microsatellites




Genetic Variations - Humans

Two genomes are roughly 99.9% identical to each other.

If human genome is 3.0 billion bp then, there are 3.0 million differences between any two genomes.

**59 Basic techniques**

The solutions: basic techniques

The fragments of DNA can be inserted to a suitable replicons-cloning vehicles

Plasmids and bacteriophages are most suitable vectors

Isolation of genomic DNA from E. coli

Harvesting of cell pellet by centrifugation

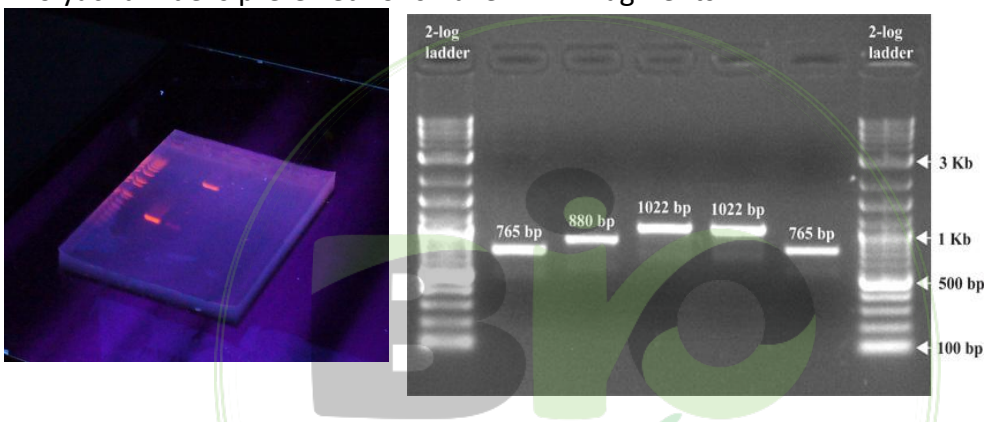
Cell lyses by SDS and proteinase K
DNA extraction with phenol and chloroform
Precipitation with isoprepenol

**60 Agarose gel electrophoresis**

Agarose is convenient for separating DNA fragments ranging in size from a few hundred base pairs to about 20 kb.

Polyacrlamide gel electrophoresis

Polyacrlamide is preferred for smaller DNA fragments




**Nucleic acid blotting**

Blotting describes the immobilization of sample nucleic acids or proteins on to a solid support, generally nylon or nitrocellulose membranes

Southern blotting
Northern blotting
Western blotting

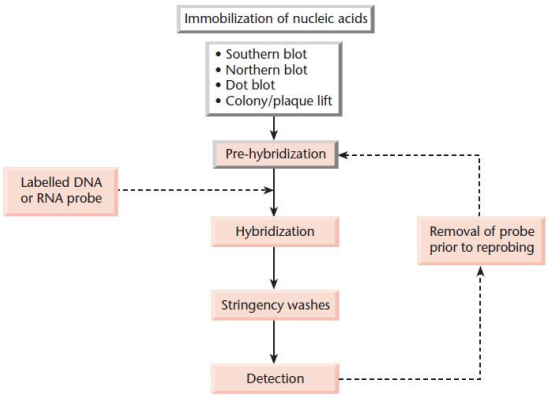
**Nucleic acid Hybridization**

The hybridization of nucleic acids on membranes is a widely used technique in gene manipulation

**Stringency control**

Stringency can be regarded as the specificity with which a particular target sequence is detected by hybridization probe

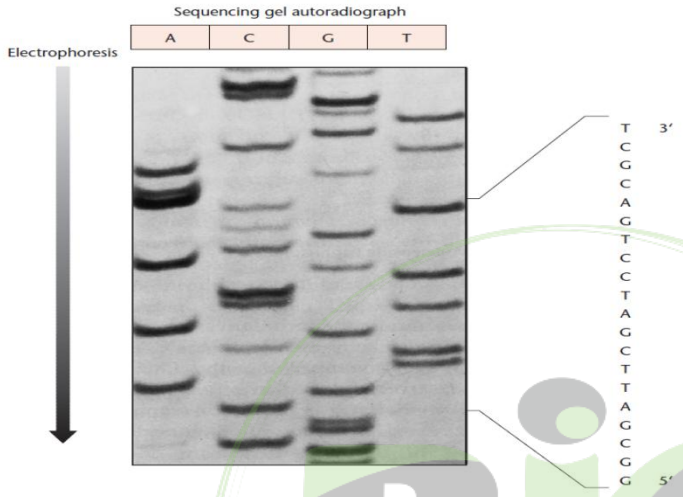
Probe is a fragment of DNA or RNA which is used to detect the sequences of target DNA



**Autoradiography**

A technique using X-ray film to visualize fragments of molecules that have been radioactively labeled

For example, it can be used to analyze the length and number of DNA fragments separated by gel electrophoresis




**Southern blotting**

A Southern blot is a laboratory method used to detect specific DNA fragment from a mixture of DNA molecules

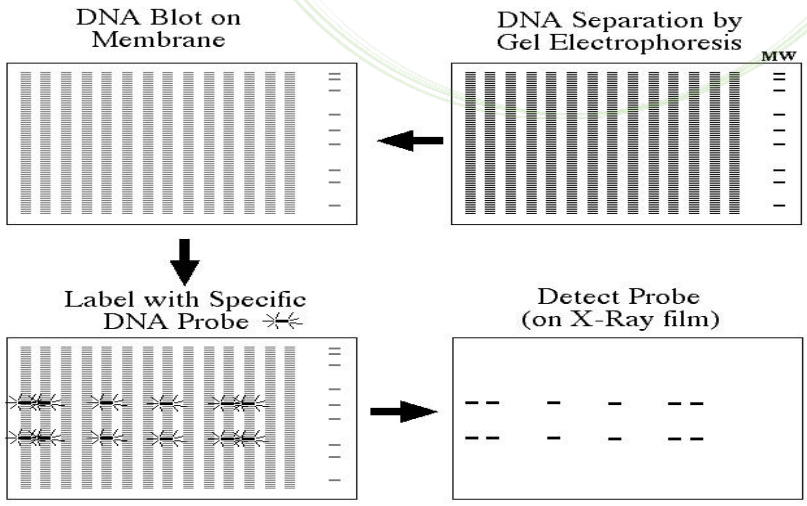
The technique was named after its inventor, Edward Southern (1975)

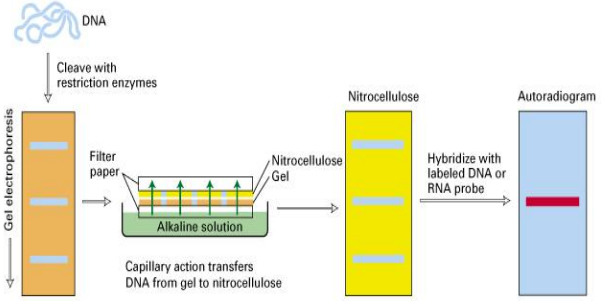
For efficient blotting, gel pretreatment is important

After transfer, the nucleic acid needs fixation

After fixation, the membrane is placed in a solution of labelled probe

After hybridization reaction, membrane is washed and hybridization are detected by autoradiography

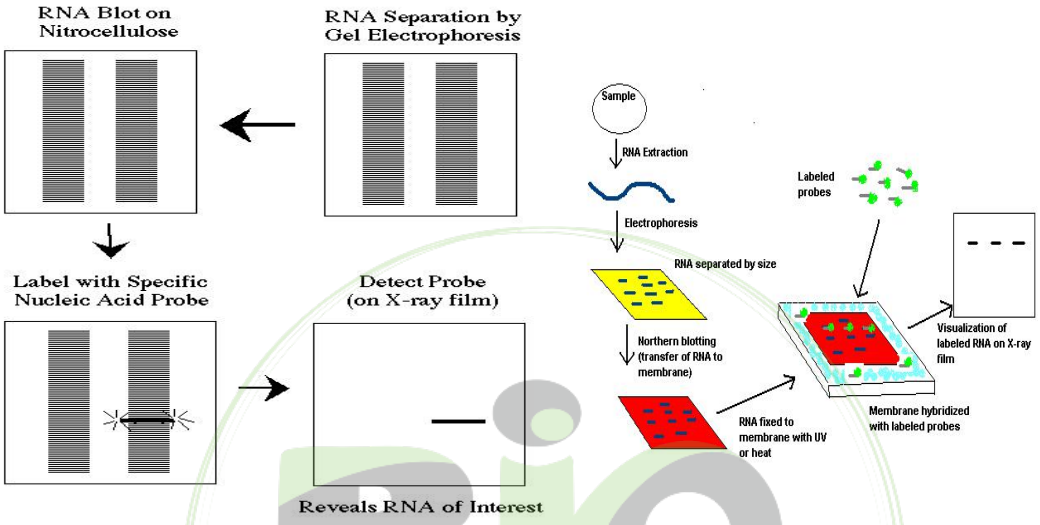


**Northern blotting**

A northern blot is a method used to detect specific RNA molecules among a mixture of RNA

Northern blotting can be used to analyze a sample of RNA from a particular tissue or cell type in order to measure the RNA expression of particular genes

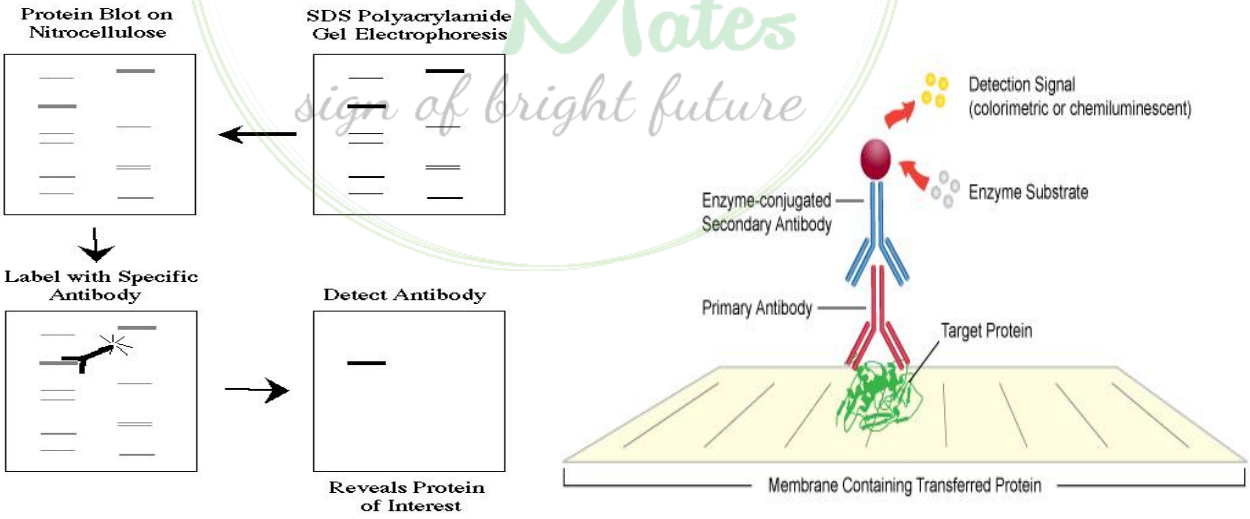



**Western blotting**

It involves the transfer of electrophoresed protein bands from a gel on to a membrane

The bounded proteins are then available for analysis

Most commonly, antibodies are used to detect specific antigens




**67 Transformation of E. coli**

Transformation is the process of getting the recombinant vector from a reaction mixture or vector solution into E. coli cells

To enable the cells to take up circular vector DNA they have to be made competent

Electroporation

A rapid and simple technique for introducing cloned genes into a wide variety of microbial, plant and animal cell, including E. coli is electroporation

In electroporation, electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

**Transformation with other organisms**

E. coli often remains the host of choice for cloning

Bacillus subtilis is a particularly attractive alternative cloning host

Animal cells, protoplasts of yeast and plant are susceptible to transformation by liposomes

**Polymerase chain reaction (PCR)**

The impact of PCR upon molecular biology has been profound

The reaction is easily performed and leads to the amplification of specific DNA by an enormous factor

| Cycle number | Number of double-stranded target molecules |
|---|---|
| 1 | 0 |
| 2 | 0 |
| 3 | 2 |
| 4 | 4 |
| 5 | 8 |
| 6 | 16 |
| 7 | 32 |
| 8 | 64 |
| 9 | 128 |
| 10 | 256 |
| 11 | 512 |
| 12 | 1024 |
| 13 | 2048 |
| 14 | 4096 |
| 15 | 8192 |
| 16 | 16,384 |
| 17 | 32,768 |
| 18 | 65,536 |
| 19 | 131,072 |
| 20 | 262,144 |
| 21 | 524,288 |
| 22 | 1,048,576 |
| 23 | 2,097,152 |
| 24 | 4,194,304 |
| 25 | 8,388,608 |
| 26 | 16,777,216 |
| 27 | 33,554,432 |
| 28 | 67,108,864 |
| 29 | 134,217,728 |
| 30 | 268,435,456 |

**Cutting DNA molecules**

Before 1970 there was no method of cleaving DNA at discrete points

Mechanical shearing was used for DNA fragmentation

During 1960s phage biologist elucidated the phenomenon of restriction and modification

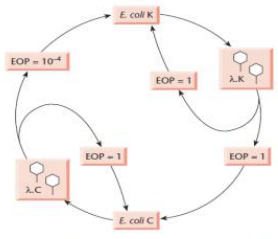
**Host-controlled restriction/ modification**

Restriction system allow bacteria to monitor the origin of incoming DNA

When the incoming DNA is a bacteriophage genome, the effect is to reduce the efficiency of plating i.e. to reduce the number of plaques in plating test

The phenomenon of restriction and modification were well illustrated and studied by the behavior of phage ʎ on two E. coli strains

E. coli C and E. coli K




Host-controlled restriction and modification of phage λ

Restrictive host must, of course protects its own DNA from restriction endonucleases by modification

Modification involves methylation of certain bases that constitute recognition sites

**Types of restriction and modification (R-M) system**

At least four R-M systems are known

Type I

Type II

Type III
Type IIs

Type I
Type I systems were the first to be characterized from E. coli K12
The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit
Type I systems are of little value for gene manipulation

Type II
Most of the useful R-M system is Type II
Type II enzymes recognize a defined sequence and cut within it

Type III
Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value

Type IIs
Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

**Cutting DNA molecules**
**Nomenclature**
A suitable system was proposed by Smith and Nathans (1973)
The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet
E. coli = Eco
H. influenzae = Hin
Strain identification is written as EcoK
In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae
HindI, HindII, HindIII etc
All restriction enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of H. influenzae
R. HindIII or M. HindIII

**Examples of restriction endonuclease nomenclature**

| Enzyme | Enzyme source | Recognition sequence |
|---|---|---|
| *SmaI* | *Serratia marcescens*, 1st enzyme | CCCGGG |
| *HaeIII* | *Haemophilus aegyptius*, 3rd enzyme | GGCC |
| *HindII* | *H. influenzae*, strain d, 2nd enzyme | GTPyPuAC |
| *HindIII* | *H. influenzae*, strain d, 3rd enzyme | AAGCTT |
| *HamHI* | *Bacillus amyloliquefaciens*, strain H, 1st enzyme | GGATCC |

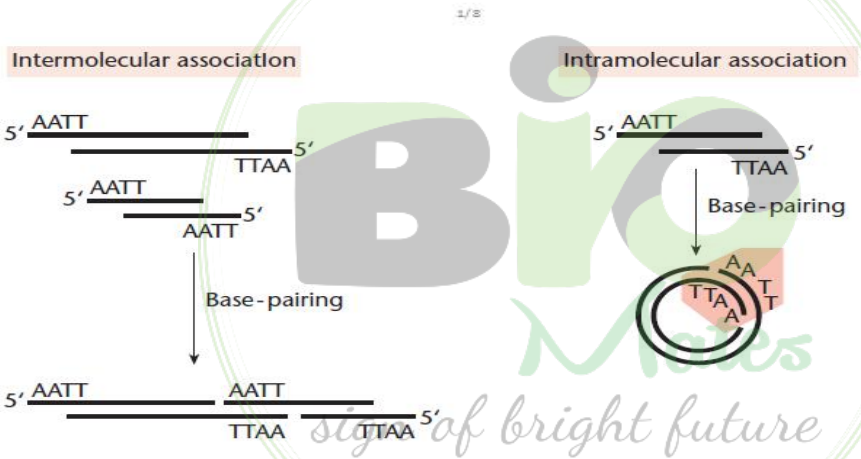
Target sites
Type II endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of rotational symmetry i.e. referred as palindromes
5'-GAATTC-3'
5'-CTTAAG-3'

5′-G A A T T C-3′
3′-C T T A A G-5′

5′-G A A* T T C-3′
3′-C T T A* A G-5′

5′-G 5′- A A T T C-3′
3′-C T T A A-5′ G-5′

Single stranded breaks by *Eco*R1




**Number and size of restriction fragments**

The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut

Four base recognition site occurs every 44 (256) bp

Six base recognition site occurs every 46 (4096) bp

Eight base recognition site occurs 48 (65,536) bp

Average fragment size (bp) produced by different enzymes

| Enzyme | Target | Arabidopsis | E. Coli | Human |
|---|---|---|---|---|
| *Apa*I | GGGCCC | 25000 | 15000 | 2000 |
| *Bam*HI | GGATCC | 6000 | 5000 | 5000 |
| *Spe*I | ACTAGT | 8000 | 60000 | 10000 |

**Summary of restriction endonucleases**

Type II restriction enzymes are heavily responsible for the current explosion in the field of gene manipulation in that they are essential in forming recombinant DNA molecules

**77 Joining DNA molecules**

DNA modifying enzymes

- Nucleases
- DNA Polymerase
- Reverse transcriptases
- DNA ligases

**Nucleases**

Nucleases or DNases are the enzymes that degrade DNA
Two broad classes of nucleases
i). Exonucleases
ii). Endonucleases

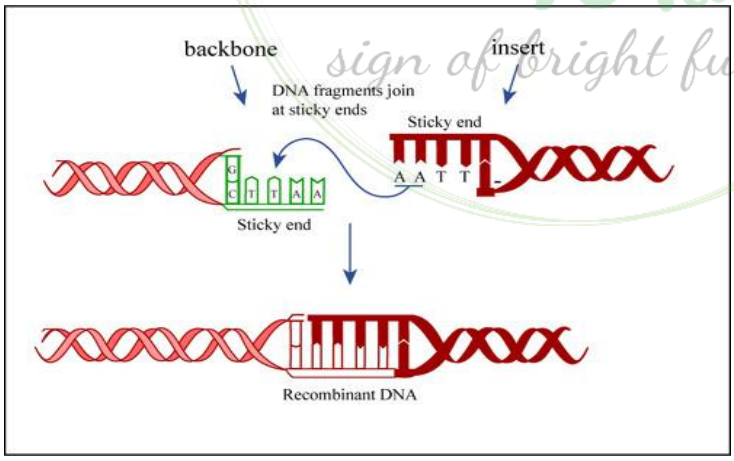
**DNA Polymerase**
An enzyme that catalyzes template-dependent synthesis of DNA
**Reverse transcriptases**
An RNA-directed DNA polymerase in retroviruses; capable of making DNA complementary to an RNA
**DNA ligase**
An enzyme that creates a phosphodiester bond between the 3' end of one DNA segment and the 5' of another

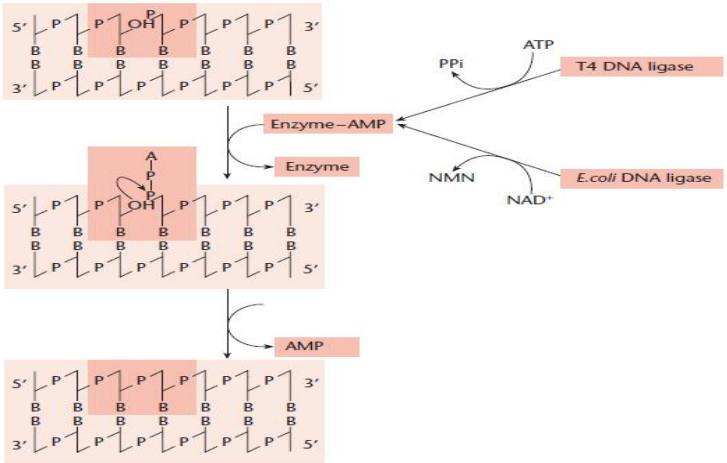
**78 Methods of joining DNA fragments**
Mainly three methods are used for joining DNA in vitro

i). Joining covalently annealed cohesive ends by DNA ligase

ii). Joining blunt-ended fragments by DNA ligase from phage T4 infected E. coli

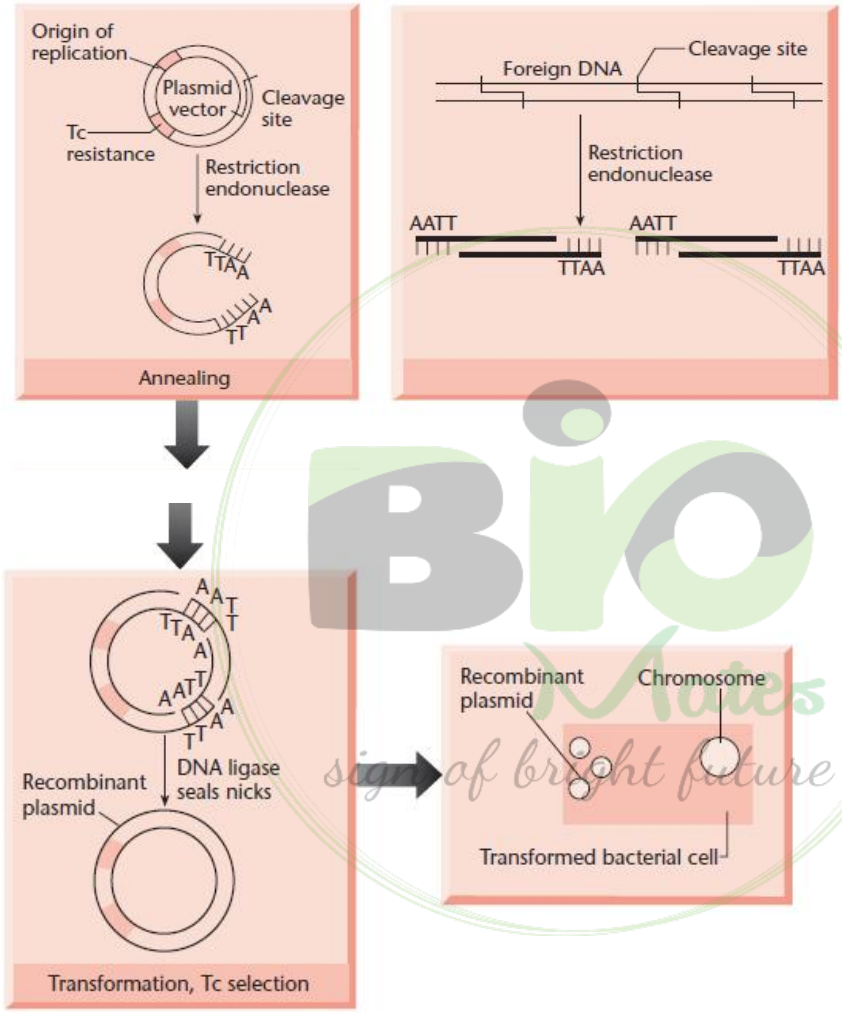
iii). The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric 3' single stranded tails at the ends of fragments
**DNA ligase**
E. coli and phage T4 encode enzyme, DNA ligase, which seals single stranded nicks between adjacent nucleotides in a duplex DNA chain



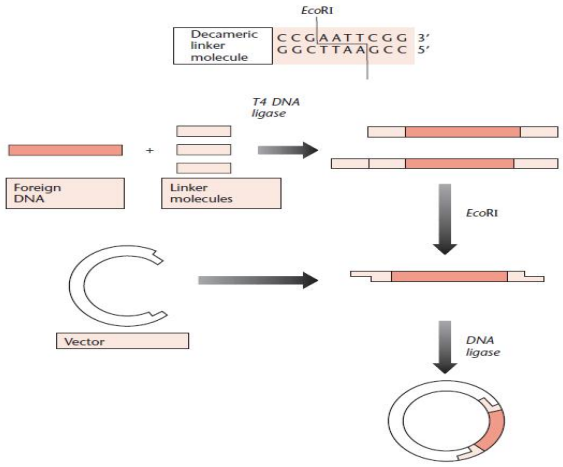
**79 DNA ligase to create covalent recombinant DNA**

DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands



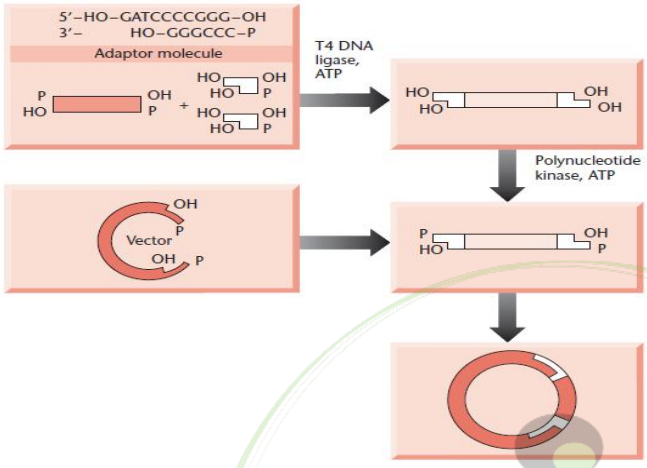

**80 Blunt end ligation via linker molecules**

Joining DNA fragments with cohesive ends by DNA ligase is a relatively efficient process
T4 DNA ligase has been used to joint blunt-ended DNA molecules



**82 Adaptors**

Chemically synthesized adaptor molecules which have a preformed cohesive end can be used to insert blunt ended foreign DNA into vector

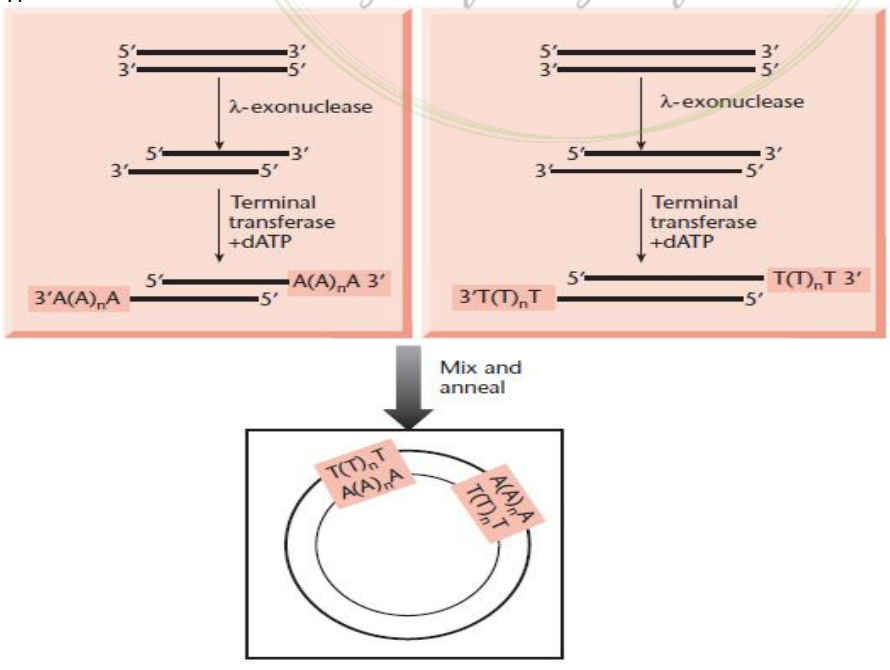



The main difference between linker and adaptor is that former having blunt ends while later have one cohesive end

**83 Homopolymer tailing**

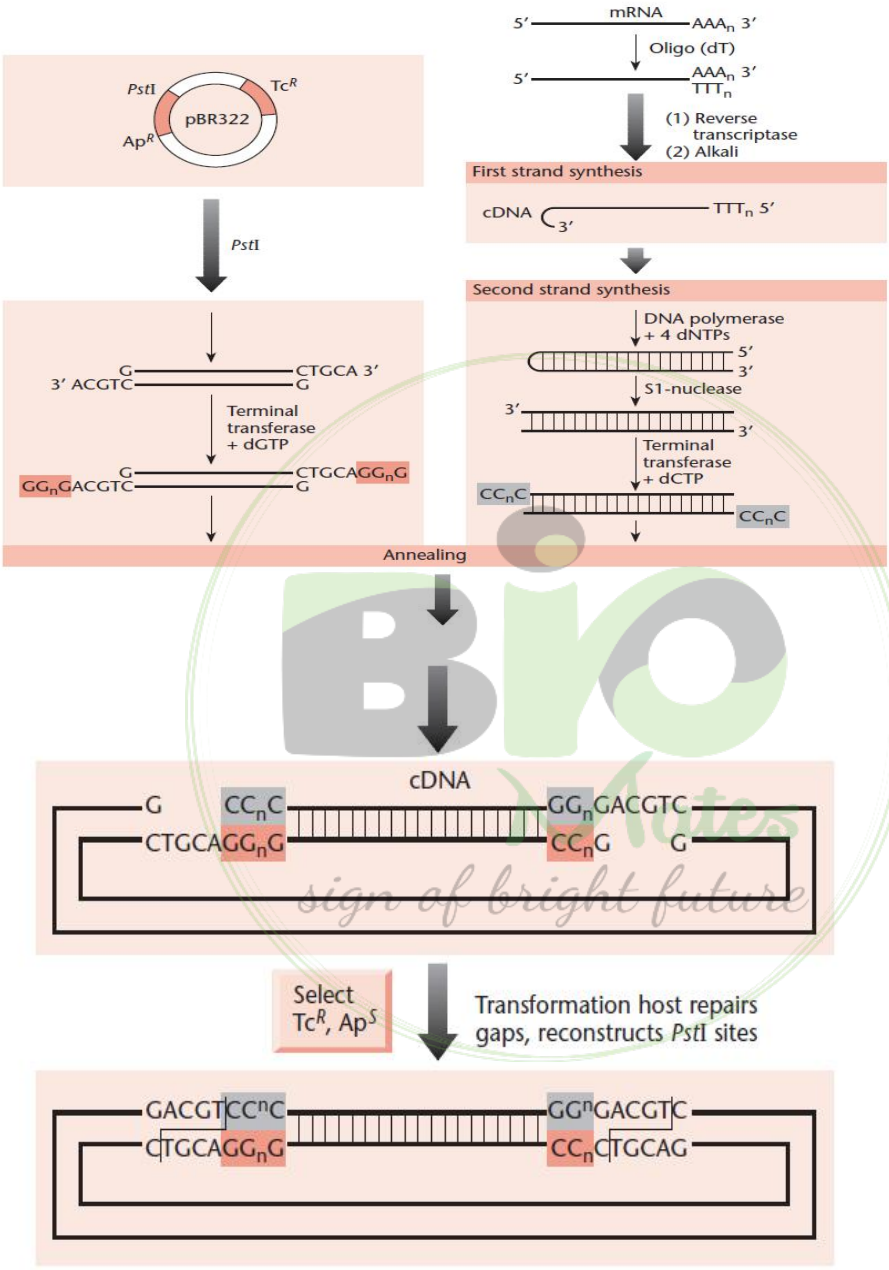
A general methods for joining DNA molecules makes use of the annealing of complementary homopolymer sequences by adding oligo(dA) sequences to 3′ ends of one population of DNA molecules and oligo(dT) blocks to the 3′ ends of another population

An enzyme purified from calf thymus, terminal deoxynucleotidyltransferase, provides the means by which the homopolymeric extensions at 3′ OH termini can be synthesized, if presented with a single type of nucleotide




**84 Cloning of cDNA by Homopolymer tailing**

Eukaryotic mRNA can be cloned in vector after converting it to cDNA by using an enzyme reverse transcriptase



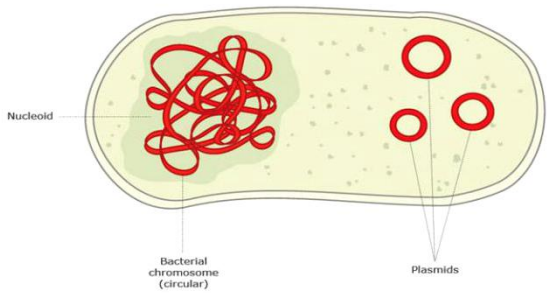

cutting and joining DNA molecules is basic step in gene manipulation

**85 Plasmids**

Plasmids are mostly circular DNA molecules that replicate separately from the host chromosome

Naturally occurring bacterial plasmids range in size from 5,000 to 400,000 bp.

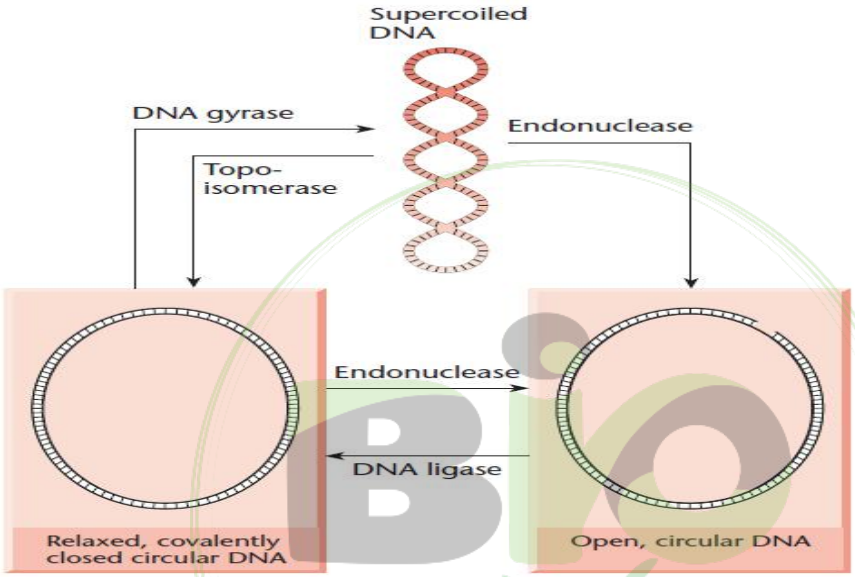
Plasmid help bacteria to survive stress



**86 Interconversion of plasmid DNA**

Most plasmids exist as double-stranded circular DNA molecules

If both strands of DNA are intact circles the molecules are describes as covalently closed circles or CCC DNA. Open circle or OC DNA have one strand intact

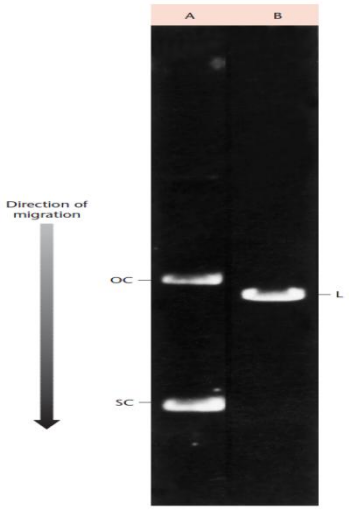



87 Effect of ethidium bromide on supercoiling of DNA

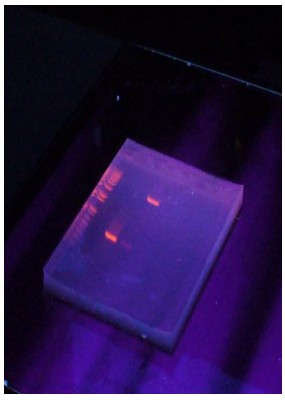
DNA bands can be visualized by soaking the gel in a solution of ethidium bromide

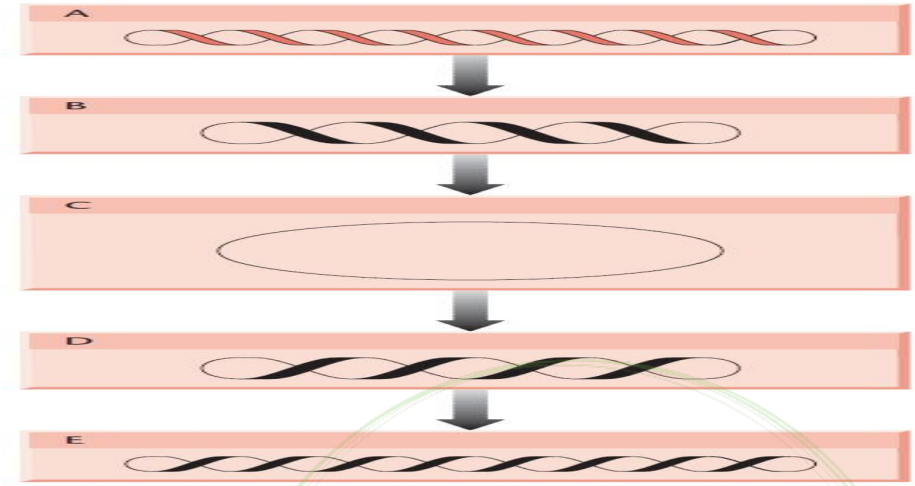
As the amount of intercalating agent i.e. ethidium bromide increases, the double helix untwists until the open form of circular DNA is produced

Electrophoresis of DNA in agarose gel




Visualization of gel under UV light after staining with ethidium bromide






Effect of ethidium bromide on supercoiling of DNA
Ethidium bromide may be a mutagen, a carcinogen, or a teratogen

**88 Plasmid Biology**
**Phenotypic traits exhibited by plasmids**

Plasmids are widely distributed throughout the prokaryotes, vary in size from less than 1 × 106 to greater than 200 × 106 daltons

| |
|---|
| Antibiotic resistance |
| Antibiotic production |
| Degradation of aromatic compounds |
| Haemolysin production |
| Sugar fermentation |
| Enterotoxin production |
| Heavy-metal resistance |
| Bacteriocin production |
| Induction of plant tumours |
| Hydrogen sulphide production |
| Host-controlled restriction and modification |
| |

**89 Properties of Conjugative and non-conjugative plasmids**

Plasmids can be categorized into conjugative or non-conjugative depending whether or not they carry a set of transfer genes, called tra genes, which promoted bacterial conjugation

| Plasmid | Size (Mda) | Conjugative | No. of plasmid copies | Phenotype |
|---|---|---|---|---|
| ColE1 | 4.2 | No | 10-15 | ColE1 production |
| RSF1030 | 5.6 | No | 20-40 | Ampicillin resistance |
| cloDF13 | 6 | No | 10 | Cloacin production |
| R6k | 25 | Yes | 13-38 | Ampicillin and streptomycin resistance |
| F | 62 | Yes | 1-2 | - |
| RI | RI | Yes | 3-6 | Multiple drug resistance |
| EntP307 | 65 | Yes | 1-3 | Enterotoxin production |

Properties of some conjugative and non-conjugative plasmids of Gram-negative organisms

**90 Host range of plasmids**

Plasmids encode only a few of the proteins required for their own replication
Host range of plasmid is determined by ori region
Plasmids whose ori derived from Col E1 have restricted host range

Host range of plasmids

RP4 and RSF1010 and many plasmids from Staphylococcus aureus have broad host range

**91 Partitioning and segregative stability of plasmids**

The loss of plasmids due to defective partioning is called segregative instability
Naturally occurring plasmids are stably maintained because they contain par regions

**92 Incompatibility of plasmids**

The inability of two different plasmids to coexist in the same cell in the absence of selection pressure

**93Desirable properties of plasmid cloning vehicles**

An ideal cloning vehicle would have the following three properties

i). Low molecular weight

ii). Ability to confer readily selectable phenotypic traits on host cells

Desirable properties of plasmid cloning vehicles

iii). Single sites for number of restriction endonucleases

95 Natural plasmids as cloning vehicles

Plasmids which were not constructed in vitro for the sole purpose of cloning are called natural plasmids
Col E1 is a natural occurring plasmid
RSF2124 is a derivative of Col E1
pSC101 is natural plasmid from Salmonella panama

| Plasmid | Size (Mda) | Sites of endonucleases | Selectable markers | Insertional inactivation |
|---|---|---|---|---|
| pSC101 | 5.8 | *Xho*I, *Eco*RI, *Pvu*II, *Hinc*II, *Hpa*I | Tetracycline resistance | --- |
| | | *Hind*III, *Bam*HI, *Sal*I | __ | Tetracycline resistance |
| Col E1 | 4.2 | *Eco*RI | Immunity to colicin E1 | Colicin E1 production |
| RSF2124 | 7.4 | *Eco*RI, *Bam*HI | Ampicillin resistance | Colicin E1 production |

Cloning in pSC101,
ColiE1 and RSF2124

**96 Use of pSC101 for cloning**

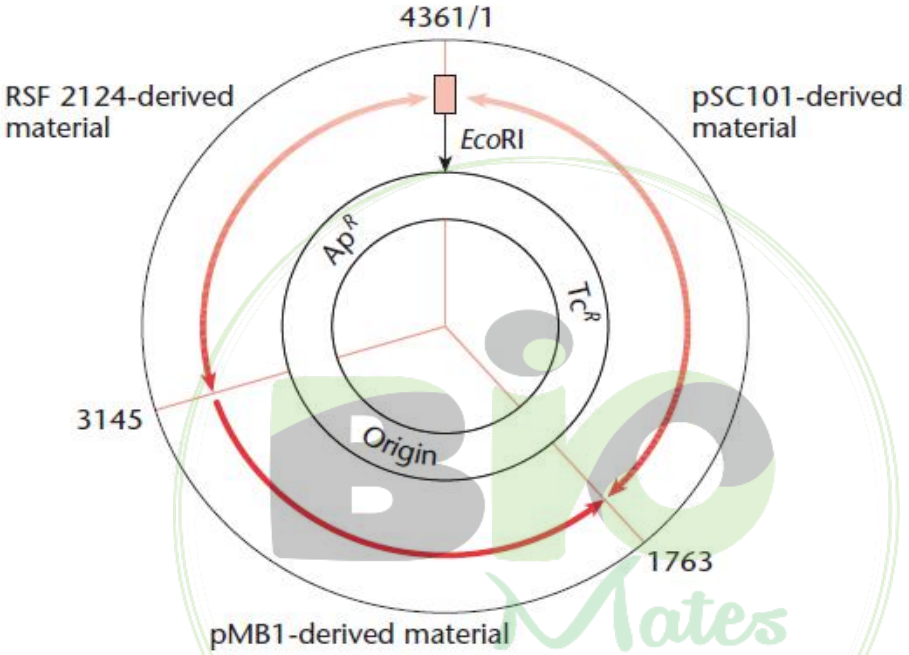
Expression of Staphylococcus plasmid genes in E. coli

pSC101 contain
Tetracycline resistance
P1248 from
Staphylococcus

**97 pBR322, a purpose-built cloning vehicle**

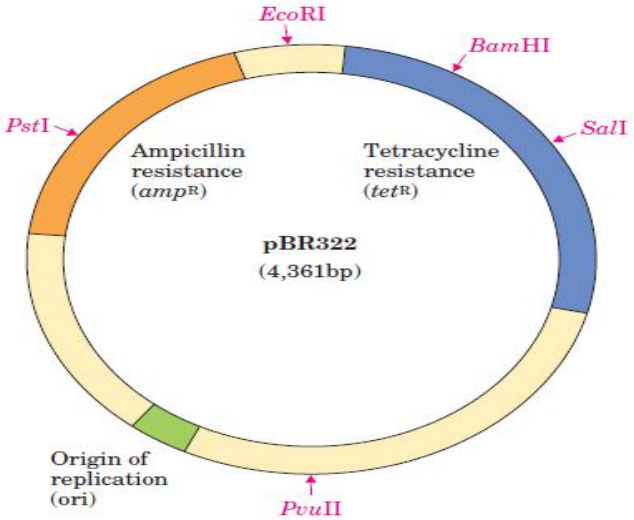
pBR322 is an example of in vitro constructed cloning vehicle

pBR322 contains the ApR and TcR genes of RSF2124 and pSC101, combined with replication element of pMB1




# The origin of plasmid pBR322

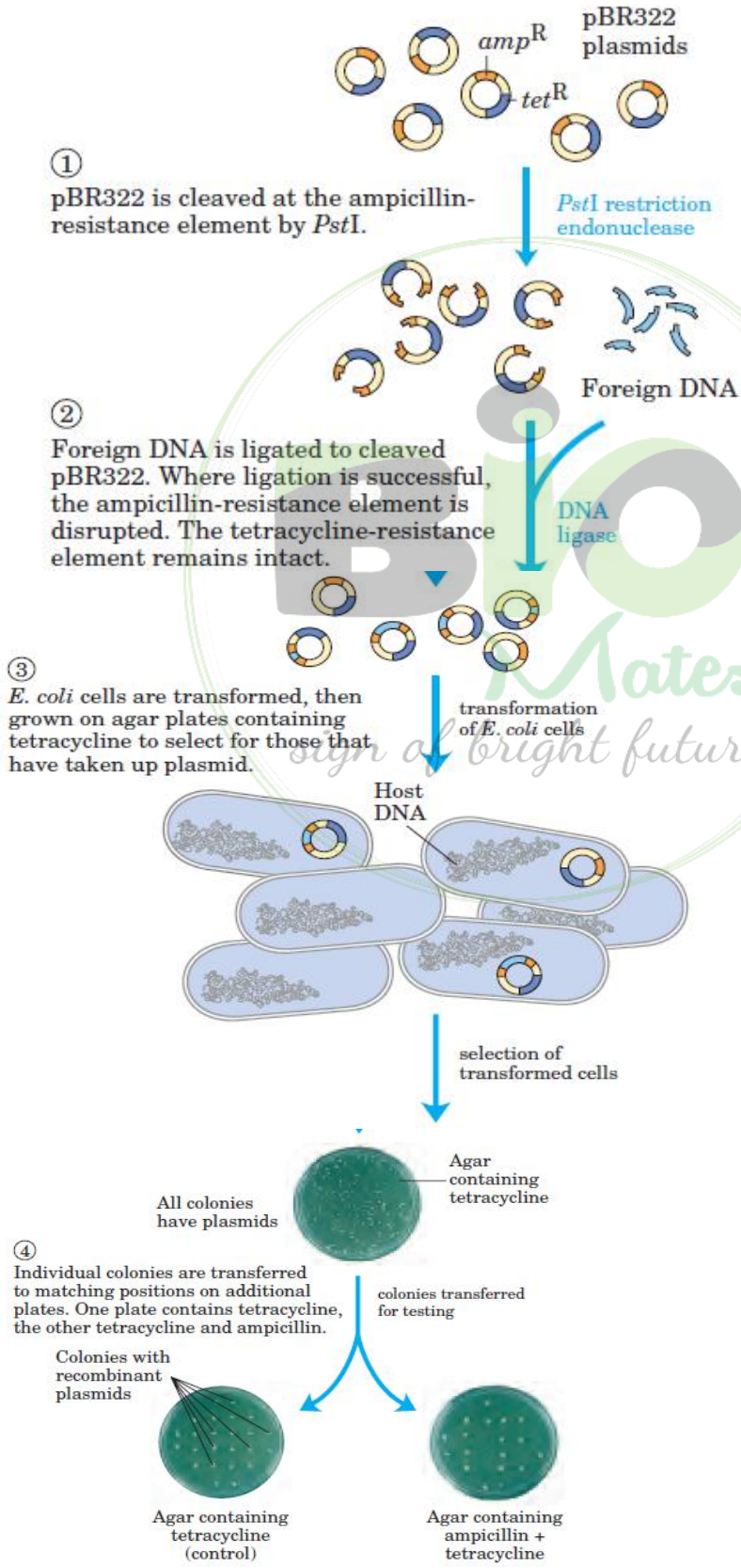
It has 40 target sites for different restriction enzym




**The constructed E. coli plasmid pBR322**

**98 Cloning with pBR322**

Plasmid pBR322 has a number of characteristics that make it suitable as cloning vehicle



**99 Improved vectors derived from pBR322**

- Over the years numerous different derivatives of pBR322 have been constructed
- pBR325 encodes chloramphenicol resistance in addition to ampicillin and tetracycline
- pUC19 carrying sites for many different restriction endonucleases
- pAT153 another derivative of pBR322
- Loss of copy number control –runaway replication

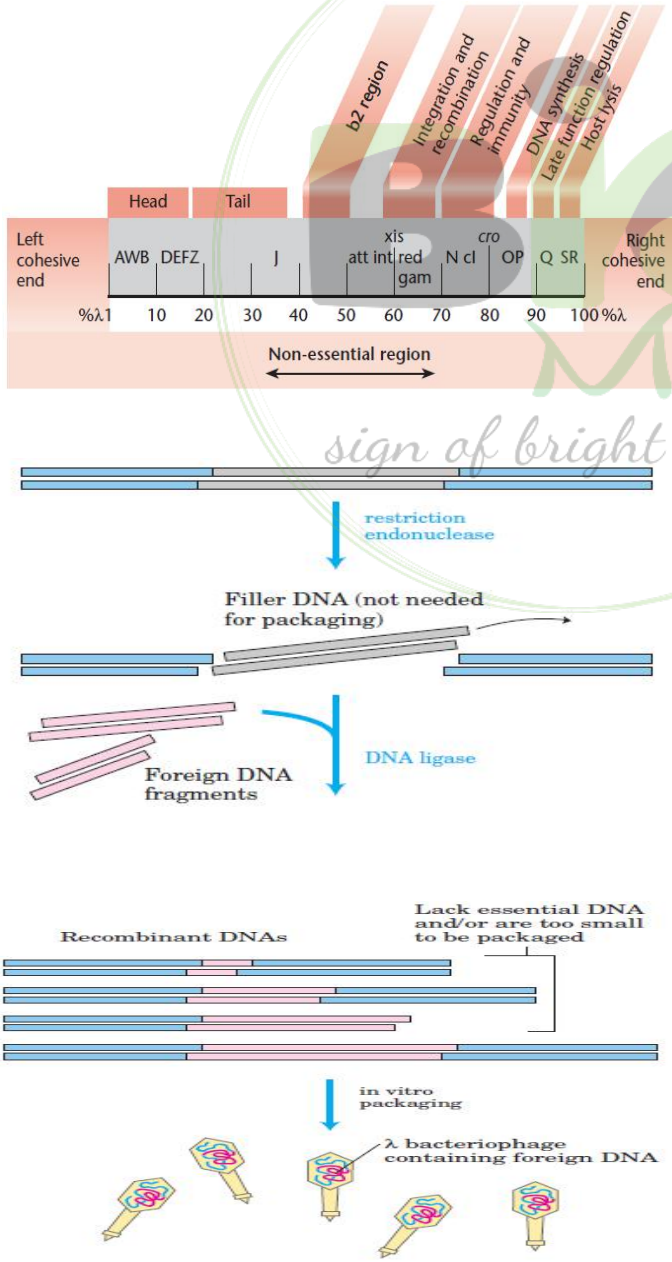
**100 Runaway plasmid vectors**

Loss of copy number control –runaway replication

101 Bacteriophage ʎ as cloning vector

Bacteriophage ʎ has a very efficient mechanism for delivering its 48,502 bp of DNA into bacterium

About one-third of the ʎ genome is nonessential and can be replaced with foreign DNA

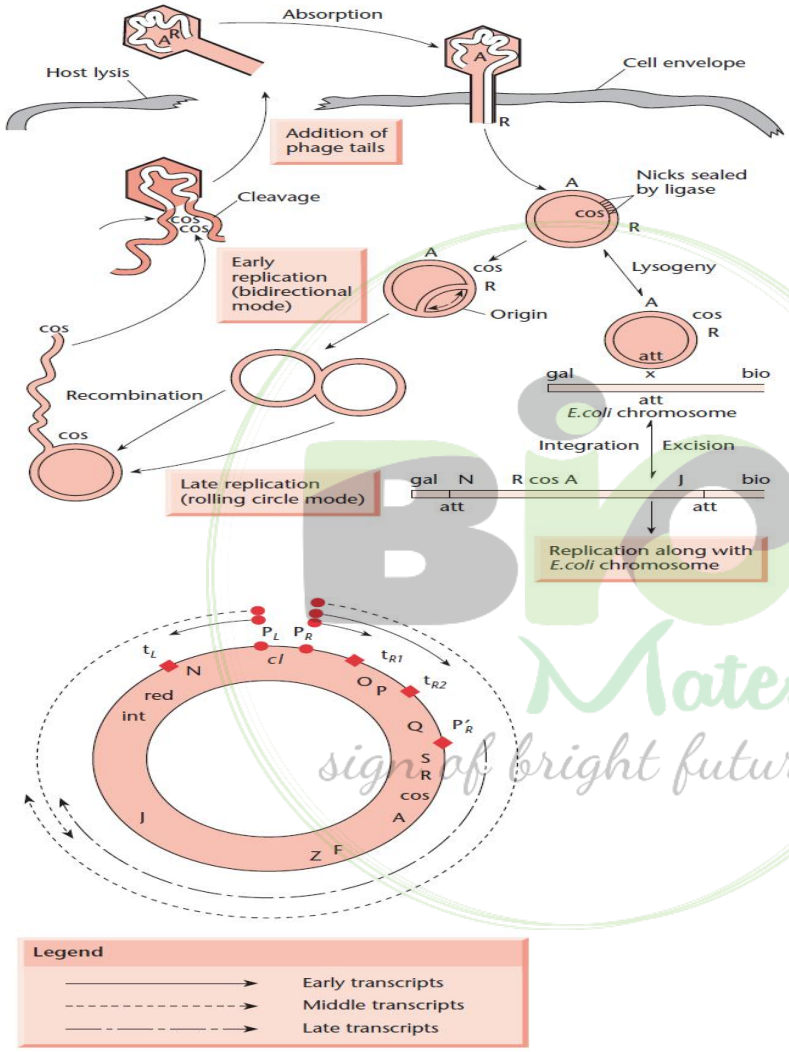



Bacteriophage ʎ permits the cloning of DNA fragments of up to 23,000 bp

**102 Replication of phage-λ DNA in lytic and lysogenic cycles**

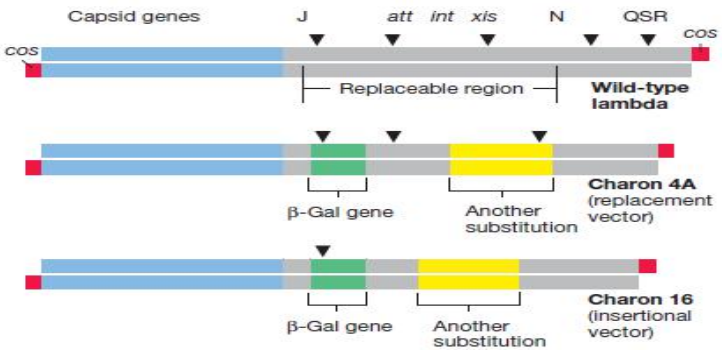
The lytic cycle results in the destruction of the infected bacterial cells

Viral DNA exists as separate molecule within the bacterial cell and replicates independently of the host DNA Lysogeny is characterized by the integration of the λ DNA into the host bacterium genome




**103 Modified λ phages**

Wild type is not suitable as a cloning vector because it has too many restriction enzymes sites




Insertional vectors

Replacement vectors

Charon 4A and 16 are both derivatives of lambda with various substitutions and deletions in the nonessential region

Each has the lacZ gene, encoding enzyme β-galactosidase, which permits detection of phage containing inserted clones

**104Steps in cloning with λ**

Cloning with lambda replacement vectors involves the following steps

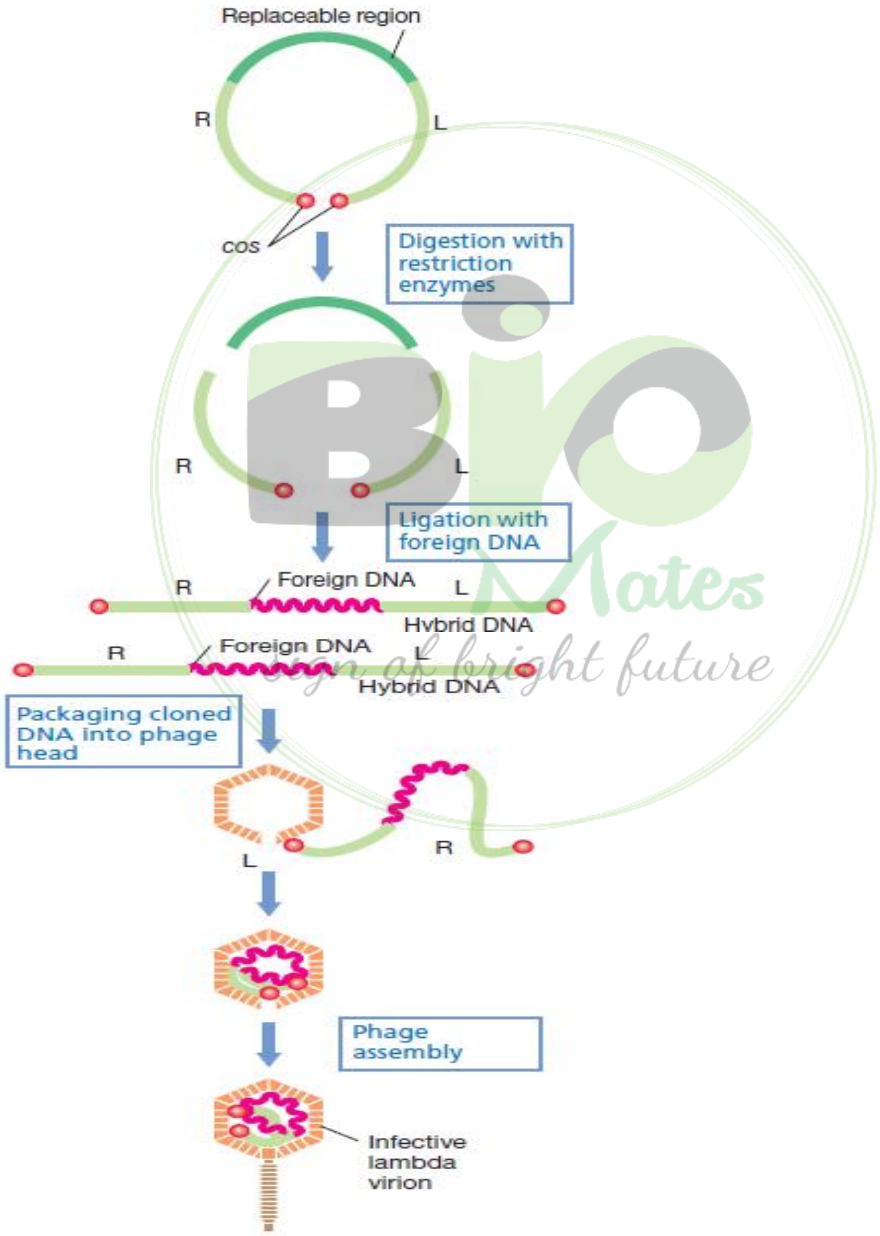
(i). Isolation and cutting vector DNA with appropriate restriction enzymes

(ii). Connecting the two lambda fragments to foreign DNA by using DNA ligase

(iii). In vitro packaging of recombinant DNA

(iv). Infection of E. coli cells and isolation of phage clones by picking plaques on a host strain

(v). Checking recombinant phage for the presence of foreign DNA

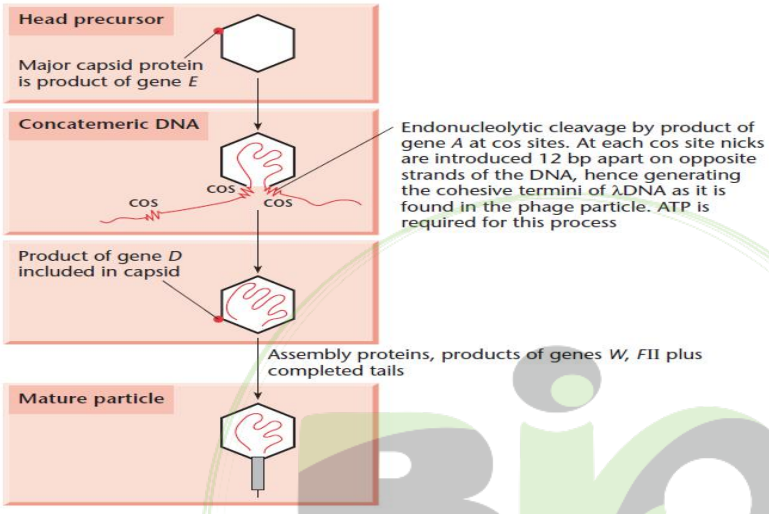



Both Charon vectors are also engineered to contain reporter genes such as for β-galactosidase for detection of recombinant phages

**105 Packaging phage-ʎ DNA in vitro**

Recombinant DNA in a phage coat allows it to be introduced into the host bacteria by the normal processes of phage infection

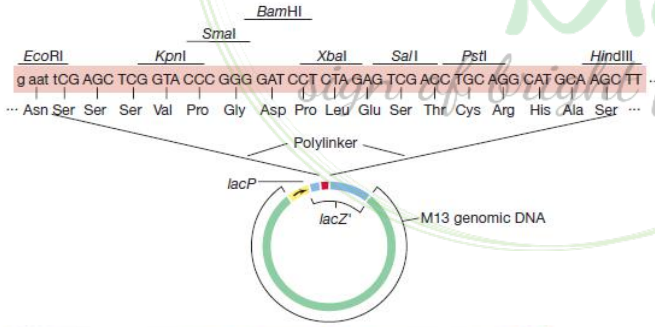
Packaging in vitro yields about 106 plaques/ µg of vector DNA

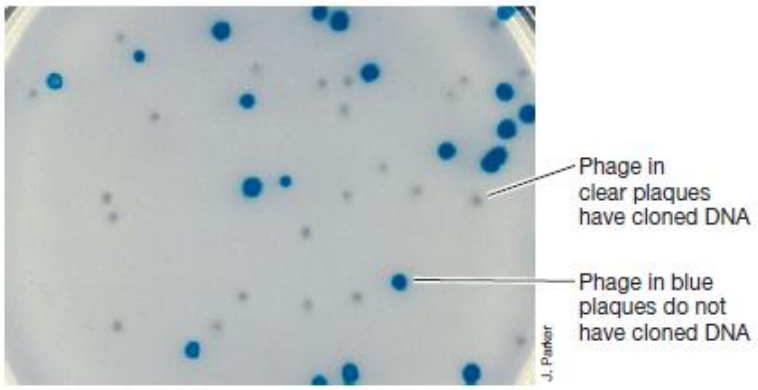



**106 Vectors for DNA sequencing: bacteriophage M13**

M13 is a filamentous bacteriophage with single stranded DNA that replicates without killing its host

Variable lengths of DNA upto about 5 kbp, can be cloned



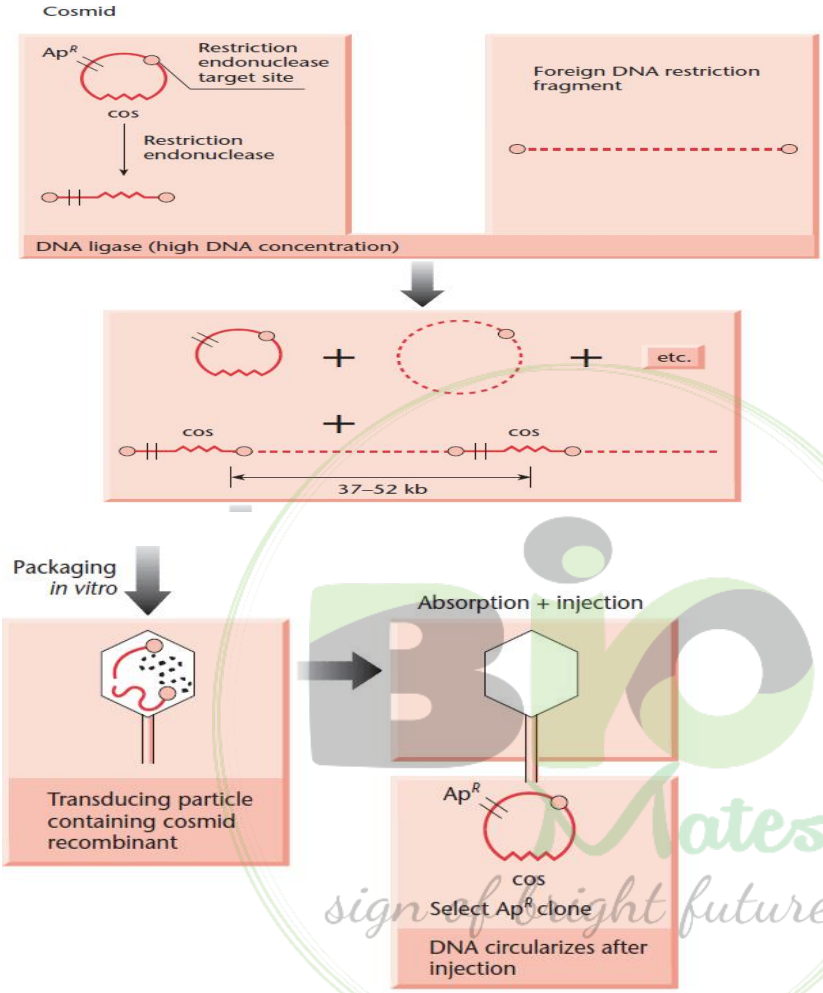




Use of M13 in molecular cloning

**107 Cosmid vectors**

Plasmids have been constructed which contain a fragment of ʎ DNA including 'cos' site

These plasmids have been termed cosmids and can be used as gene-cloning vectors in conjunction with the in vitro packaging system



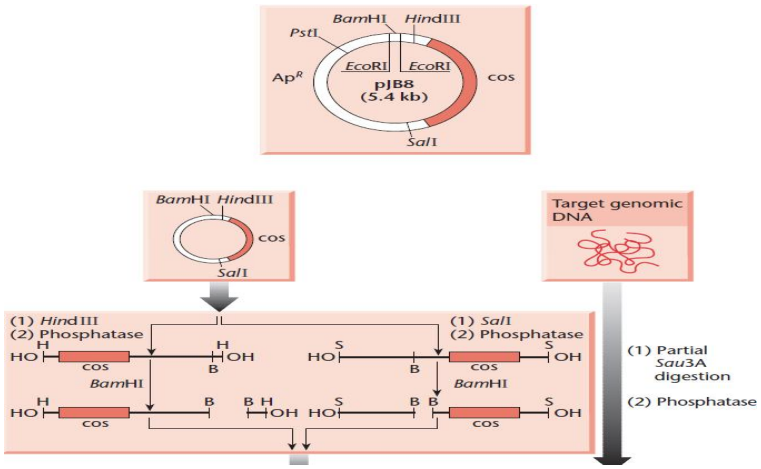

Cosmid vectors

Cosmids provide an efficient means of cloning large pieces of foreign DNA

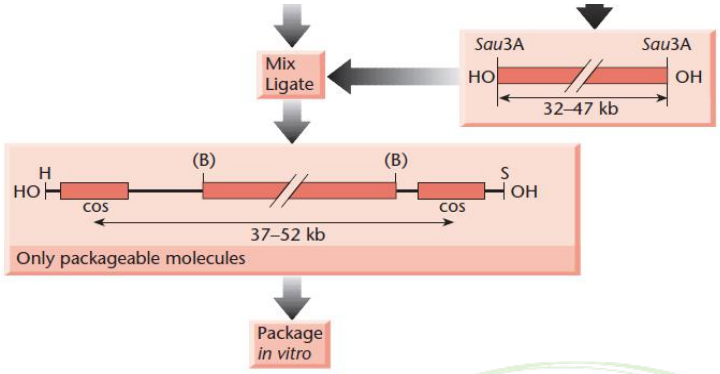
Cosmids are particularly attractive vectors for constructing libraries of eukaryotic genomic fragments

108 Modified schemes for cloning in Cosmid vectors

Difficulties in cloning with cosmids have been overcome in a cosmid cloning procedure devised by Ish-Horowicz and Burke (1981)




Plasmid cloning scheme of Ish-Horowicz and Burke (1981)




Plasmid cloning scheme of Ish-Horowicz and Burke (1981)

An alternate solution to these problems has been devised by Bates and Swift (1983) Who constructed cosmid c2XB

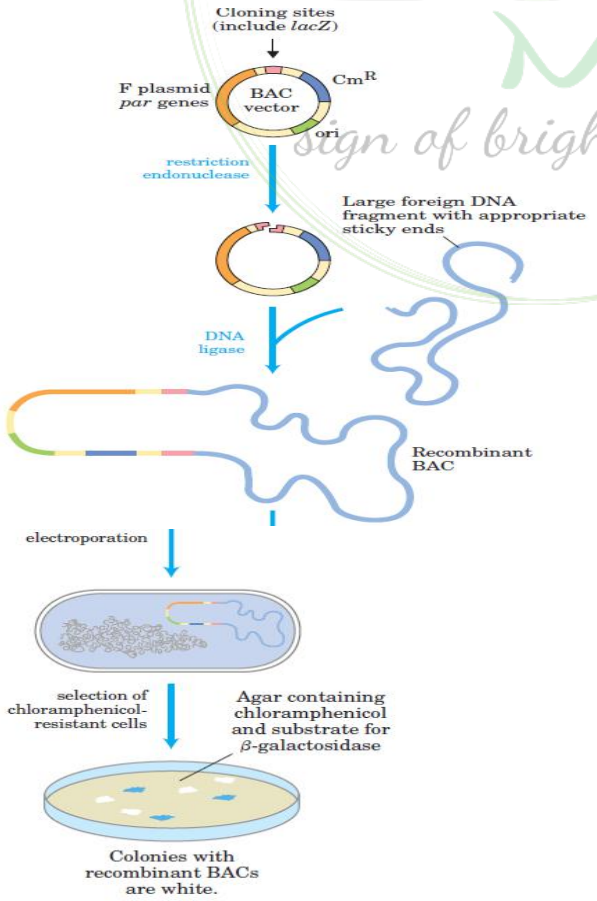
**109 Phasmid vectors**

Phasmid vectors are combination of plasmid and ʎ phage sequences

It consists of a plasmid vector carrying a ʎ attachment (ʎatt) site

**110 Bacterial artificial chromosomes (BACs)**

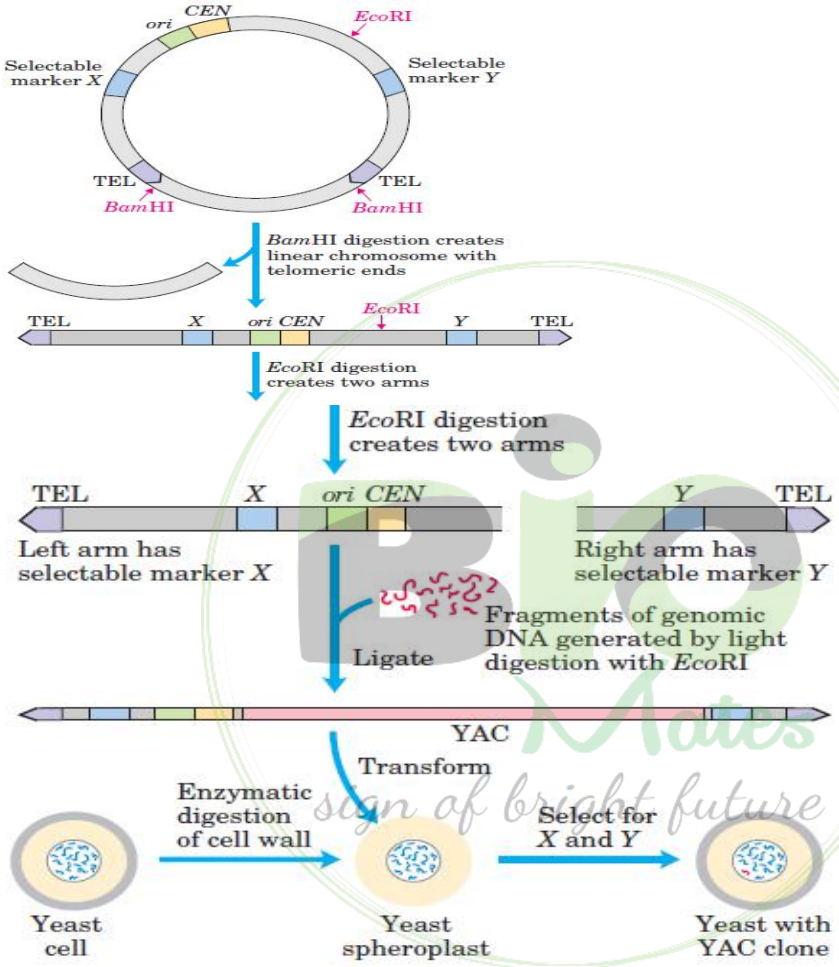
Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA (100,000 to 300,000 bp)




Bacterial artificial chromosome as cloning vectors

**111 Yeast artificial chromosomes (YACs)**

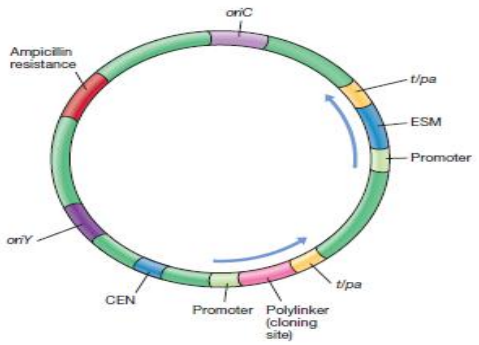
These vectors replicate in yeast like normal chromosome but they have sites where very large DNA fragments can be inserted




YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosomes

**112 Shuttle and Expression vectors**

Vectors that can replicate and are stably maintained in two or more unrelated host organisms are called shuttle vectors




To control the expression of cloned genes expression vectors are used

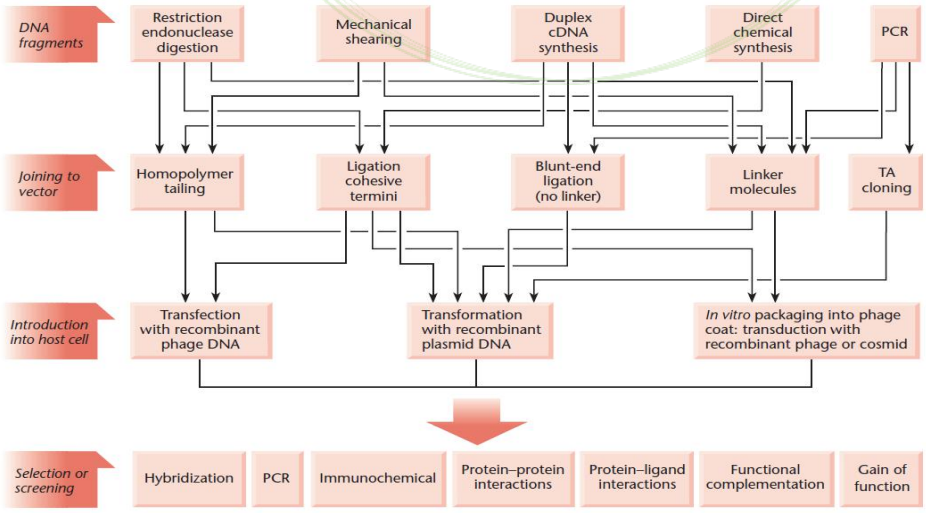
**113 Comparison of different cloning vectors-summary**

Different cloning vectors can carry variable sizes of target DNA into host organisms

| Vector | Host | Insert size |
|---|---|---|
| pBR322 | *E. coli* | 10-15 kb |
| ʎ phage | *E. coli* | 2-25 kb |
| ʎ Cosmids | *E. coli* | 35-45 kb |
| P1 phage | *E. coli* | 70-100 kb |
| PACs | *E. coli* | 100-300 kb |
| BACs | *E. coli* | ≤ 300 kb |
| YACs | *S. cerevisiae* | 200-2000 kb |

114
OVERVIEW OF CLONING STRATEGIES
Any cell based cloning procedure has four essential parts
i). A method of generating DNA fragment for cloning
ii). A reaction that inserts that fragment into the chosen cloning vectors

Introduction
iii). A means for introducing that recombinant vector into a host cell wherein it is replicated
iv). A method of selecting recipient cells that have acquired the recombinant




Introduction
There are two major strategies for isolating sequences from complex sources
i). Cell based cloning strategy, is to divide the source DNA into manageable fragments and clone everything

115 Genomic DNA libraries

Producing representative genomic libraries in ʎ cloning vector

| Organism | Genome size (kb) (haploid where appropriate) |
|---|---|
| *Escherichia coli* | $4.0 \times 10^3$ |
| Yeast (*Saccharomyces cerevisiae*) | $1.35 \times 10^4$ |
| *Arabidopsis thaliana* (higher plant) | $1.25 \times 10^5$ |
| Tobacco | $1.6 \times 10^6$ |
| Wheat | $5.9 \times 10^6$ |
| *Zea mays* | $1.5 \times 10^7$ |
| *Drosophila melanogaster* | $1.8 \times 10^5$ |
| Mouse | $2.3 \times 10^6$ |
| Human | $2.8 \times 10^6$ |
| *Xenopus laevis* | $3.0 \times 10^6$ |

Genomic DNA libraries

Cloning of large DNA fragments of 20 kb in ʎ replacement vectors, fewer clones are required for nearly complete library

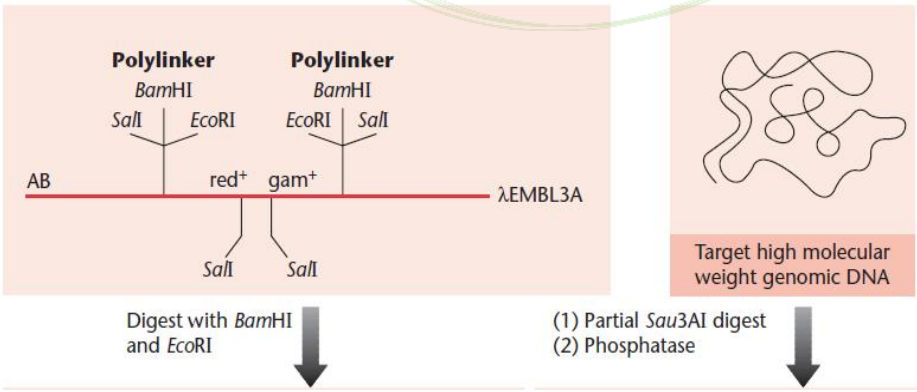
How can appropriately sized random fragments be produced?

i). Average fragment size can be controlled by mechanical shearing

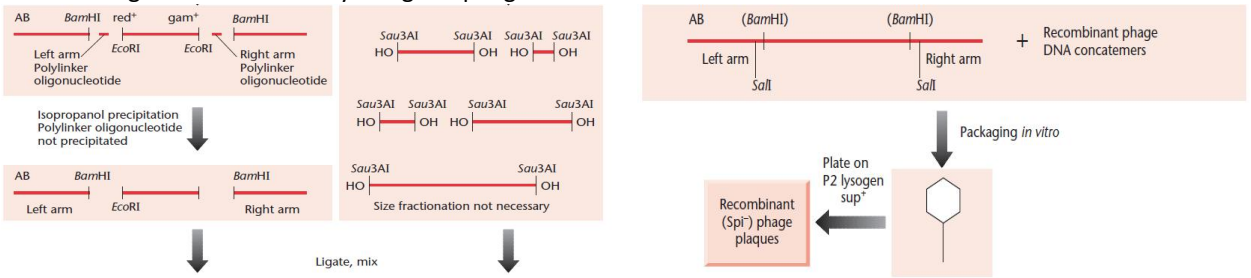
ii). The more commonly used procedure is restriction endonucleases

116 ʎEMBL vectors for genomic library construction

High molecular weight genomic DNA is digested with Sau3AI and subsequently ligated in ʎEMBL vector digested with BamH1




Creation of genomic DNA library using the phage-ʎ vector EMBL3A



**117 Genomic libraries in high-capacity vectors**

In place of phage-λ derivatives, a number of high capacity cloning vectors such as cosmids, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs) are available for construction of genomic libraries

**118 PCR as an alternative to genomic DNA cloning**

PCR with specific primers could be used to isolate genes directly from genomic DNA, obviating the need for the production of genomic libraries

**119 Properties of cDNA**

cDNA is prepared by reverse-transcribing cellular RNA
cDNA lack introns and other non-coding sequences present in the corresponding genomic DNA
Introns are rare in bacteria but occur in genes of higher eukaryotes

**120 cDNA libraries**

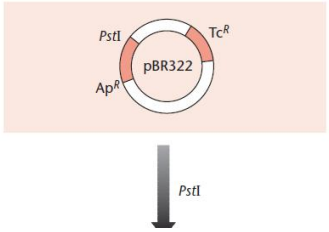
cDNA library is a combination of cloned cDNA fragments inserted into a collection of host cells
cDNA library is representative of the RNA population from which it was derived

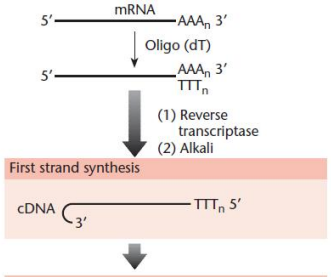
| Source | Number of different mRNAs | Abundance (molecules/cell) |
|---|---|---|
| Mouse liver cytoplasmic poly(A)$^+$ | 9 | 12 000 |
| | 700 | 300 |
| | 11 500 | 15 |
| Chick oviduct polysomal poly(A)$^+$ | 1 | 100 000 |
| | 7 | 4 000 |
| | 12 500 | 5 |

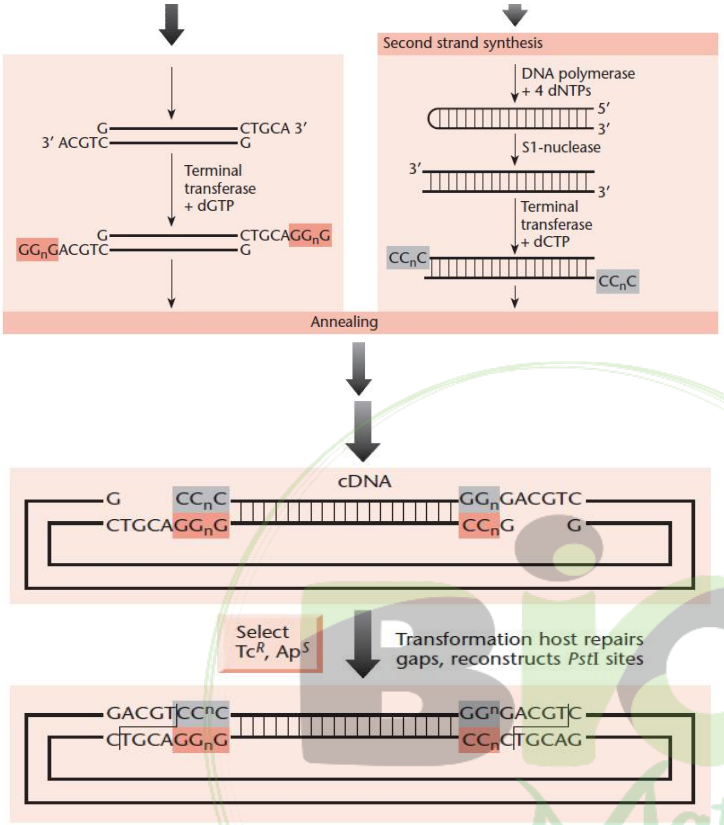
References: mouse (Young *et al.* 1976); chick oviduct (Axel *et al.* 1976).

cDNA libraries
Phage-λ vectors for cDNA cloning and expression
λgt10 and λgt11 vectors

121 Preparation of cDNA for library construction
The cDNA synthesis reaction
Development of cDNA cloning strategies



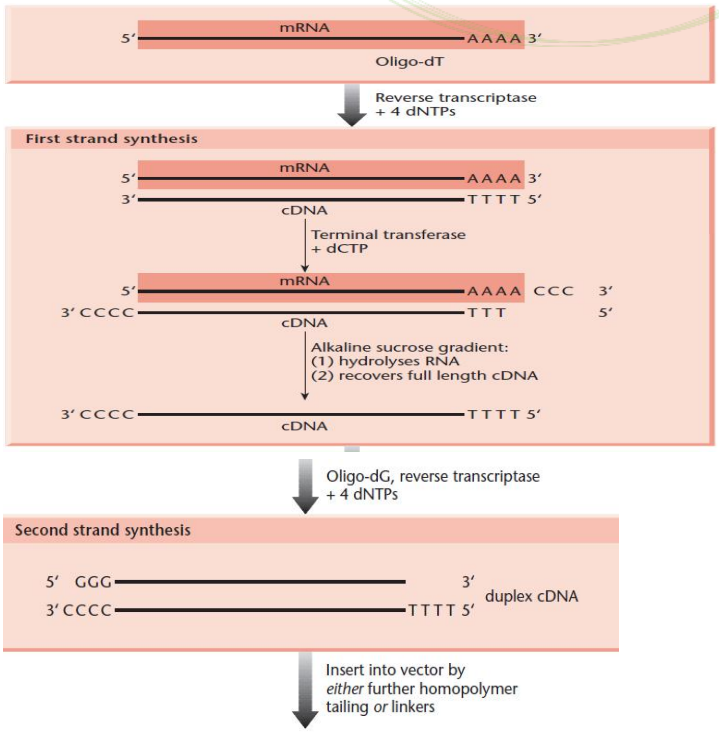






**122 Improved methods for cDNA cloning**

A serious disadvantage of the hairpin method is that cleavage with S1 nuclease results in the loss of sequences at the 5' end of the clone

This strategy has therefore been superseded with other methods




For cDNA expression libraries, it is advantageous if the cDNA can be inserted into the vector in the correct orientation

This can be achieved by self-priming method by adding linker molecule to double stranded cDNA

**123 PCR as an alternative for cDNA cloning**

Reverse transcription followed by the PCR (RT-PCR) leads to the amplification of RNA sequences in cDNA form

RT-PCR is a sensitive means for detecting, quantifying and cloning specific cDNA molecules

**124**

Major screening strategies involve

1). Genetic methods

2). Sequence-dependent screening

3). Screening expression libraries

All useful vector molecules carry a selectable genetic marker or property

Plasmid and cosmid vectors carry drug resistance or nutritional marker

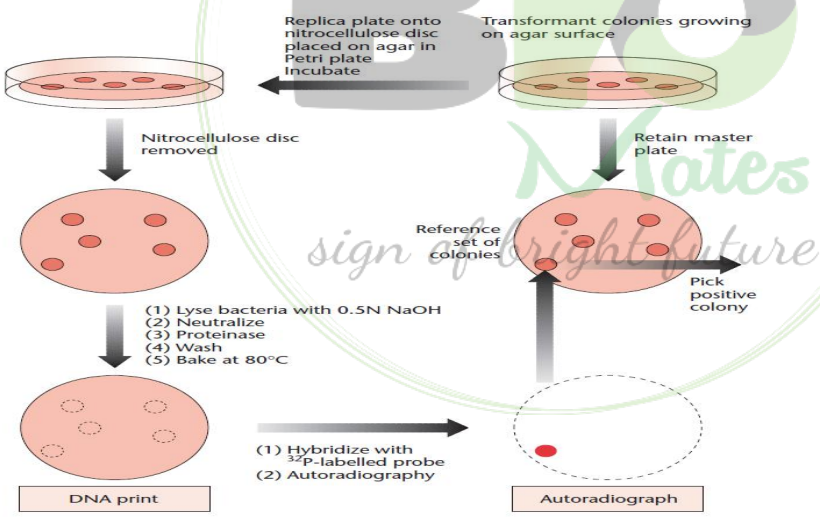
In phage vectors, plaque formation is itself the selected property

**125 Sequence-dependent screening**

Screening by hybridization

Nucleic acid hybridization is the most commonly used method of library screening

Grunstein and Hogness (1975) developed a screening procedure to detect DNA sequences in transformed colonies by in situ hybridization



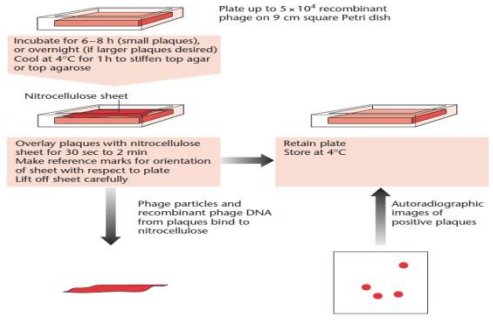

Screening by hybridization

The results of the hybridization can be monitored by autoradiography

**126 Sequence-dependent screening**

Benton and Davis' plaque lift procedure

Benton and Davis (1977) devised a method called plaque lift, in which nitrocellulose filter is applied to the upper surface of agar plates, making direct contact between plaques and filter



Benton and Davis' plaque lift procedure

Number of alternative labeling methods are available that avoid the use of radioactivity. It included digoxigenin or biotin

**127 Probe design**

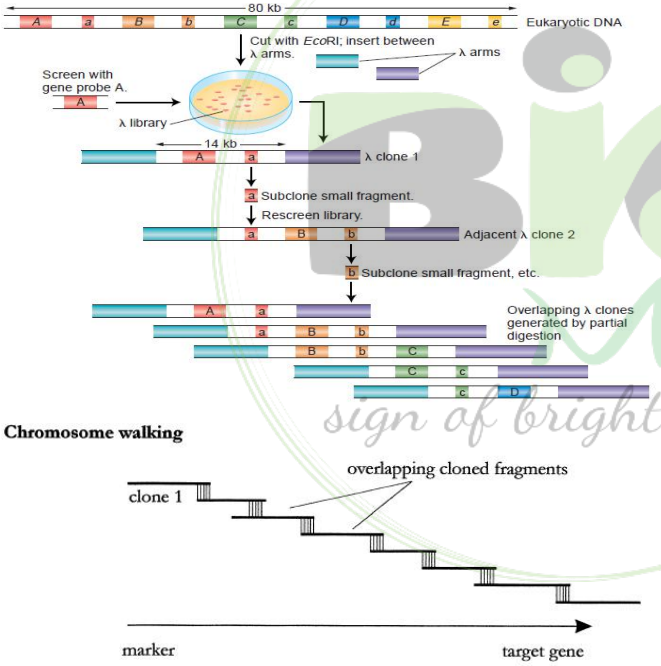
A great advantage of hybridization for library screening is that it is extremely versatile

Conditions can be used in which hybridization is very stringent, so that only sequences identical to the probe are identified

**128 Chromosome walking**

Walking along the chromosome is a term used to describe an approach which allows the isolation of gene sequences whose function is quite unknown but whose genetic location is known

A cloned genomic fragment must be found as a starting point for the walk In human genome, the starting point may be a restriction fragment length polymorphism (RFLP) sequence that is closely linked to disease locus





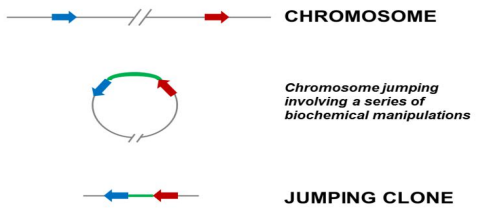


**129 Chromosome jumping**

In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and circularized

It brings together DNA sequence that were originally located a considerable distance apart in the genome

These cloned DNAs from the closure sites make up a jumping library




One of the application of chromosome walking and jumping is in the cloning of human cystic fibrosis gene

**130 Screening by PCR**

The PCR is widely used to isolate specific DNA sequences from uncloned genomic DNA or cDNA but it also a useful technique for library screening

To isolate specific clone, PCR is carried out with gene-specific primers that flank a unique sequence in the target

There are several applications where the use of degenerate primes is favorable

A degenerate primer is a mixture of primers, all of similar sequences but with variations at one or more positions

**131 Expression cloning**

If DNA library is established using expression vectors, each individual clone can be expressed to yield a polypeptide

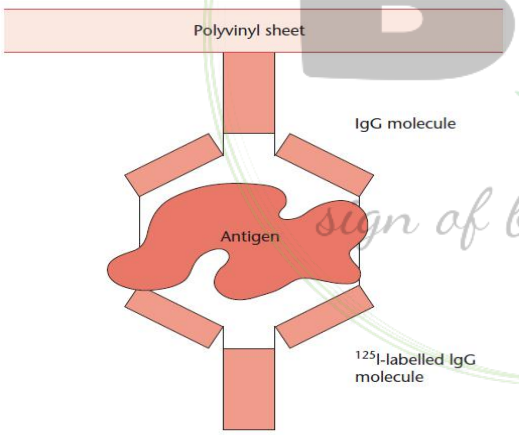
Expression libraries are useful because they allow a range of alternative techniques to be employed

**132 Immunochemical screening**

Immunochemical screening involves the use of antibodies that specifically recognize antigenic determinants on the polypeptides synthesized by a target clone

The molecular target for recognition is generally an epitope

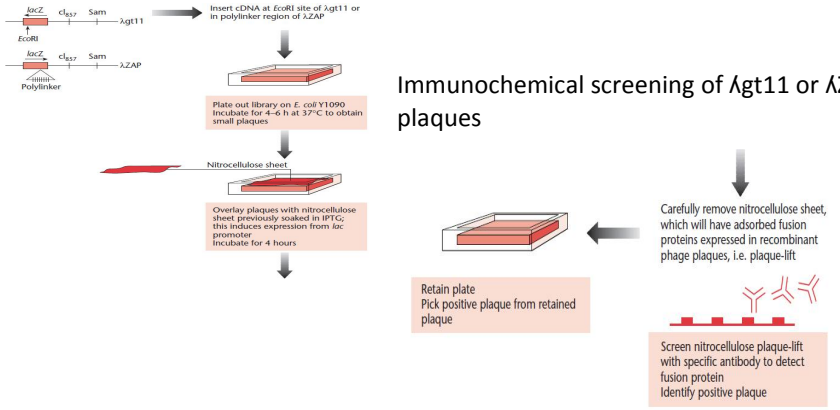
The method of Broome and Gilbert (1978) exploited the fact that antibodies adsorb very strongly to certain types of plastic such as polyvinyl and that IgG antibodies can be readily labelled with 125I by iodination in vitro




Immunochemical detection method of Broome and Gilbert

**133 Immunochemical screening of λgt11**

It is much more convenient to use bacteriophage-λ insertion vectors, because these have a higher capacity and the efficiency of in vitro packaging allows large numbers of recombinants to be prepared and screened




Immunochemical screening of λgt11 or λZAP recombinant plaques

The original detection method using iodinated antibodies has been superseded by more convenient methods using non-isotopic labels, which are also more sensitive and have a lower background of non-specific signal

**134 South-western and north-western blotting**

It involves the screening and isolation of clones expressing sequence specific DNA or RNA-binding proteins

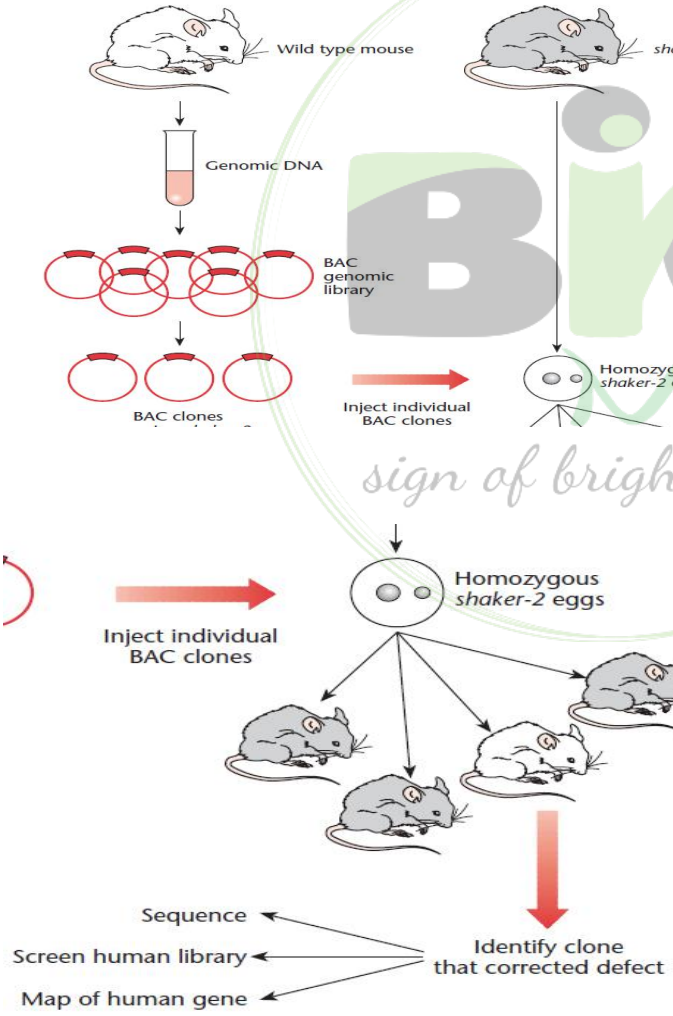
Screening is carried out without using an antibody, by incubating the membranes with radiolabelled double stranded DNA probe

**135 Screening by functional complementation**

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype

Functional complementation is also possible in transgenic animals and plants

It has been used for complementation in transgenic mice to isolate the Shaker-2 gene




**136 Requirement for expression in E. coli**

Synthesis of functional protein depends upon transcription of appropriate gene, efficient translation of mRNA and in many cases, post-translational processing and compartmentalization of nascent polypeptide

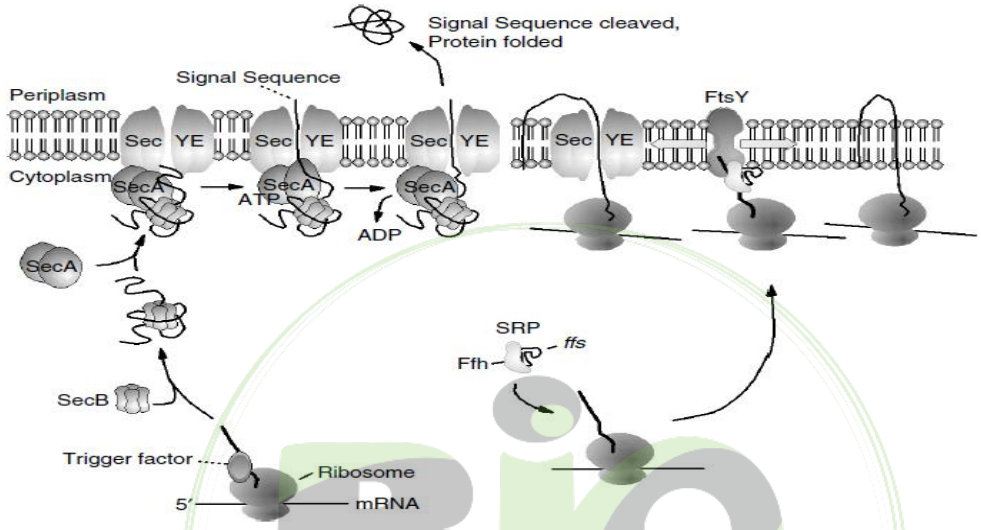
**137 Secretion of proteins**

Gram-negative bacteria such as E. coli have a complex wall-membrane structure comprising an inner, cytoplasmic membrane separated from an outer membrane by a cell wall and periplasmic space

**138 Protein trafficking**

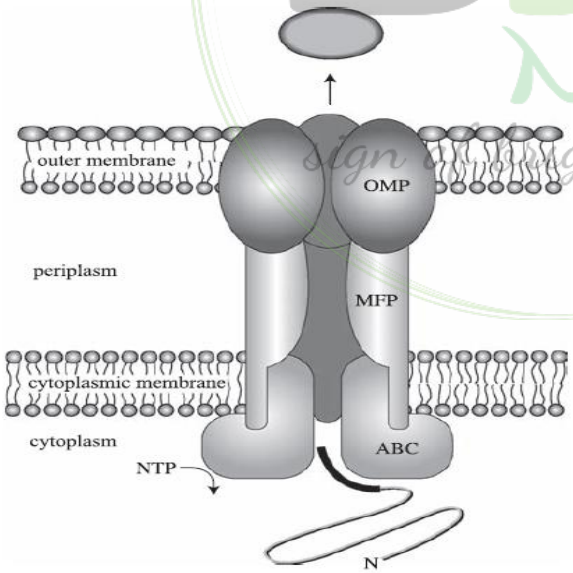
The bacterial inner membrane, periplasmic space and outer membrane all contain proteins not found in the cytoplasm

All of these proteins are first synthesized in the cytoplasm but somehow find their way out




Protein trafficking

There are actually several general mechanisms of protein export in bacteria




ABC pathway for protein export

**139 Stability of foreign proteins in E. coli**

Various strategies have been developed to cope with the instability of foreign proteins in E. coli

In the case of somatostatin, degradation was prevented by producing a fused protein consisting of somatostatin and β-galactosidase

**140 Constructing the optimal promoter**

Large number of promoters for E. coli have been analysed

Many promoters has led to the formulation of a consensus sequence which consists of the -35 region (5'-TTGACA-) and -10 region or Pribnow box (5'-TATAAT-)




Figure. The base sequence of the -10 and -35 regions for natural and hybrid promoters
Constructing the optimal promoter
Expression from a strong promoter can represent 20-40% of cloned gene product of toral cell protein

**141 Optimizing translation initiation**
Complementarity of Shine-Dalgarno (S-D) sequence with 16s rRNA can affect the rate of translation
Composition of triplet immediately preceding the AUG start codon also affects the efficiency of t translation

**142 Stability of mRNA and codon choice**
The rate of synthesis of a particular protein will depend on the steady-state of mRNA in the cell
Degradation of mRNA usually proceeds by a combination of endonuclease and 3'-exonuclease attack

**143 The effect of plasmid copy number**
The number of ribosomes in a cell far exceeds any one class of mRNA
One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcript

**144 Plasmid stability**
Having maximized the expression of a particular gene it is important to consider what effects this will have on the bacterium harbouring the recombinant plasmid
The loss of plasmids due to defective partitioning is called segregative instability

**145 Structural instability**
Structural instability of plasmids may arise by deletions or rearrangements of DNA

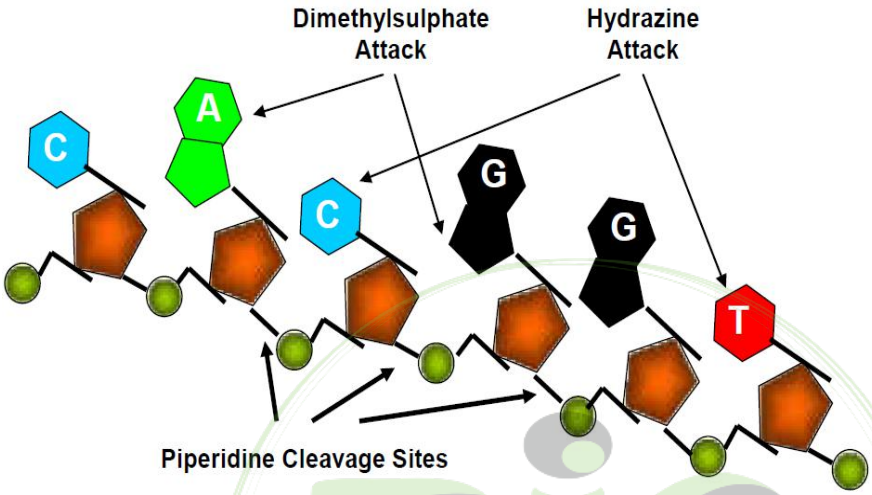
**146 Host cell physiology can affect the level of expression**
Factors which will be important include the choice of nutrients and environmental parameters

**147 DNA sequencing: Benefits and Applications**
DNA sequence information is a prerequisite for planning any substantial manipulation of the DNA
Information is useful to various fields i.e. molecular and evolutionary biology, metagenomics, medicine, forensics etc.

DNA sequencing: Benefits and Applications
Techniques for DNA sequencing became available in the late 1970s

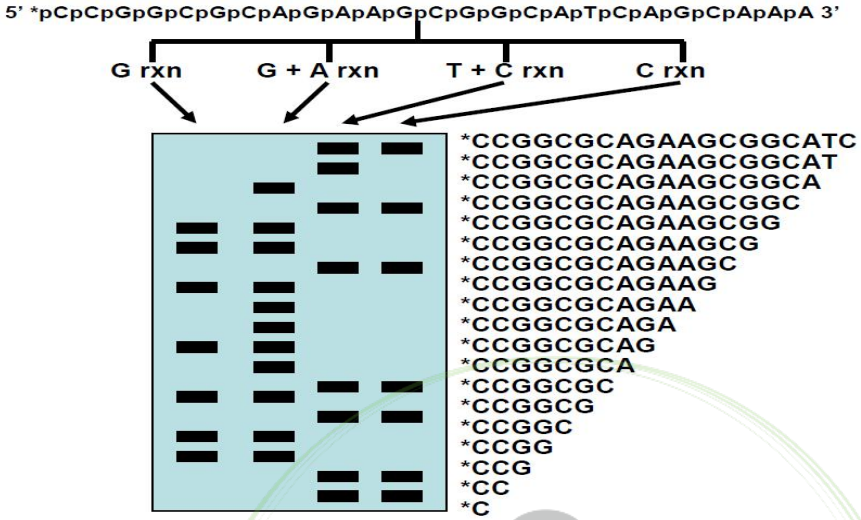
**148 Maxam-Gilbert method**

This method for DNA sequencing makes use of chemical reagents to bring about base-specific cleavage of the DNA




| Base specificity | Base reaction | Altered base removal | Strand cleavage |
|---|---|---|---|
| G | Dimethylsulphate | Pepridine | Pepridine |
| G+A | Acid | Acid-catalysed depurination | Pepridine |
| T+C | Hydrazine | Pepridine | Pepridine |
| C | Hydrazing+NaCl | Pepridine | Pepridine |
| A > C | NaOH | Pepridine | Pepridine |

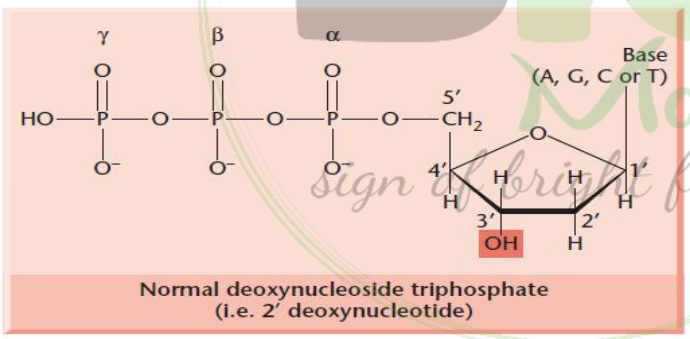
Table. Reagents for Maxam and Gilbert DNA sequencing

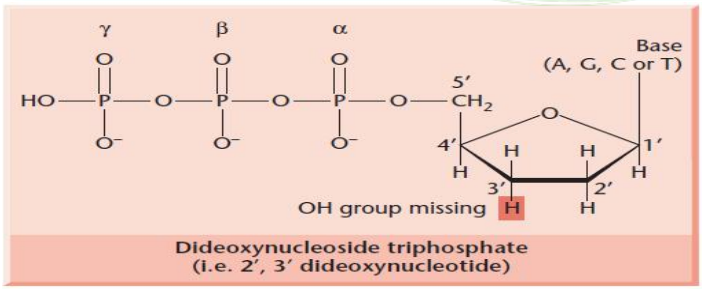



**149 Chain termination or dideoxy procedure**

This method was developed by Sanger et al. (1977)

It capitalizes on the ability of DNA polymerase to incorporate dideoxynucleoside triphosphates as substrate

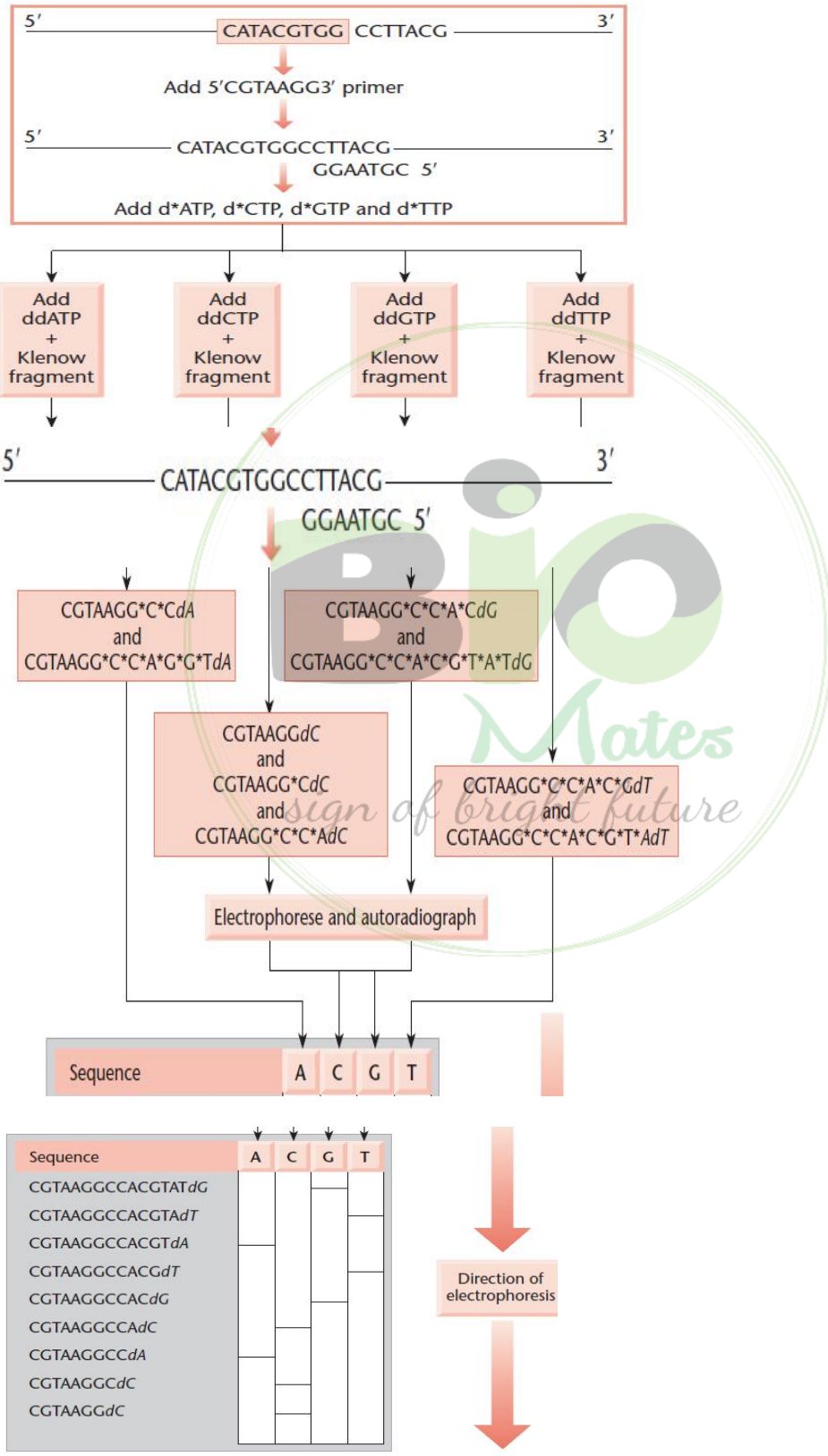


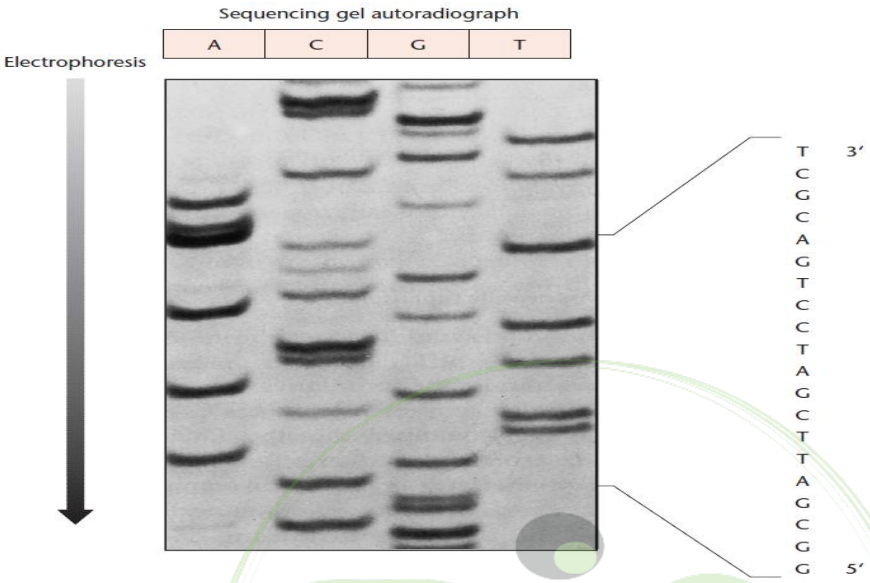




Dideoxynucleoside triphosphate act as chain terminator

Chain termination or dideoxy procedure

DNA synthesis is carried out in the presence of the four deoxynucleoside triphospahtes, one or more of which is labelled with 32P and in four separate incubation mixes


5′ CATACGTGG CCTTACG 3′
Add 5′CGTAAGG3′ primer
5′ CATACGTGGCCTTACG 3′
GGAATGC 5′
Add d*ATP, d*CTP, d*GTP and d*TTP
Add ddATP + Klenow fragment
Add ddCTP + Klenow fragment
Add ddGTP + Klenow fragment
Add ddTTP + Klenow fragment
5′ CATACGTGGCCTTACG 3′
GGAATGC 5′
CGTAAGG*C*CdA and CGTAAGG*C*C*A*G*T*dA
CGTAAGG*C*C*A*CdG and CGTAAGG*C*C*A*C*G*T*A*TdG
CGTAAGGdC and CGTAAGG*CdC and CGTAAGG*C*C*AdC
CGTAAGG*C*C*A*C*GdT and CGTAAGG*C*C*A*C*G*T*AdT
Electrophorese and autoradiograph
Sequence A C G T
Sequence A C G T
CGTAAGGCCACGTATdG
CGTAAGGCCACGTAdT
CGTAAGGCCACGTdA
CGTAAGGCCACGdT
CGTAAGGCCACdG
CGTAAGGCCAdC
CGTAAGGCCdA
CGTAAGGCdC
CGTAAGGdC
Direction of electrophoresis



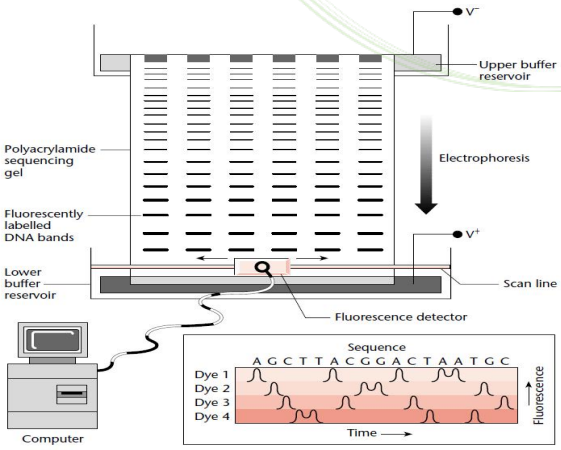

**150 Modifications of chain terminator sequencing**

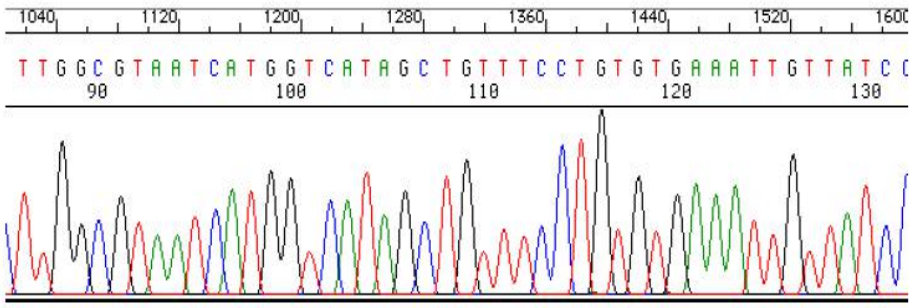
Improvements to original Sanger's method have been made by replacing the Klenow fragment of E. coli DNA polymerase I

The combination of chain terminator and M13 vectors to produce single stranded DNA is very powerful

**151 Automated DNA sequencing**

To automate the process, it is desirable to acquire sequence data in real time by detecting the DNA bands within the gel during the electrophoretic separation



Automated DNA sequencing

Automated DNA sequencing offers a number of advantages

**152 Sequencing accuracy**

Sequences that were read beyond 4oo bp contained an average of 3.2% error, while those less than 400 bp had 2.8% error

**153 DNA sequence databases**

Since the development of current DNA sequencing technology large amount of sequence data has accumulated that is maintained in 3 data bases

i). National center for Biotechnology Information
ii). DNA Databank of Japan
iii). European Bioinformatic institute-UK

**154 Mutagenesis**

Mutagenesis is a process to change the genetic information of an organism

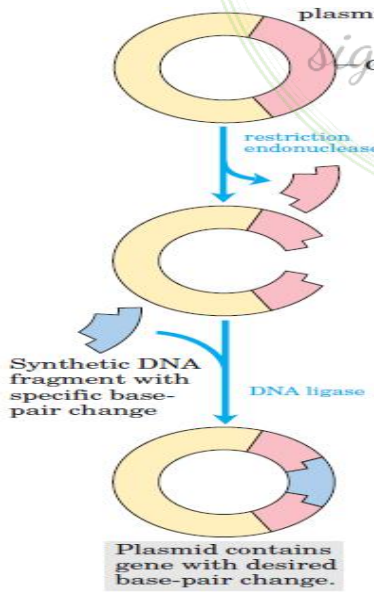
It can occur naturally, or as a result of exposure to mutagens or induced experimentally in laboratory Mutagenesis

Three different methods of site-directed mutagenesis have been devised

i). Cassette mutagenesis
ii). Primer extension
iii). Procedures based on PCR

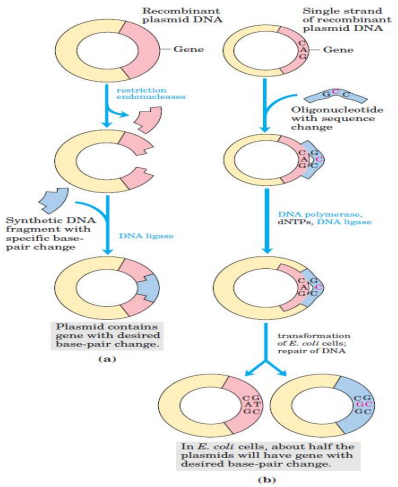
**155 Cassette mutagenesis**

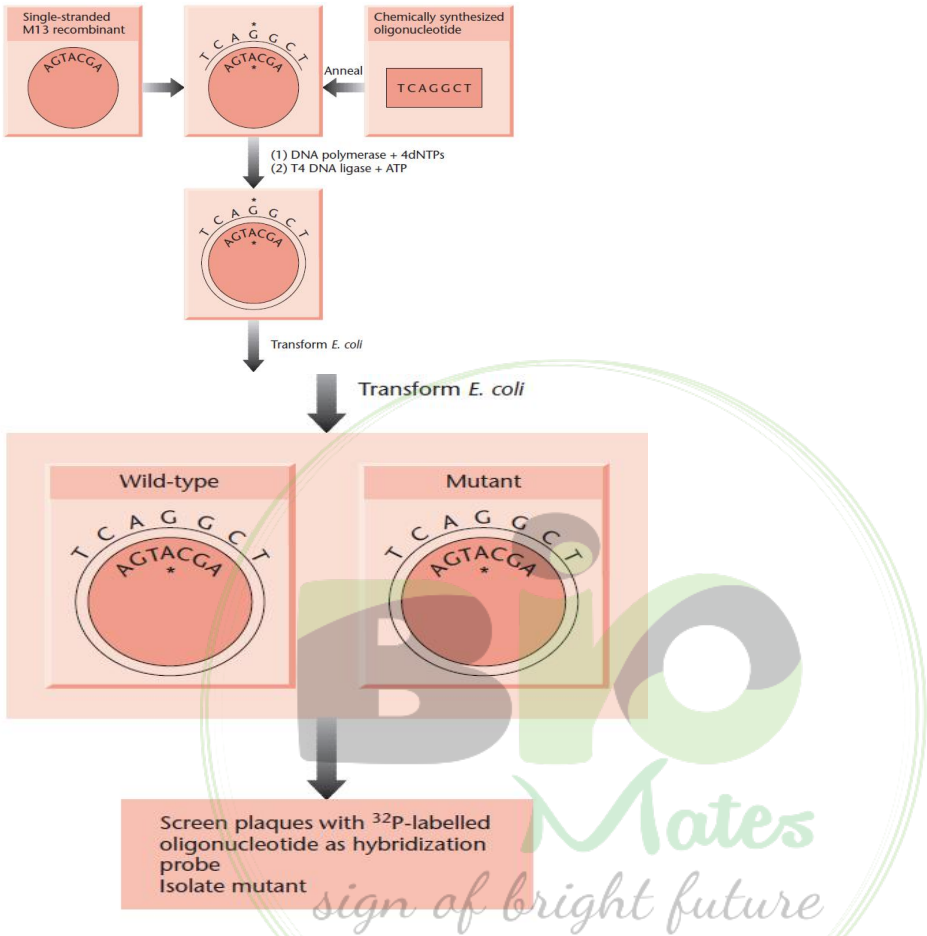
A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene




**156 Primer extension: the single primer method**

It involves priming in vitro DNA synthesis with a chemically synthesized oligonucleotide (7-20 bp) that carries a base mismatch with the complementary sequence

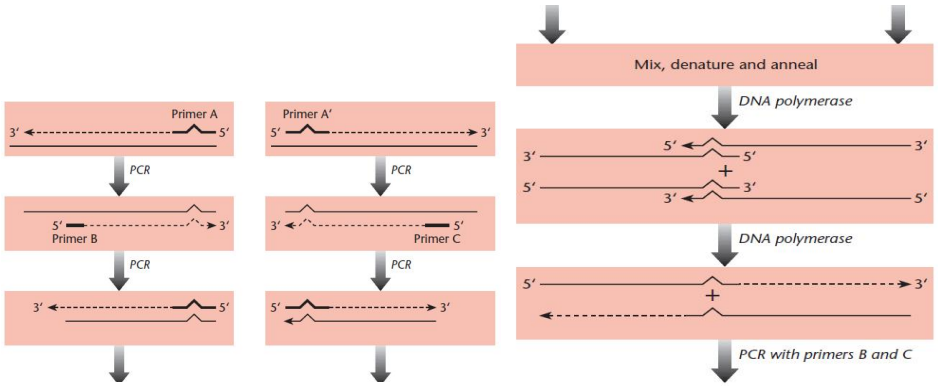





After E. coli transformation, heteroduplex that are formed are either that of original wild type DNA or that containing the mutated base

**157 PCR methods for site-directed mutagenesis**

Single bases mismatched between the amplification primer and the template become incorporated into the template sequence as a result of amplification




Site directed mutagenesis by means of the PCR

Site directed mutagenesis by means of the PCR

In megaprimer method, the product of the first PCR is used for the second PCR

**160 DNA polymerases**

DNA polymerases are enzymes that synthesize DNA molecules by adding nucleotide, the building blocks of DNA

Table. Comparison of DNA polymerases of E. coli

| | DNA polymerase | | |
|---|---|---|---|
| | I | II | III |
| Structural gene* | *polA* | *polB* | *polC* (*dnaE*) |
| Subunits (number of different types) | 1 | 7 | ≥10 |
| $M_r$ | 103,000 | 88,000† | 791,500 |
| 3′→5′ Exonuclease (proofreading) | Yes | Yes | Yes |
| 5′→3′ Exonuclease | Yes | No | No |
| Polymerization rate (nucleotides/s) | 16-20 | 40 | 250-1,000 |
| Processivity (nucleotides added before polymerase dissociates) | 3-200 | 1,500 | ≥500,000 |

Taq DNA polymerase

Taq polymerase is a thermostable DNA polymerase isolated from thermophilic bacterium Thermus aquaticus and is used in PCR to amply DNA fragments

| DNA polymerase | Source |
|---|---|
| *Tma* | *Thermotoga maritima* |
| Deep Vent™ | *Pyrococcus* sp. |
| *Tli* | *Thermococcus litoralis* |
| *Pfu* | *Pyrococcus furiosus* |
| *Pwo* | *Pyrococcus woesi* |

Table. Sources of thermostable DNA polymerases

162 Degenerate primers

A degenerate primer is actually a mixture of primers, all of similar sequence but with variations at one or more positions

| One Codon | Two Codons | Three Codons | Four Codons | Six Codons |
|---|---|---|---|---|
| Met (M) | Cys (C) | Ile (I) | Ala (A) | Leu (L) |
| Trp (W) | Asp (D) | | Gly (G) | Arg (R) |
| | Glu (E) | | Pro (P) | Ser (S) |
| | Phe (F) | | Thr (T) | |
| | His (H) | | Val (V) | |
| | Lys (K) | | | |
| | Asn (N) | | | |
| | Gln (Q) | | | |
| | Tyr (Y) | | | |

| Symbol | Base represented |
|---|---|
| A | A |
| C | C |
| G | G |
| T | T |
| R | A, G |
| Y | C, T |
| M | A, C |
| K | G, T |
| H | A, C, T |
| B | C, G, T |
| V | A, C, G |
| N | A, C, G, T |

‘’

Degenerate primers

A degenerate primer is actually a mixture of primers, all of similar sequence but with variations at one or more positions

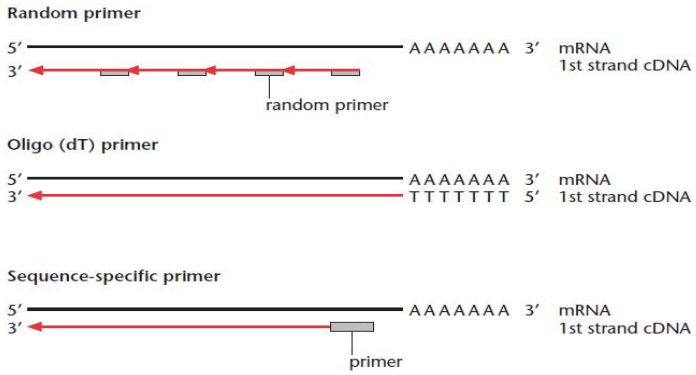
**163 Types of PCR**

Recently, variants have been developed from basic PCR method to improve performance and specificity and to amplify other molecules such as RNA

**RT-PCR**

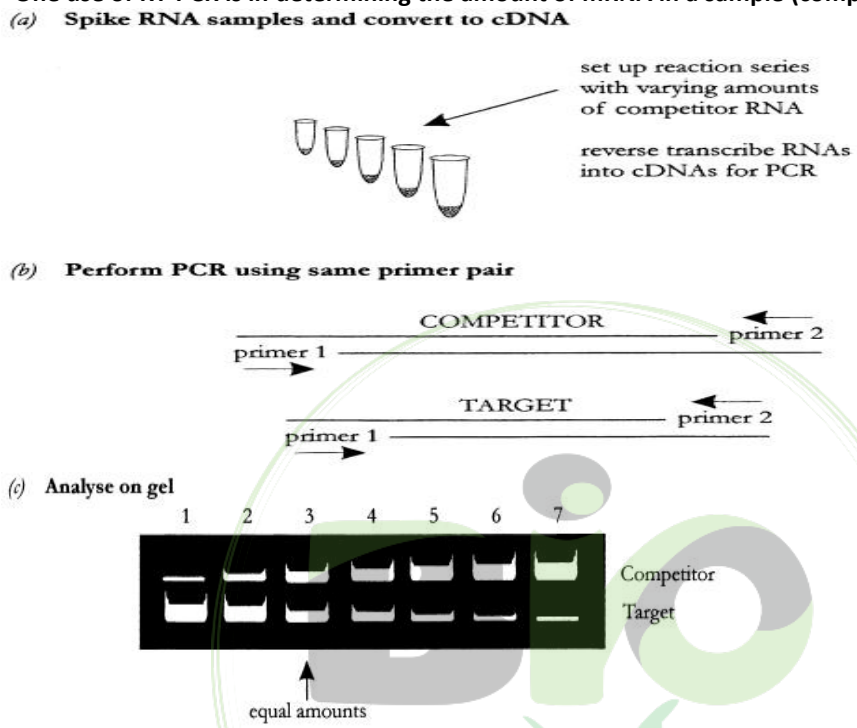
Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase

Various strategies can be adopted for first-strand cDNA synthesis



**164Competitor RT-PCR**

**One use of RT-PCR is in determining the amount of mRNA in a sample (competitor RT-PCR)**



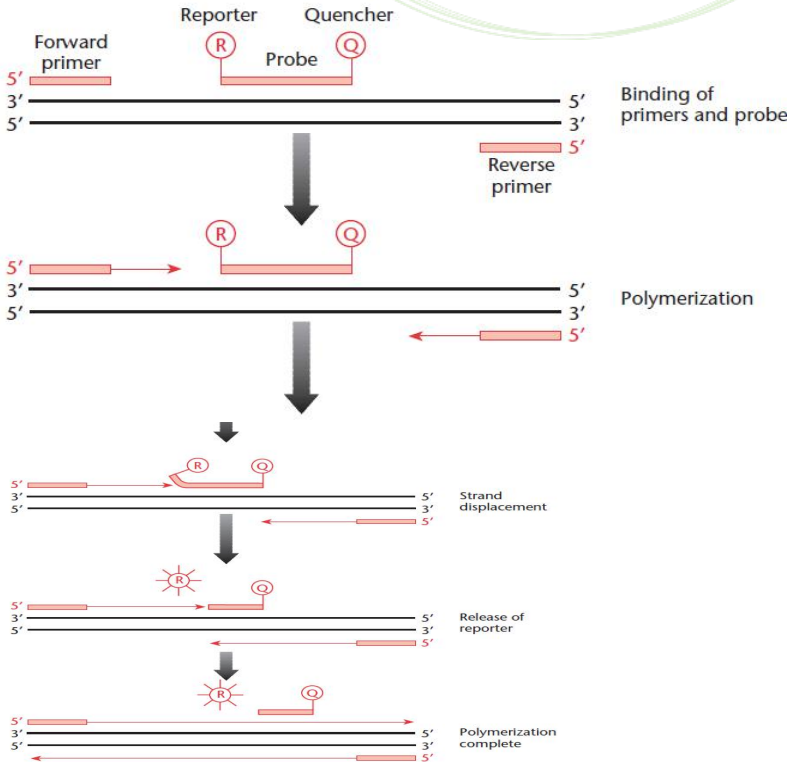

RT-PCR

Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase

**165 Real-time quantitative PCR (qPCR)**

qPCR is used to measure the quantity of a target sequence

It is used to measure starting amounts of DNA, cDNA, or RNA



Real-time quantitative PCR (qPCR)

qPCR is used to measure the quantity of a target sequence

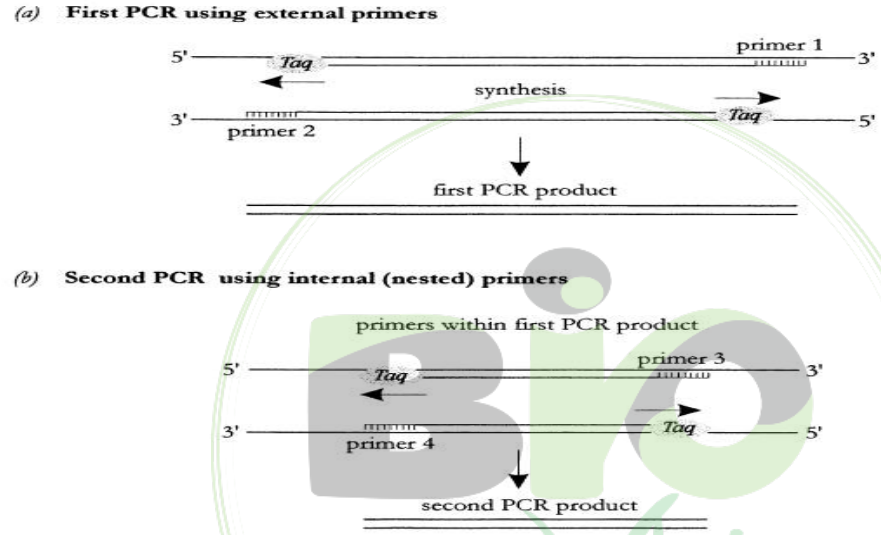
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**166 Nested PCR**

In nested PCR, two sets of primers are used in two successive PCRs

First set of primers allows a first amplification

Product of first PCR is subjected to a second PCR using the second set of primers




Nested PCR

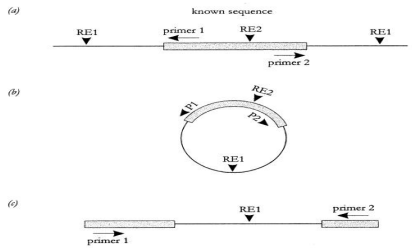
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First set of primers allows a first amplification

Product of first PCR is subjected to a second PCR using the second set of primers

**167 Inverse PCR**

Inverse PCR is used to identify the flanking sequences around genomic inserts




168 Multiplex PCR

Multiplex PCR is used to amplify several different DNA sequences simultaneously

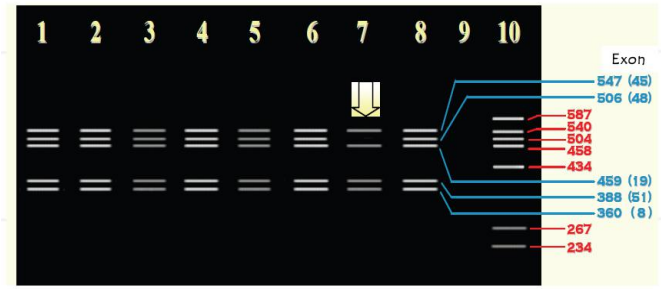
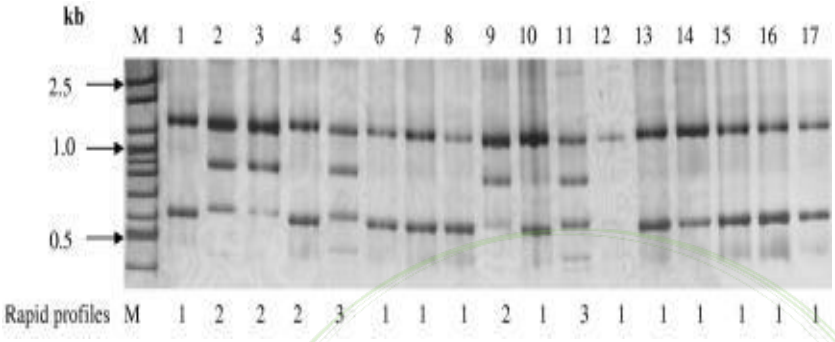



Fig. Results of multiplex PCR in a patient with Duchenne Muscular Dystrophy Inverse PCR
Hernandex-Rodriguez et al., 2000;
Hernandex-Rodriguez & Restrepo, 2002
(www.intechopen.com)

**169 RAPD**

RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random

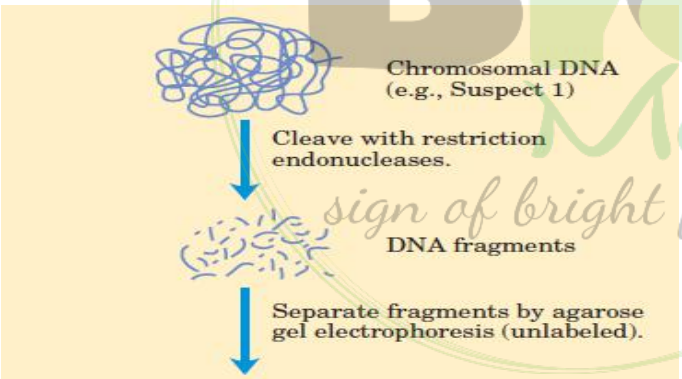



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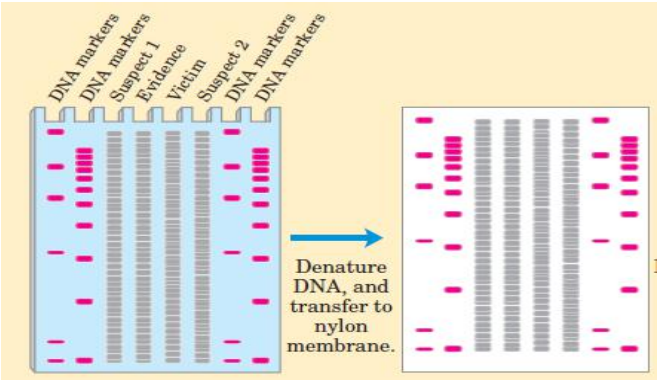
**170 RFLP**

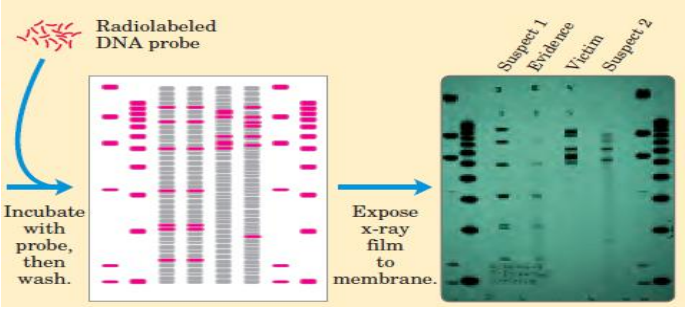
RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences

In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis




DNA fingerprinting



**171 AFLP-PCR**

AFLPs-Amplified fragment length polymorphisms are differences in restriction fragments length caused by SNPs or INDELS

AFLP is a PCR based tool used in genetic engineering.

**172 Applications of PCR**

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

PCR has been used in gene cloning and screening of genomic libraries

**173 PCR-Medicine**

PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis

Molecular tools have also allowed to perform prenatal genetic diagnosis

**174 PCR-Forensic sciences**

Forensic science is the application of scientific procedures to solve criminal and legal matters

Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

PCR-DNA profiling

DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

**175 PCR-Agricultural sciences and environment**

PCR has also facilitated research in detection of pathogens in plants, animals and environment

**176 PCR-Molecular paleontology**

Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains

**177 Genome mapping**

Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome

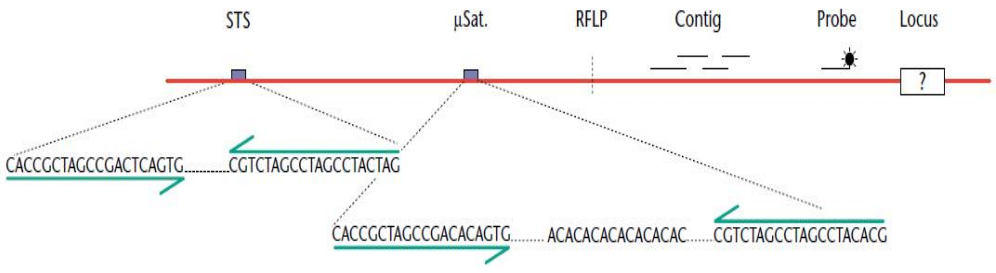
Types of maps:

1. Linkage maps
2. Physical maps

**178 Markers for genome mapping**

Any identifiable feature of the genome can serve as a marker in mapping

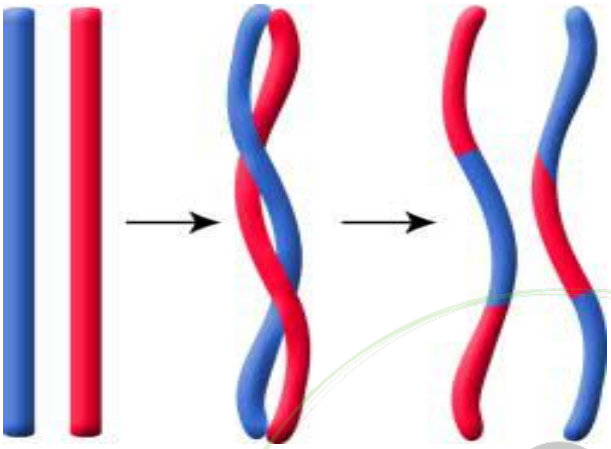
The landmarks on genome map might include short DNA sequences, regulatory sites and genes themselves




179 Genetic linkage mapping

Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes

Figure. Crossing over in chromosomes
Source: Genomic News Network (GNN)



Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes

**180 Physical mapping**

Physical maps give the DNA base pair distances from one landmark to another

**181 Physical versus linkage maps**

Chromosomal maps of bacteria has been obtained by combining the mapping techniques of interrupted mating, recombination mapping, transformation and transduction

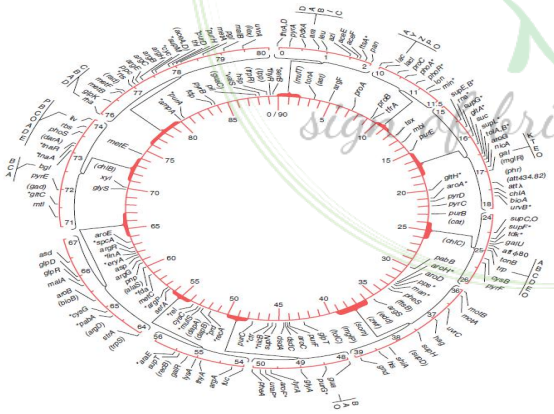


Figure. The 1963 map of E. coli

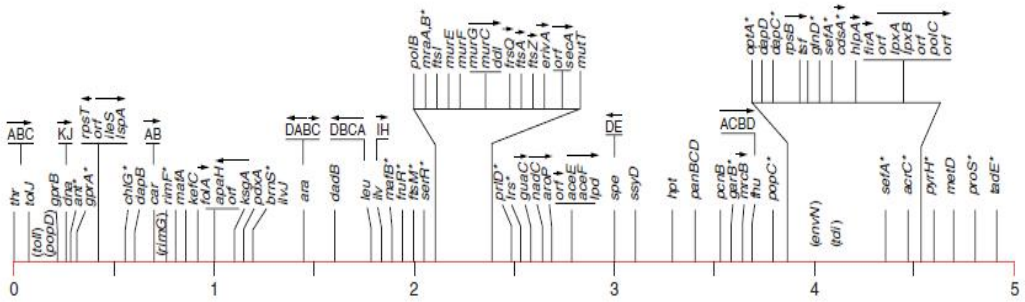



Figure. Linear scale drawing of a 5-minute section of the 100-minute 1990 E. coli linkage map

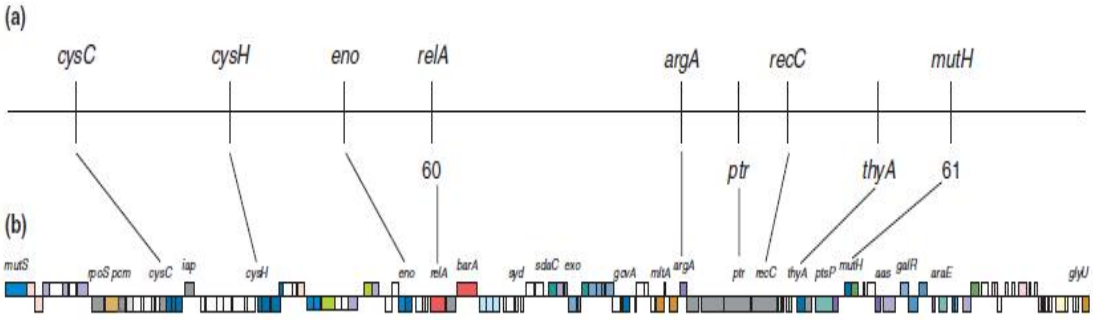


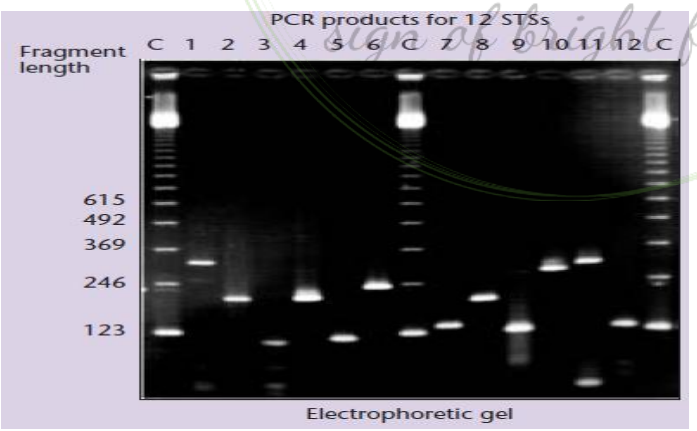

Figure. Correlation of the genetic and physical maps

**183 STS in physical maps**

Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting




**Figure. Examples of a sequence-tagged site (STS)**




STS in physical maps

Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting

**184 SNPs as physical markers**

Single nucleotide polymorphisms (SNPs) are single base-pair positions in genomic DNA at which different sequence alternatives (alleles) exist in a population

SNPs probably are the most important sequence markers for physical mapping of genomes

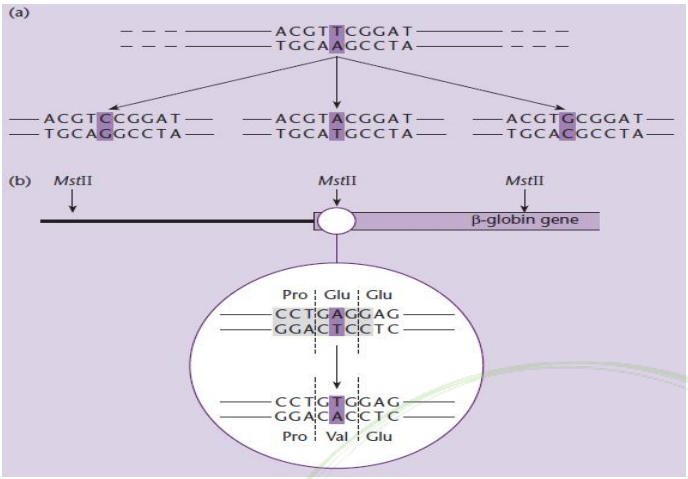



Figure. Examples of single nucleotide polymorphisms

SNPs as physical markers

Different methods to detect SNPs have been developed that based on enzymatic, electrophoretic, solid phase or chromatographic analysis

**185 Polymorphic DNA detection in the absence of sequence information**

Polymorphic DNA can be detected by amplification in the absence of the target DNA sequence information used to generate STSs

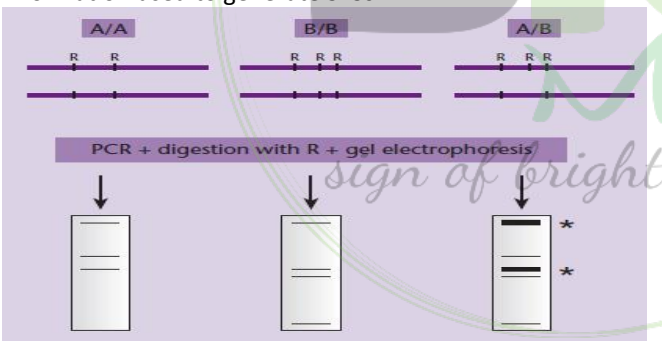
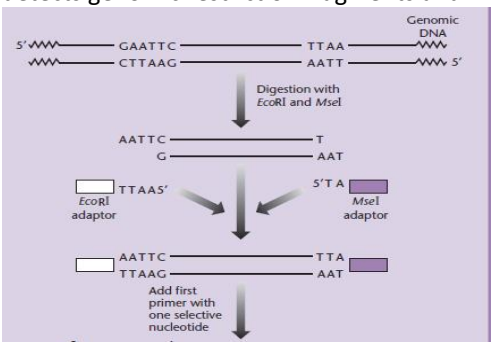
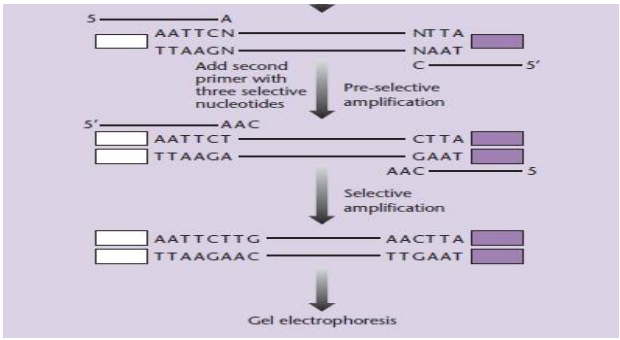



Figure. Generation and visualization of CAPS markers

**186 AFLPs detection in the absence of sequence information**

Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic restriction fragments and in that respect resembles the RFLP technique




AFLPs detection in the absence of sequence information

Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic restriction fragments and in that respect resembles the RFLP technique

**187Fluorescence in situ hybridization (FISH)**

FISH is a cytogenetic method that used fluorescent probes that bind to only those parts of the chromosomes with a high degree of sequence complementarity

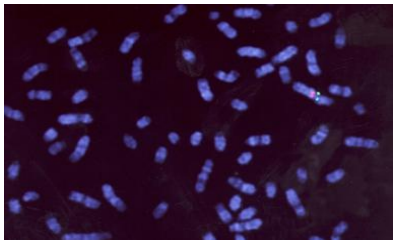


Figure. Dual-color fluorescence in situ hybridization

**188 Radiation hybrid (RH) mapping**

RH mapping used X-ray breakage of chromosomes to determine the distances between DNA markers as well as their order on the chromosome

**189 Happy mapping**

RH mapping have a few problems that arise from the in vivo step. Happy mapping overcomes these limitations, being an entirely in vitro technique

**190 Map integration**

It is essential that the different mapping methods are integrated as each has its own strengths and weaknesses

**191 Sequencing genome**

The process of determining the exact order of nucleotides within a DNA or RNA molecule is called sequencing

In 1970s, DNA sequencing was a very tedious process which involved determining only a few hundred nucleotides at a time

**192 High-throughput sequencing**

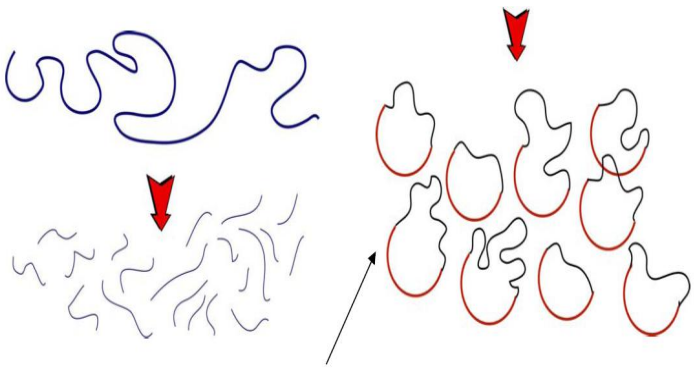
Recent advances in DNA sequencing have make it possible to sequence data very rapidly and at a substantially lower cost

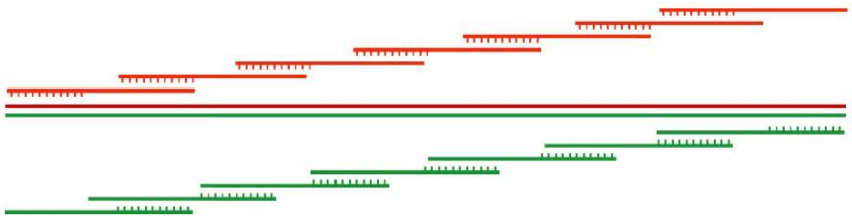
It mainly describe a number of different modern sequencing technologies to sequence genome

**193 High-throughput sequencing**

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Shotgun library

**194 Clone-by-clone sequencing**

In clone-by-clone sequencing, a map of each chromosome of the genome is made before the DNA is split up into fragments for sequencing

**196 Comparative genomics of bacteria**

Genome of bacteria may be of variable in size. For example, it may vary from 0.49 Mb (Nanoarchaeum equitans) to 9.1 Mb (Bradyrhizobium japonicum or Streptomyces)

**197 Comparative genomics of organelles**

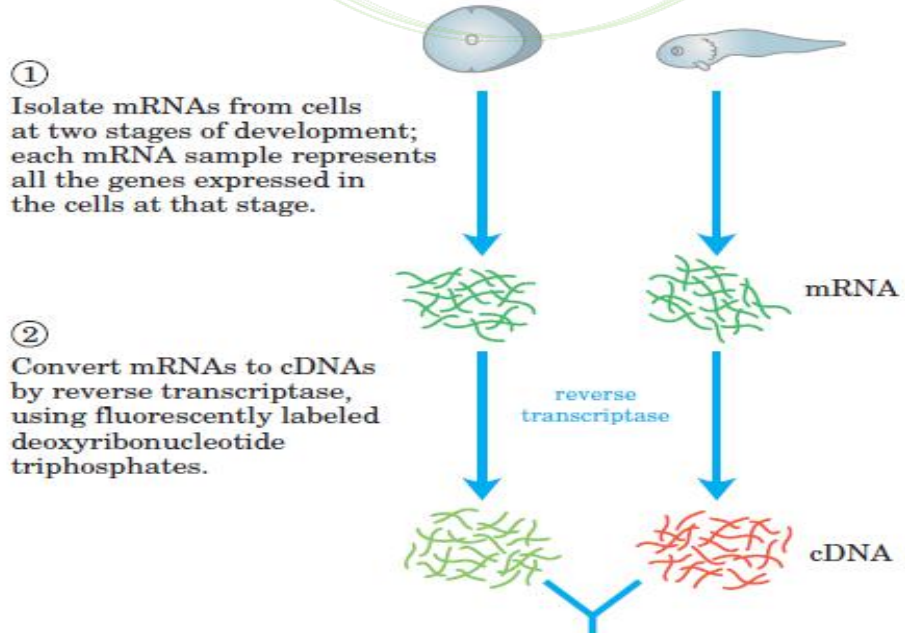
Mitochondrial genomes exhibit an amazing structural diversity
Gene transfer has occurred between mtDNA and nuclear DNA

**198 Comparative genomics of eukaryotes**

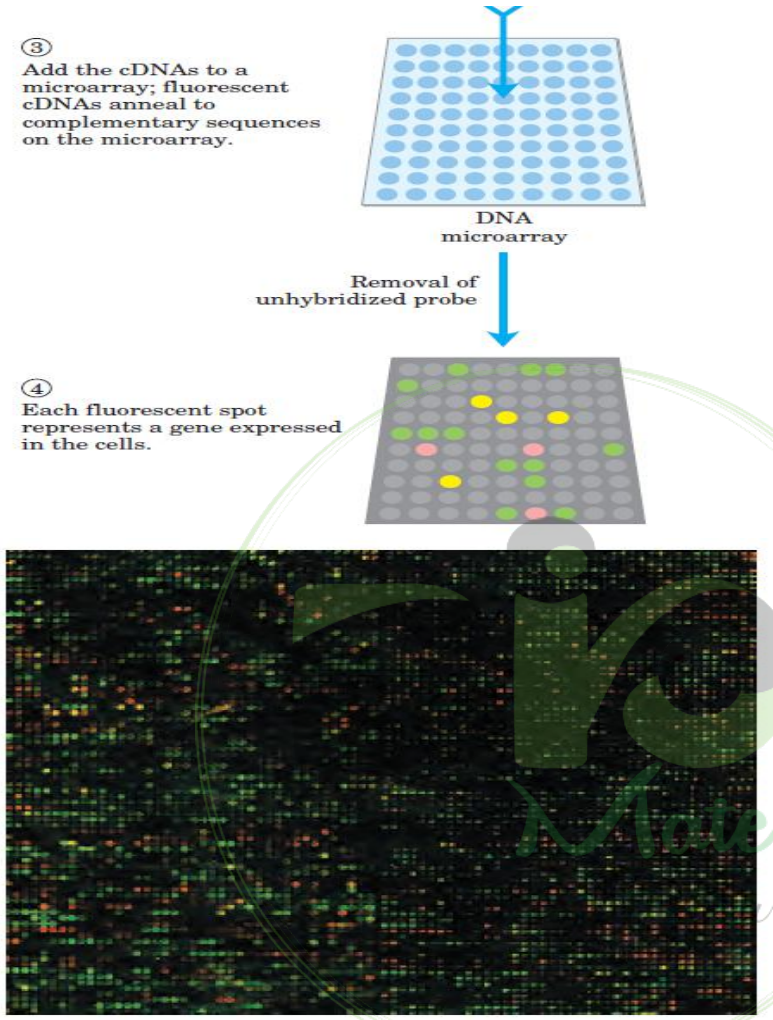
The minimal eukaryotic genome is smaller than many bacterial genomes
Comparative genomics can be used to identify genes and regulatory elements
It gives insight into the evolution of key proteins

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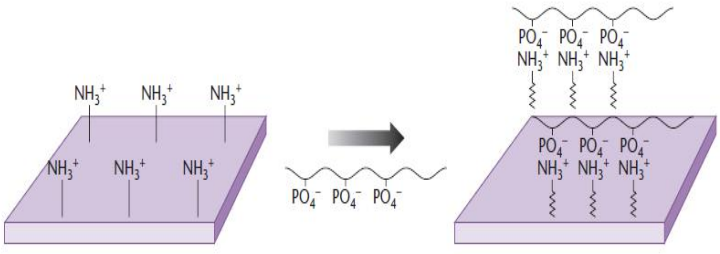

DNA Microarray



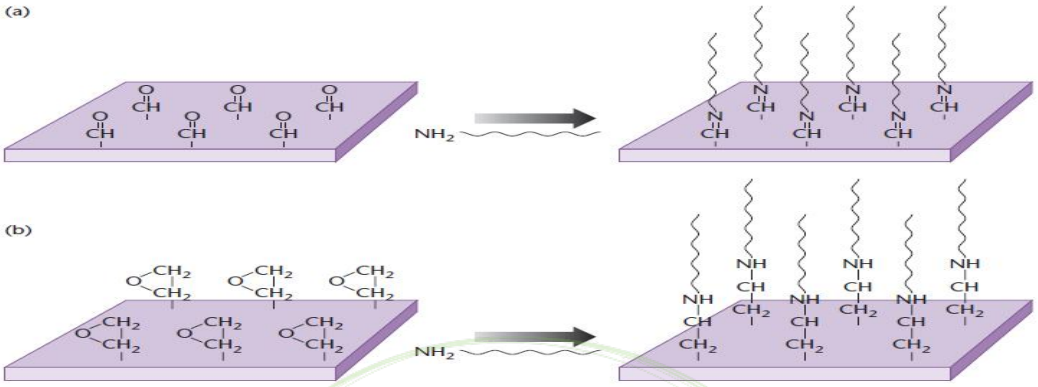

Enlarged image of a DNA Microarray

**200 Spotted DNA arrays**

Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized




DNA binding by electrostatic interactions



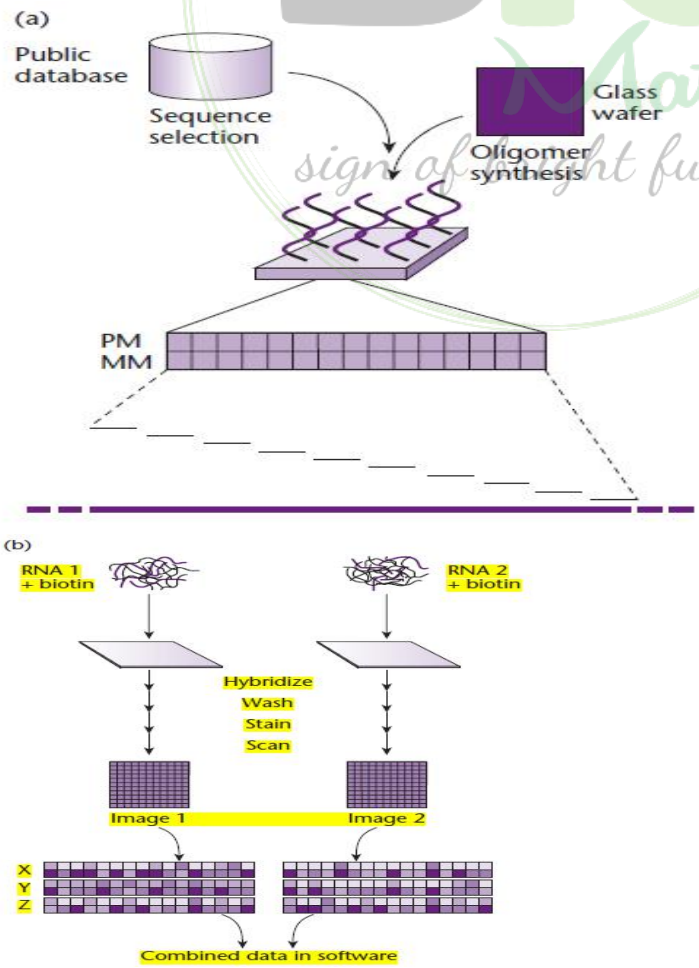

b)- Binding of DNA molecules to epoxy-derivatized surfaces

Spotted DNA arrays

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**201Oligonucleotide chips**

Oligonucleotide chips are manufactured by in situ oligonucleotide synthesis




Principle of chip hybridization

**202 Microbial gene expression analysis**

The predominate application of DNA microarray has been to measure gene expression levels

**203 Profiling in human disease**

Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets

Genotyping

Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms

**204 Phage display**

Phage display is the technology that allows expression of exogenous polypeptides on the surface of phage particles

**205 Screening phage display libraries**

The most common screening method is based on enriching the phage clones with binding affinity for the target by a process called biopanning

**206 Applications of phage display**

Applications of phage display includes determination of protein to protein interactions, to determine enzyme specificity and to generate target specific antibodies

**207 Knock outs**

A gene knock out is a genetic technique in which one of an organism's gene is made inoperative that is simply called as knocked out

In molecular cloning, a knock-in refers to a gene manipulation method that involves the insertion of a protein coding cDNA sequence at a particular locus in an organism's chromosome

**208 siRNA technology**

Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short-term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation

**209 Applications of siRNA**

siRNA) is the method of choice to target specific genes for silencing and has provided immense potential as therapeutic tools